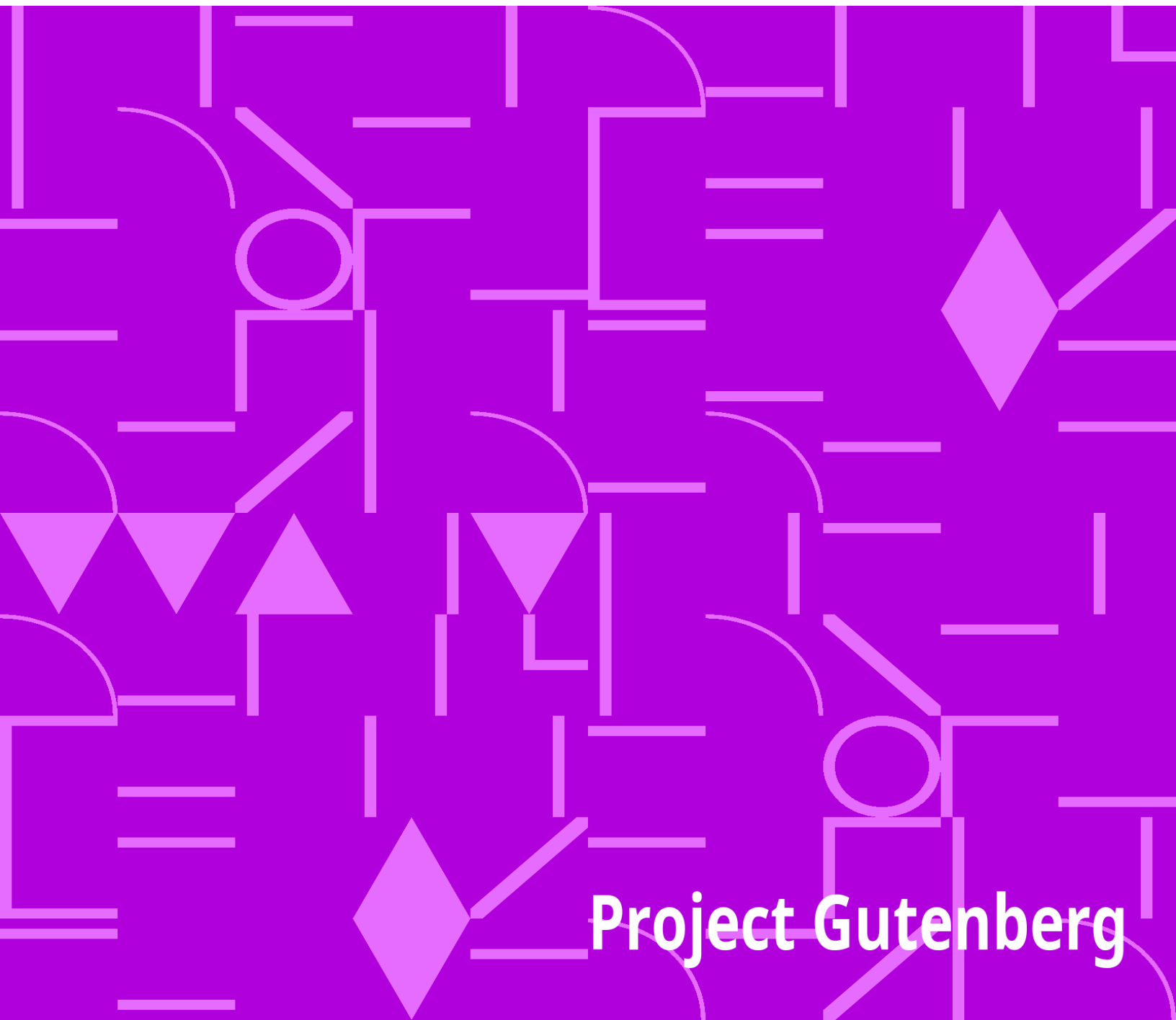


The Birth-Time of the World and Other Scientific Essays

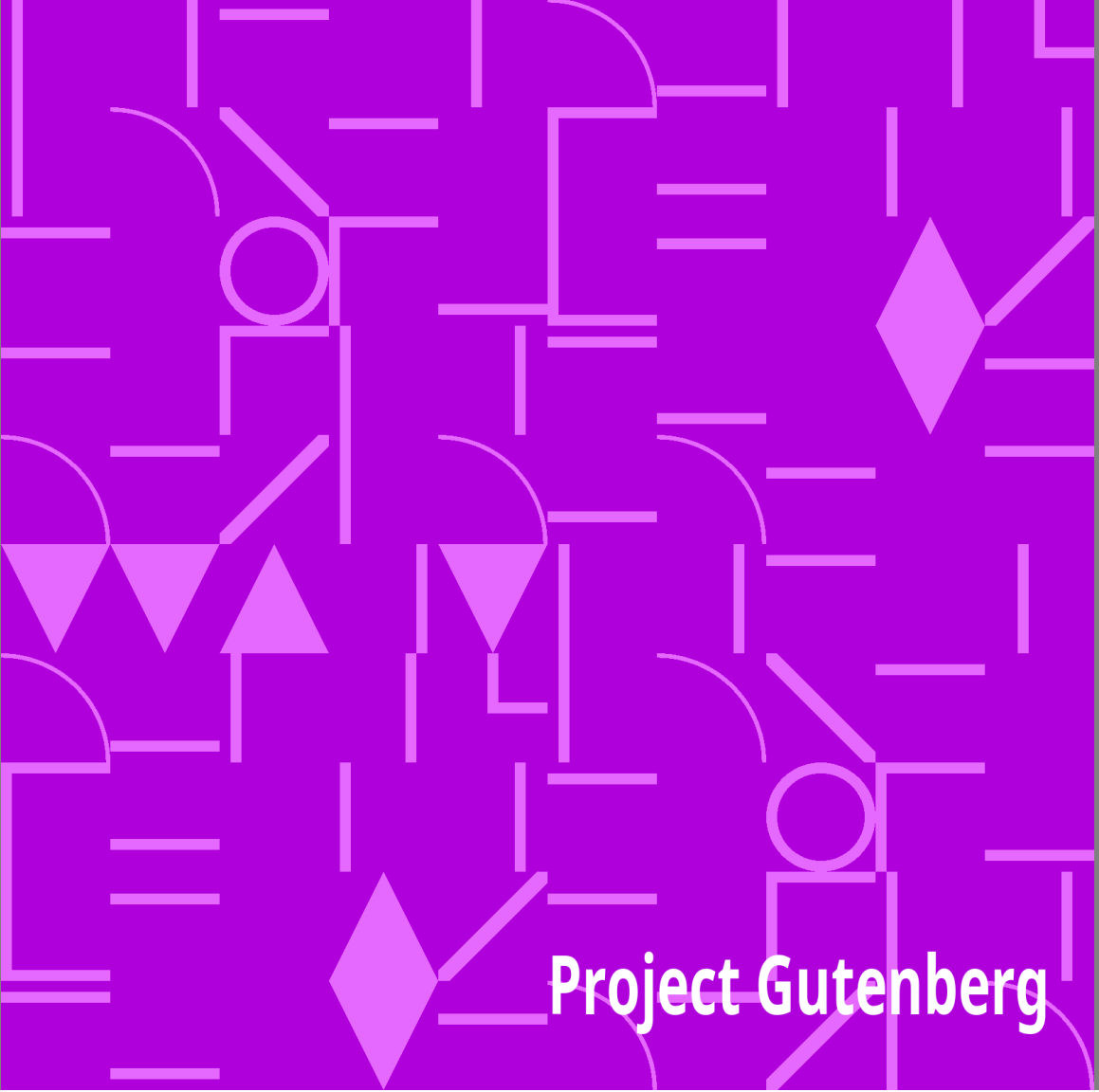
John Joly



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The Birth-Time of the World and Other Scientific Essays

John Joly



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*** START OF THE PROJECT GUTENBERG EBOOK THE
BIRTH-TIME OF THE WORLD AND OTHER SCIENTIFIC
ESSAYS ***

THE BIRTH-TIME OF THE WORLD AND OTHER SCIENTIFIC ESSAYS

by

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Cover

Title page

CONTENTS PAGE

I. THE BIRTH-TIME OF THE WORLD - - - - -	- 1
II. DENUDATION - - - - -	- 30
III. THE ABUNDANCE OF LIFE - - - - -	- 60
IV. THE BRIGHT COLOURS OF ALPINE FLOWERS - - - - -	102
V. MOUNTAIN GENESIS - - - - -	116
VI. ALPINE STRUCTURE - - - - -	146
VII. OTHER MINDS THAN OURS - - - - -	162

VIII. THE LATENT IMAGE - - - - -
202

IX. PLEOCHROIC HALOES - - - - -
214

X. THE USE OF RADIUM IN MEDICINE - - - - -
244

XI. SKATING - - - - -
260

XII. A SPECULATION AS TO A PRE-MATERIAL UNIVERSE -
288

LIST OF ILLUSTRATIONS

PLATE I. LAKE OF LUCERNE, LOOKING WEST FROM
BRUNNEN -
Frontispiece

PLATE II. "UPLIFTED FROM THE SEAS." CLIFFS OF THE
TITLIS,
SWITZERLAND - to face p. 4

PLATE III. AN ALPINE TORRENT AT WORK—VAL D'HERENS,
SWITZERLAND -
to face p. 31

PLATE IV. EARTH PILLARS—VAL D'HERENS - to face p.
34

PLATE V. "SCENES OF DESOLATION." THE WEISSHORN
SEEN FROM BELLA
TOLA, SWITZERLAND - to face p. 40

PLATE VI. ALLUVIAL CONE—NICOLAI THAL, SWITZERLAND.
MORAINE ON
ALETSCHE GLACIER SWITZERLAND - to face p. 50

PLATE VII. IN THE REGION OF THE CROCI; DOLOMITES.
THE ROTHWAND
SEEN FROM MONTE PIANO - to face p. 60

PLATE VIII. FIRS ASSAILING THE HEIGHTS OF THE
MADERANER THAL,
SWITZERLAND - to face p. 73

PLATE IX. LIFE NEAR THE SNOW LINE; THE BOG-COTTON
IN POSSESSION.
NEAR THE TSCHINGEL PASS, SWITZERLAND - to face p.
80

PLATE X. THE JOY OF LIFE. THE AMPEZZO THAL;
DOLOMITES - to face
p. 93

PLATE XI. "PINES SOLEMNLY QUIET." DÜSSISTOCK;
MADERANER THAL - to
face p. 100

PLATE XII. ALPINE FLOWERS IN THE VALLEYS - to face
p. 105

PLATE XIII. ALPINE FLOWERS ON THE HEIGHTS - to
face p. 106

PLATE XIV. MOUNTAIN SOLITUDES; VAL DE ZINAL. FROM
LEFT TO RIGHT
ROTHHORN; BESSO; OBERGABELHORN; MATTERHORN; PIC DE
ZINAL (THROUGH
CLOUD); DENT BLANCHE - to face p. 116

ix

PLATE XV. SECTOR OF THE EARTH RISE OF ISOGEOTHERMS
INTO A DEPOSIT
EVOLVING RADIOACTIVE HEAT - to face p. 118

PLATE XVI. "THE MOUNTAINS COME AND GO." THE DENT
BLANCHE SEEN

FROM THE SASSENEIRE - to face p. 133

PLATE XVII. DIAGRAMMATIC SECTIONS OF THE HIMALAYA
- to face p.
140

PLATE XVIII. RESIDUES OF DENUDATION. THE
MATTERHORN SEEN FROM THE
SUMMIT OF THE ZINAL ROTHORN - to face p. 148

PLATE XIX. THE FOLDED ROCKS OF THE MATTERHORN,
SEEN FROM NEAR
HÖHBALM. SKETCH MADE IN 1906 - to face p. 156

PLATE XX. SCHIAPARELLI'S MAP OF MARS OF 1882, AND
ADDITIONS (IN
RED) OF 1892 - to face p. 166

PLATE XXI. GLOBE OF MARS SHOWING PATH OF IN-
FALLING SATELLITE -
to face p. 188

PLATE XXII. CANALS MAPPED BY LOWELL COMPARED WITH
CANALS FORMED
BY IN-FALLING SATELLITES - to face p. 192

PLATE XXIII. HALOES IN MICA; CO. CARLOW. HALO IN
BIOTITE
CONTAINED IN GRANITE - to face p. 224

PLATE XXIV. RADIUM HALO, MUCH ENLARGED. THORIUM
HALO AND RADIUM
HALO IN MICA - to face p. 228

PLATE XXV. HALO ROUND CAPILLARY IN GLASS TUBE.
HALOES ROUND
TUBULAR PASSAGES IN MICA - to face p. 230

PLATE XXVI. ALETSCHE GLACIER, SWITZERLAND - to face
p. 282

PLATE XXVII. THE MIDDLE ALETSCHE GLACIER JOINING
THE GREAT ALETSCHE
GLACIER. GLACIERS OF THE LAUTERBRUNNEN THAL - to
face p. 285

PLATE XXVIII. PERCHED BLOCK ON THE ALETSCHE
GLACIER. GRANITE
ERRATIC NEAR ROUNDWOOD, CO. WICKLOW; NOW BROKEN UP
AND REMOVED -
to face p. 286

And Fifteen Illustrations in the Text.

X

PREFACE

This volume contains twelve essays written at
various times
during recent years. Many of them are studies
contributed to
Scientific Reviews or delivered as popular
lectures. Some are
expositions of views the scientific basis of which
may be
regarded as established. Others—the greater number
—may be
described as attempting the solution of problems
which cannot be
approached by direct observation.

The essay on The Birth-time of the World is based
on a lecture
delivered before the Royal Dublin Society. The
subject has
attracted much attention within recent years. The
age of the
Earth is, indeed, of primary importance in our
conception of the

longevity of planetary systems. The essay deals with the evidence, derived from the investigation of purely terrestrial phenomena, as to the period which has elapsed since the ocean condensed upon the Earth's surface. Dr. Decker's recent addition to the subject appeared too late for inclusion in it. He finds that the movements (termed isostatic) which geologists recognise as taking place deep in the Earth's crust, indicate an age of the same order of magnitude

xi

as that which is inferred from the statistics of denudative history.[1]

The subject of Denudation naturally arises from the first essay.

In thinking over the method of finding the age of the ocean by the accumulation of sodium therein, I perceived so long ago as

1899, when my first paper was published, that this method

afforded a means of ascertaining the grand total of denudative

work effected on the Earth's surface since the beginning of

geological time; the resulting knowledge in no way involving any

assumption as to the duration of the period comprising the

denudative actions. This idea has been elaborated in various publications since then, both by myself and by others.

"Denudation," while including a survey of the subject generally, is mainly a popular account of this method and its results. It closes with a reference to the fascinating problems presented by the inner nature of sedimentation: a branch of science to which I endeavoured to contribute some years ago.

Mountain Genesis first brings in the subject of the geological intervention of radioactivity. There can, I believe, be no doubt as to the influence of transforming elements upon the developments of the surface features of the Earth; and, if I am right, this source of thermal energy is mainly responsible for that local accumulation of wrinkling which we term mountain chains. The

[1] Bull. Geol. Soc. America, vol. xxvi, March 1915.

xii

paper on Alpine Structure is a reprint from "Radioactivity and Geology," which for the sake of completeness is here included. It is directed to the elucidation of a detail of

mountain genesis: a detail which enters into recent theories of Alpine development.

The weakness of the theory of the "horst" is manifest, however, in many of its other applications; if not, indeed, in all.

The foregoing essays on the physical influences affecting the surface features of the Earth are accompanied by one entitled _The Abundance of Life._ This originated amidst the overwhelming presentation of life which confronts us in the Swiss Alps. The subject is sufficiently inspiring. Can no fundamental reason be given for the urgency and aggressiveness of life? Vitality is an ever-extending phenomenon. It is plain that the great principles which have been enunciated in explanation of the origin of species do not really touch the problem. In the essay—which is an early one (1890)—the explanation of the whole great matter is sought—and as I believe found—in the attitude of the organism towards energy external to it; an attitude which results in its evasion of the retardative and dissipatory effects which prevail in lifeless dynamic systems of all kinds.

Other Minds than Ours? attempts a solution of the vexed question

of the origin of the Martian "canals." The essay is an abridgment of two popular lectures on the subject. I had previously written an account of my views which carried the enquiry as far as it was in

xiii

my power to go. This paper appeared in the "Transactions of the Royal Dublin Society, 1897." The theory put forward is a purely physical one, and, if justified, the view that intelligent beings exist in Mars derives no support from his visible surface features; but is, in fact, confronted with fresh difficulties.

Pleochroic Haloes is a popular exposition of an inconspicuous but very beautiful phenomenon of the rocks. Minute darkened spheres—a microscopic detail—appear everywhere in certain of the rock minerals. What are they? The discoveries of recent radioactive research—chiefly due to Rutherford—give the answer. The measurements applied to the little objects render the explanation beyond question. They turn out to be a quite extraordinary record of radioactive energy; a record accumulated since remote geological times, and assuring us, indirectly, of

the stability
of the chemical elements in general since the
beginning of the
world. This assurance is, without proof, often
assumed in our
views on the geological history of the Globe.

Skating is a discourse, with a recent addition
supporting the
original thesis. It is an illustration of a common
experience—the
explanation of an unimportant action involving
principles the
most influential considered as a part of Nature's
resources.

The address on _The Latent Image_ deals with a
subject which had
been approached by various writers before the time
of my essay;
but, so far as I know, an explanation

xiv

based on the facts of photo-electricity had not
been attempted.
Students of this subject will notice that the
views expressed are
similar to those subsequently put forward by
Lenard and Saeland
in explanation of phosphorescence. The whole
matter is of more
practical importance than appears at first sight,
for the
photoelectric nature of the effects involved in
the radiative
treatment of many cruel diseases seems to be
beyond doubt.

It was in connection with photo-electric science that I was led to take an interest in the application of radioactivity in medicine. The lecture on _The Use of Radium in Medicine_ deals with this subject. Towards the conclusion of this essay reference will be found to a practical outcome of such studies which, by improving on the methods, and facilitating the application, of radioactive treatment, has, in the hands of skilled medical men, already resulted in the alleviation of suffering.

Leaving out much which might well appear in a prefatory notice, a word should yet be added respecting the illustrations of scenery.

They are a small selection from a considerable number of photographs taken during my summer wanderings in the Alps in company with Henry H. Dixon. An exception is Plate X, which is by the late Dr. Edward Stapleton. From what has been said above, it will be gathered that these illustrations are fitly included among pages which owe so much to Alpine inspiration. They illustrate the

xv

subjects dealt with, and, it is to be hoped, they will in some

cases recall to the reader scenes which have in
past times
influenced his thoughts in the same manner; scenes
which in their
endless perspective seem to reduce to their proper
insignificance
the lesser things of life.

My thanks are due to Mr. John Murray for kindly
consenting to the
reissue of the essay on _The Birth-time of the
World_ from the
pages of _Science Progress_; to Messrs. Constable
& Co. for leave
to reprint _Pleochroic Haloes_ from _Bedrock_, and
also to make some
extracts from _Radioactivity and Geology_; and to
the Council of
the Royal Dublin Society for permission to
republish certain
papers from the Proceedings of the Society.

_Iveagh Geological Laboratory, Trinity College,
Dublin._

July, 1915.

xvi

THE BIRTH-TIME OF THE WORLD [1]

LONG ago Lucretius wrote: "For lack of power to
solve the
question troubles the mind with doubts, whether
there was ever a
birth-time of the world and whether likewise there
is to be any
end." "And if" (he says in answer) "there was no
birth-time of

earth and heaven and they have been from everlasting, why before the Theban war and the destruction of Troy have not other poets as well sung other themes? Whither have so many deeds of men so often passed away, why live they nowhere embodied in lasting records of fame? The truth methinks is that the sum has but a recent date, and the nature of the world is new and has but lately had its commencement." [2]

Thus spake Lucretius nearly 2,000 years ago. Since then we have attained another standpoint and found very different limitations.

To Lucretius the world commenced with man, and the answer he would give to his questions was in accord with his philosophy: he would date the birth-time of the world from the time when

[1] A lecture delivered before the Royal Dublin Society, February 6th, 1914. *_Science Progress_*, vol. ix., p. 37

[2] *_De Rerum Natura_*, translated by H. A. J. Munro (Cambridge, 1886).

1

poets first sang upon the earth. Modern Science has along with the theory that the Earth dated its beginning with the advent of

man, swept utterly away this beautiful imagining.
We can, indeed,
find no beginning of the world. We trace back
events and come to
barriers which close our vista—barriers which, for
all we know,
may for ever close it. They stand like the gates
of ivory and of
horn; portals from which only dreams proceed; and
Science cannot
as yet say of this or that dream if it proceeds
from the gate of
horn or from that of ivory.

In short, of the Earth's origin we have no certain
knowledge; nor
can we assign any date to it. Possibly its
formation was an event
so gradual that the beginning was spread over
immense periods. We
can only trace the history back to certain events
which may with
considerable certainty be regarded as ushering in
our geological
era.

Notwithstanding our limitations, the date of the
birth-time of
our geological era is the most important date in
Science. For in
taking into our minds the spacious history of the
universe, the
world's age must play the part of time-unit upon
which all our
conceptions depend. If we date the geological
history of the
Earth by thousands of years, as did our
forerunners, we must

shape our ideas of planetary time accordingly; and the duration of our solar system, and of the heavens, becomes comparable with that of the dynasties of ancient nations. If by millions of years, the sun and stars are proportionately venerable. If by hundreds or thousands of millions of

2

years the human mind must consent to correspondingly vast epochs for the duration of material changes. The geological age plays the same part in our views of the duration of the universe as the Earth's orbital radius does in our views of the immensity of space. Lucretius knew nothing of our time-unit: his unit was the life of a man. So also he knew nothing of our space-unit, and he marvels that so small a body as the sun can shed so much, heat and light upon the Earth.

A study of the rocks shows us that the world was not always what it now is and long has been. We live in an epoch of denudation. The rains and frosts disintegrate the hills; and the rivers roll to the sea the finely divided particles into which they have been resolved; as well as the salts which have been leached from them.

The sediments collect near the coasts of the continents; the dissolved matter mingles with the general ocean. The geologist has measured and mapped these deposits and traced them back into the past, layer by layer. He finds them ever the same; sandstones, slates, limestones, etc. But one thing is not the same. _Life_ grows ever less diversified in character as the sediments are traced downwards. Mammals and birds, reptiles, amphibians, fishes, die out successively in the past; and barren sediments ultimately succeed, leaving the first beginnings of life undecipherable. Beneath these barren sediments lie rocks collectively differing in character from those above: mainly volcanic or poured out from fissures in

3

the early crust of the Earth. Sediments are scarce among these materials.[1]

There can be little doubt that in this underlying floor of igneous and metamorphic rocks we have reached those surface materials of the earth which existed before the long epoch of sedimentation began, and before the seas came into being. They

formed the floor of a vaporised ocean upon which the waters condensed here and there from the hot and heavy atmosphere. Such were the probable conditions which preceded the birth-time of the ocean and of our era of life and its evolution.

It is from this epoch we date our geological age. Our next purpose is to consider how long ago, measured in years, that birth-time was.

That the geological age of the Earth is very great appears from what we have already reviewed. The sediments of the past are many miles in collective thickness: yet the feeble silt of the rivers built them all from base to summit. They have been uplifted from the seas and piled into mountains by movements so slow that during all the time man has been upon the Earth but little change would have been visible. The mountains have again been worn down into the ocean by denudation and again younger mountains built out of their redeposited materials. The contemplation of such vast events

[1] For a description of these early rocks, see especially the monograph of Van Hise and Leith on the pre-

Cambrian Geology of
North America (Bulletin 360, U.S. Geol. Survey).

4

prepares our minds to accept many scores of millions of years or hundreds of millions of years, if such be yielded by our calculations.

THE AGE AS INFERRED FROM THE THICKNESS OF THE SEDIMENTS

The earliest recognised method of arriving at an estimate of the Earth's geological age is based upon the measurement of the collective sediments of geological periods. The method has undergone much revision from time to time. Let us briefly review it on the latest data.

The method consists in measuring the depths of all the successive sedimentary deposits where these are best developed. We go all over the explored world, recognising the successive deposits by their fossils and by their stratigraphical relations, measuring their thickness and selecting as part of the data required those beds which we believe to most completely represent each formation. The total of these measurements would tell us the age of the Earth if their tale was indeed complete,

and if we knew
the average rate at which they have been
deposited. We soon,
however, find difficulties in arriving at the
quantities we
require. Thus it is not easy to measure the real
thickness of a
deposit. It may be folded back upon itself, and so
we may measure
it twice over. We may exaggerate its thickness by
measuring it
not quite straight across the bedding or by
unwittingly including
volcanic materials. On the other hand, there

5

may be deposits which are inaccessible to us; or,
again, an
entire absence of deposits; either because not
laid down in the
areas we examine, or, if laid down, again washed
into the sea.
These sources of error in part neutralise one
another. Some make
our resulting age too long, others make it out too
short. But we
do not know if a balance of error does not still
remain. Here,
however, is a table of deposits which summarises a
great deal of
our knowledge of the thickness of the
stratigraphical
accumulations. It is due to Sollas.[1]

Feet.

Recent and Pleistocene - - 4,000

Pliocene	- -	13,000
Miocene	- -	14,000
Oligocene	- -	2,000
Eocene	- -	<u>20,000</u>
		63,000
Upper Cretaceous	- -	24,000
Lower Cretaceous	- -	20,000
Jurassic	- -	8,000
Trias	- -	<u>7,000</u>
		69,000
Permian	- -	2,000
Carboniferous	- -	29,000
Devonian	- -	<u>22,000</u>
		63,000
Silurian	- -	15,000
Ordovician	- -	17,000
Cambrian	- -	<u>6,000</u>
		58,000
Algonkian—Keeweenawan	- -	50,000
Algonkian—Animikian	- -	14,000
Algonkian—Huronian	- -	<u>18,000</u>
		82,000
Archæan	- -	?
Total	- -	<u>335,000</u> feet.

[1] Address to the Geol. Soc. of London, 1509.

6

In the next place we require to know the average rate at which these rocks were laid down. This is really the

weakest link in the chain. The most diverse results have been arrived at, which space does not permit us to consider. The value required is most difficult to determine, for it is different for the different classes of material, and varies from river to river according to the conditions of discharge to the sea. We may probably take it as between two and six inches in a century.

Now the total depth of the sediments as we see is about 335,000 feet (or 64 miles), and if we take the rate of collecting as three inches in a hundred years we get the time for all to collect as 134 millions of years. If the rate be four inches, the time is 100 millions of years, which is the figure Geikie favoured, although his result was based on somewhat different data. Sollas most recently finds 80 millions of years.[1]

THE AGE AS INFERRED FROM THE MASS OF THE SEDIMENTS

In the above method we obtain our result by the measurement of the linear dimensions of the sediments. These measurements, as we have seen, are difficult to arrive at. We may, however, proceed by measurements of the mass of the sediments, and then the method

becomes more definite. The new method is pursued as follows:

[1] Geikie, Text Book of Geology (Macmillan, 1903), vol. i., p. 73, et seq. Sollas, loc. cit. Joly, Radioactivity and Geology (Constable, 1909), and Phil. Mag., Sept. 1911.

7

The total mass of the sediments formed since denudation began may be ascertained with comparative accuracy by a study of the chemical composition of the waters of the ocean. The salts in the ocean are undoubtedly derived from the rocks; increasing age by age as the latter are degraded from their original character under the action of the weather, etc., and converted to the sedimentary form. By comparing the average chemical composition of these two classes of material—the primary or igneous rocks and the sedimentary—it is easy to arrive at a knowledge of how much of this or that constituent was given to the ocean by each ton of primary rock which was denuded to the sedimentary form. This, however, will not assist us to our object unless the ocean has retained the salts shed into it. It has not generally done so. In the case of every substance but one the ocean

continually gives
up again more or less of the salts supplied to it
by the rivers.

The one exception is the element sodium. The great
solubility of
its salts has protected it from abstraction, and
it has gone on
collecting during geological time, practically in
its entirety.

This gives us the clue to the denudative history
of the
Earth.[1]

The process is now simple. We estimate by chemical
examination of
igneous and sedimentary rocks the amount of sodium
which has been
supplied to the ocean per ton of sediment produced
by denudation.

We also calculate

[1] _Trans. R.D.S._, May, 1899.

8

the amount of sodium contained in the ocean. We
divide the one
into the other (stated, of course, in the same
units of mass),
and the quotient gives us the number of tons of
sediment. The
most recent estimate of the sediments made in this
manner affords

56×10^{16} tonnes.[1]

Now we are assured that all this sediment was
transported by the
rivers to the sea during geological time. Thus it
follows that,

if we can estimate the average annual rate of the river supply of sediments to the ocean over the past, we can calculate the required age. The land surface is at present largely covered with the sedimentary rocks themselves. Sediment derived from these rocks must be regarded as, for the most part, purely cyclical; that is, circulating from the sea to the land and back again. It does not go to increase the great body of detrital deposits. We cannot, therefore, take the present river supply of sediment as representing that obtaining over the long past. If the land was all covered still with primary rocks we might do so. It has been estimated that about 25 per cent. of the existing continental area is covered with archæan and igneous rocks, the remainder being sediments.[2] On this estimate we may find valuable

[1] Clarke, A Preliminary Study of Chemical Denudation (Washington, 1910). My own estimate in 1899 (loc. cit.) made as a test of yet another method of finding the age, showed that the sediments may be taken as sufficient to form a layer 1.1 mile deep if spread uniformly over the continents; and

would amount to
64 x 10¹⁸ tons.

[2] Van Tillo, _Comptes Rendues_ (Paris), vol. cxiv., 1892.

9

major and minor limits to the geological age. If we take 25 per cent. only of the present river supply of sediment, we evidently fix a major limit to the age, for it is certain that over the past there must have been on the average a faster supply. If we take the entire river supply, on similar reasoning we have what is undoubtedly a minor limit to the age.

The river supply of detrital sediment has not been very extensively investigated, although the quantities involved may be found with comparative ease and accuracy. The following table embodies the results obtained for some of the leading rivers.[1]

of sediment water weight.	Mean annual	Total annual	Ratio
	discharge in	sediment in	
	cubic feet	thousands	to
	per second.	of tons.	by
Potomac -	20,160	5,557	1 :

3.575			
Mississippi -	610,000	406,250	1 :
1,500			
Rio Grande -	1,700	3,830	1 :
291			
Uruguay -	150,000	14,782	1 :
10,000			
Rhone -	65,850	36,000	1 :
1,775			
Po -	62,200	67,000	1 :
900			
Danube -	315,200	108,000	1 :
2,880			
Nile -	113,000	54,000	1 :
2,050			
Irrawaddy -	475,000	291,430	1 :
1,610			
Mean -	201,468	109,650	1 :
2,731			

We see that the ratio of the weight of water to the

[1] Russell, River Development (John Murray, 1888).

10

weight of transported sediment in six out of the nine rivers does not vary widely. The mean is 2,730 to 1. But this is not the required average. The water-discharge of each river has to be taken into account. If we ascribe to the ratio given for each river the weight proper to the amount of water it

discharges, the proportion of weight of water to weight of sediment, for the whole quantity of water involved, comes out as 2,520 to 1.

Now if this proportion holds for all the rivers of the world—which collectively discharge about 27×10^{12} tonnes of water per annum—the river-born detritus is 1.07×10^{10} tonnes. To this an addition of 11 per cent. has to be made for silt pushed along the river-bed.[1] On these figures the minor limit to the age comes out as 47 millions of years, and the major limit as 188 millions. We are here going on rather deficient estimates, the rivers involved representing only some 6 per cent. of the total river supply of water to the ocean. But the result is probably not very far out.

We may arrive at a probable age lying between the major and minor limits. If, first, we take the arithmetic mean of these limits, we get 117 millions of years. Now this is almost certainly excessive, for we here assume that the rate of covering of the primary rocks by sediments was uniform. It would not be so, however, for the rate of supply of original

sediment must have
been continually diminishing

[1] According to observations made on the
Mississippi (Russell,
loc. cit.).

11

during geological time, and hence we may assume
that the rate of
advance of the sediments on the primary rocks has
also been
diminishing. Now we may probably take, as a fair
assumption, that
the sediment-covered area was at any instant
increasing at a rate
proportionate to the rate of supply of sediment;
that is, to the
area of primary rocks then exposed. On this
assumption the age is
found to be 87 millions of years.

THE AGE BY THE SODIUM OF THE OCEAN

I have next to lay before you a quite different
method. I have
already touched upon the chemistry of the ocean,
and on the
remarkable fact that the sodium contained in it
has been
preserved, practically, in its entirety from the
beginning of
geological time.

That the sea is one of the most beautiful and
magnificent sights
in Nature, all admit. But, I think, to those who
know its story

its beauty and magnificence are ten-fold increased. Its saltiness is due to no magic mill. It is the dissolved rocks of the Earth which give it at once its brine, its strength, and its buoyancy. The rivers which we say flow with "fresh" water to the sea nevertheless contain those traces of salt which, collected over the long ages, occasion the saltiness of the ocean. Each gallon of river water contributes to the final result; and this has been going on since the beginning of our era. The mighty total of the rivers is 6,500 cubic miles of water in the year!

12

There is little doubt that the primeval ocean was in the condition of a fresh-water lake. It can be shown that a primitive and more rapid solution of the original crust of the Earth by the slowly cooling ocean would have given rise to relatively small salinity. The fact is, the quantity of salts in the ocean is enormous. We are only now concerned with the sodium; but if we could extract all the rock-salt (the chloride of sodium) from the ocean we should have enough to cover the entire dry land of the Earth to a depth of 400 feet. It is this gigantic quantity which

is going to enter into our estimate of the Earth's age. The calculated mass of sodium contained in this rock-salt is 14,130 million million tonnes.

If now we can determine the rate at which the rivers supply sodium to the ocean, we can determine the age.[1] As the result of many thousands of river analyses, the total amount of sodium annually discharged to the ocean

[1] _Trans. R.D.S._, 1899. A paper by Edmund Halley, the astronomer, in the _Philosophical Transactions of the Royal Society_ for 1715, contains a suggestion for finding the age of the world by the following procedure. He proposes to make observations on the saltiness of the seas and ocean at intervals of one or more centuries, and from the increment of saltiness arrive at their age. The measurements, as a matter of fact, are impracticable. The salinity would only gain (if all remained in solution) one millionth part in Too years; and, of course, the continuous rejection of salts by the ocean would invalidate the method. The last objection also invalidates the calculation by T. Mellard Reade (_Proc. Liverpool Geol. Soc._, 1876) of a minor limit

to the age by the calcium sulphate in the ocean. Both papers were quite unknown to me when working out my method. Halley's paper was, I think, only brought to light in 1908.

13

by all the rivers of the world is found to be probably not far from 175 million tonnes.[1] Dividing this into the mass of oceanic sodium we get the age as 80.7 millions of years. Certain corrections have to be applied to this figure which result in raising it to a little over 90 millions of years. Sollas, as the result of a careful review of the data, gets the age as between 80 and 150 millions of years. My own result[2] was between 80 and 90 millions of years; but I subsequently found that upon certain extreme assumptions a maximum age might be arrived at of 105 millions of years.[3] Clarke regards the 80.7 millions of years as certainly a maximum in the light of certain calculations by Becker.[4]

The order of magnitude of these results cannot be shaken unless on the assumption that there is something entirely misleading in the existing rate of solvent denudation. On the

strength of the
results of another and

[1] F. W. Clarke, A Preliminary Study of Chemical Denudation
(Smithsonian Miscellaneous Collections, 1910).

[2] Loc. cit.

[3] "The Circulation of Salt and Geological Time"
(Geol. Mag.,
1901, p. 350).

[4] Becker (loc. cit.), assuming that the exposed igneous and archæan rocks alone are responsible for the supply of sodium to the ocean, arrives at 74 millions of years as the geological age.

This matter was discussed by me formerly (Trans. R.D.S., 1899, pp. 54 et seq.). The assumption made is, I believe, inadmissible.

It is not supported by river analyses, or by the chemical character of residual soils from sedimentary rocks. There may be some convergence in the rate of solvent denudation, but—as I think on the evidence—in our time unimportant.

14

entirely different method of approaching the question of the Earth's age (which shall be presently referred to), it has been contended that it is too low. It is even asserted that it is from

nine to fourteen times too low. We have then to consider whether such an enormous error can enter into the method. The measurements involved cannot be seriously impugned. Corrections for possible errors applied to the quantities entering into this method have been considered by various writers. My own original corrections have been generally confirmed. I think the only point left open for discussion is the principle of uniformitarianism involved in this method and in the methods previously discussed.

In order to appreciate the force of the evidence for uniformity in the geological history of the Earth, it is, of course, necessary to possess some acquaintance with geological science. Some of the most eminent geologists, among whom Lyell and Geikie[1] may be mentioned, have upheld the doctrine of uniformity. It must here suffice to dwell upon a few points having special reference to the matter under discussion.

The mere extent of the land surface does not, within limits, affect the question of the rate of denudation. This arises from the fact that the rain supply is quite insufficient to denude the

whole existing land surface. About 30 per cent. of it does not, in fact, drain to the

[1] See especially Geikie's Address to Sect. C., Brit. Assoc. Rep., 1399.

15

ocean. If the continents become invaded by a great transgression of the ocean, this "rainless" area diminishes: and the denuded area advances inwards without diminution. If the ocean recedes from the present strand lines, the "rainless" area advances outwards, but, the rain supply being sensibly constant, no change in the river supply of salts is to be expected.

Age-long submergence of the entire land, or of any very large proportion of what now exists, is negatived by the continuous sequence of vast areas of sediment in every geologic age from the earliest times. Now sediment-receiving areas always are but a small fraction of those exposed areas whence the sediments are supplied.[1] Hence in the continuous records of the sediments we have assurance of the continuous exposure of the continents above the ocean surface. The doctrine of the permanency of the

continents has in its main features been accepted by the most eminent authorities. As to the actual amount of land which was exposed during past times to denudative effects, no data exist to show it was very different from what is now exposed. It has been estimated that the average area of the North American continent over geologic time was about eight-tenths of its existing area.[2] Restorations of other continents, so far as they have been attempted, would not

[1] On the strength of the Mississippi measurements about 1 to 18 (Magee, *Am. Jour. of Sc.*, 1892, p. 188).

[2] Schuchert, *Bull. Geol. Soc. Am.*, vol. xx., 1910.

16

suggest any more serious divergency one way or the other.

That climate in the oceans and upon the land was throughout much as it is now, the continuous chain of teeming life and the sensitive temperature limits of protoplasmic existence are sufficient evidence.[1] The influence at once of climate and of elevation of the land may be appraised at their true value by the ascertained facts of solvent denudation, as the

following table
shows.

elevation.	Tonnes removed in solution per square mile per annum.	Mean Metres.
North America -	79	700
South America -	50	650
Europe -	100	300
Asia -	84	950
Africa -	44	650

In this table the estimated number of tonnes of matter in solution, which for every square mile of area the rivers convey to the ocean in one year, is given in the first column. These results are compiled by Clarke from a very large number of analyses of river waters. The second column of the table gives the mean heights in metres above sea level of the several continents, as cited by Arrhenius.[2]

Of all the denudation results given in the table, those relating to North America and to Europe are far the

[1] See also Poulton, Address to Sect. D., Brit. Assoc. Rep., 1896.

[2] _Lehybuch dev Kosmischen Physik_, vol. i., p. 347.

most reliable. Indeed these may be described as highly reliable, being founded on some thousands of analyses, many of which have been systematically pursued through every season of the year.

These show that Europe with a mean altitude of less than half that of North America sheds to the ocean 25 per cent. more salts.

A result which is to be expected when the more important factors of solvent denudation are given intelligent consideration and we discriminate between conditions favouring solvent and detrital

denudation respectively: conditions in many cases antagonistic.[1] Hence if it is true, as has been stated, that we

now live in a period of exceptionally high continental elevation,

we must infer that the average supply of salts to the ocean by

the rivers of the world is less than over the long past, and

that, therefore, our estimate of the age of the Earth as already given is excessive.

There is, however, one condition which will operate to unduly

diminish our estimate of geologic time, and it is a condition

which may possibly obtain at the present time. If the land is, on

the whole, now sinking relatively to the ocean level, the

denudation area tends, as we have seen, to move inwards. It will thus encroach upon regions which have not for long periods drained to the ocean. On such areas there is an accumulation of soluble salts which the deficient rivers have not been able to carry to the ocean. Thus the salt content of certain of

[1] See the essay on Denudation.

18

the rivers draining to the ocean will be influenced not only by present denudative effects, but also by the stored results of past effects. Certain rivers appear to reveal this unduly increased salt supply those which flow through comparatively arid areas. However, the flowoff of such tributaries is relatively small and the final effects on the great rivers apparently unimportant—a result which might have been anticipated when the extremely slow rate of the land movements is taken into account.

The difficulty of effecting any reconciliation of the methods already described and that now to be given increases the interest both of the former and the latter.

THE AGE BY RADIOACTIVE TRANSFORMATIONS

Rutherford suggested in 1905 that as helium was continually being evolved at a uniform rate by radioactive substances (in the form of the alpha rays) a determination of the age of minerals containing the radioactive elements might be made by measurements of the amount of the stored helium and of the radioactive elements giving rise to it, The parent radioactive substances are—according to present knowledge—uranium and thorium. An estimate of the amounts of these elements present enables the rate of production of the helium to be calculated. Rutherford shortly afterwards found by this method an age of 240 millions of years for a radioactive mineral of presumably remote age. Strutt, who carried

19

his measurements to a wonderful degree of refinement, found the following ages for mineral substances originating in different geological ages:

Oligocene -	8.4 millions of years.
Eocene -	31 millions of years.
Lower Carboniferous -	150 millions of years.
Archæan -	750 millions of years.

Periods of time much less than, and very inconsistent with, these were also found. The lower results are, however, easily explained if we assume that the helium—which is a gas under prevailing conditions—escapes in many cases slowly from the mineral.

Another product of radioactive origin is lead. The suggestion that this substance might be made available to determine the age of the Earth also originated with Rutherford. We are at least assured that this element cannot escape by gaseous diffusion from the minerals. Boltwood's results on the amount of lead contained in minerals of various ages, taken in conjunction with the amount of uranium or parent substance present, afforded ages rising to 1,640 millions of years for archæan and 1,200 millions for Algonkian time. Becker, applying the same method, obtained results rising to quite incredible periods: from 1,671 to 11,470 millions of years. Becker maintained that original lead rendered the determinations indefinite. The more recent results of Mr. A. Holmes support the conclusion that "original" lead may be present and may completely falsify results derived

from minerals of low radioactivity in which the derived lead would be small in amount. By rejecting such results as appeared to be of this character, he arrives at 370 millions of years as the age of the Devonian.

I must now describe a very recent method of estimating the age of the Earth. There are, in certain rock-forming minerals, colour-changes set up by radioactive causes. The minute and curious marks so produced are known as haloes; for they surround, in ringlike forms, minute particles of included substances which contain radioactive elements. It is now well known how these haloes are formed. The particle in the centre of the halo contains uranium or thorium, and, necessarily, along with the parent substance, the various elements derived from it. In the process of transformation giving rise to these several derived substances, atoms of helium—the alpha rays—projected with great velocity into the surrounding mineral, occasion the colour changes referred to. These changes are limited to the distance to which the alpha rays penetrate; hence the halo is a spherical volume surrounding the central substance.[1]

The time required to form a halo could be found if on the one hand we could ascertain the number of alpha rays ejected from the nucleus of the halo in, say, one year, and, on the other, if we determined by experiment just how many alpha rays were required to produce the same

[1] Phil. Mag., March, 1907 and February, 1910; also Bedrock, January, 1913. See Pleochroic Haloes in this volume.

21

amount of colour alteration as we perceive to extend around the nucleus.

The latter estimate is fairly easily and surely made. But to know the number of rays leaving the central particle in unit time we require to know the quantity of radioactive material in the nucleus. This cannot be directly determined. We can only, from known results obtained with larger specimens of just such a mineral substance as composes the nucleus, guess at the amount of uranium, or it may be thorium, which may be present.

This method has been applied to the uranium haloes of the mica of County Carlow.[1] Results for the age of the halo

of from 20 to
400 millions of years have been obtained. This
mica was probably
formed in the granite of Leinster in late Silurian
or in Devonian
times.

The higher results are probably the least in
error, upon the data
involved; for the assumption made as to the amount
of uranium in
the nuclei of the haloes was such as to render the
higher results
the more reliable.

This method is, of course, a radioactive method,
and similar to
the method by helium storage, save that it is free
of the risk of
error by escape of the helium, the effects of
which are, as it
were, registered at the moment of its production,
so that its
subsequent escape is of no moment.

[1] Joly and Rutherford, Phil. Mag., April,
1913.

22

REVIEW OF THE RESULTS

We shall now briefly review the results on the
geological age of
the Earth.

By methods based on the approximate uniformity of
denudative
effects in the past, a period of the order of 100
millions of

years has been obtained as the duration of our geological age; and consistently whether we accept for measurement the sediments or the dissolved sodium. We can give reasons why these measurements might afford too great an age, but we can find absolutely no good reason why they should give one much too low.

By measuring radioactive products ages have been found which, while they vary widely among themselves, yet claim to possess accuracy in their superior limits, and exceed those derived from denudation from nine to fourteen times.

In this difficulty let us consider the claims of the radioactive method in any of its forms. In order to be trustworthy it must be true; (1) that the rate of transformation now shown by the parent substance has obtained throughout the entire past, and (2) that there were no other radioactive substances, either now or formerly existing, except uranium, which gave rise to lead. As regards methods based on the production of helium, what we have to say will largely apply to it also. If some unknown source of these elements exists we, of course, on our assumption overestimate the age.

As regards the first point: In ascribing a constant rate of change to the parent substance—which Becker (loc. cit.) describes as "a simple though tremendous extrapolation"—we reason upon analogy with the constant rate of decay observed in the derived radioactive bodies. If uranium and thorium are really primary elements, however, the analogy relied on may be misleading; at least, it is obviously incomplete. It is incomplete in a particular which may be very important: the mode of origin of these parent bodies—whatever it may have been—is different to that of the secondary elements with which we compare them. A convergence in their rate of transformation is not impossible, or even improbable, so far as we know.

As regards the second point: It is assumed that uranium alone of the elements in radioactive minerals is ultimately transformed to lead by radioactive changes. We must consider this assumption.

Recent advances in the chemistry of the radioactive elements has brought out evidence that all three lines of radioactive descent known to us—_i.e._ those beginning with uranium,

with thorium,
and with actinium—alike converge to lead.[1] There
are
difficulties in the way of believing that all the
lead-like atoms
so produced ("isotopes" of lead, as Soddy proposes
to call them)
actually remain as stable lead in the minerals.
For one

[1] See Soddy's Chemistry of the Radioactive
Elements (Longmans,
Green & Co.).

24

thing there is sometimes, along with very large
amounts of
thorium, an almost entire absence of lead in
thorianites and
thorites. And in some urano—thorites the lead may
be noticed to
follow the uranium in approximate proportionality,
notwithstanding the presence of large amounts of
thorium.[1] This
is in favour of the assumption that all the lead
present is
derived from the uranium. The actinium is present
in negligibly
small amounts.

On the other hand, there is evidence arising from
the atomic
weight of lead which seems to involve some other
parent than
uranium. Soddy, in the work referred to, points
this out. The
atomic weight of radium is well known, and uranium

in its descent
has to change to this element. The loss of mass
between radium
and uranium-derived lead can be accurately
estimated by the
number of alpha rays given off. From this we get
the atomic
weight of uranium-derived lead as closely 206. Now
the best
determinations of the atomic weight of normal lead
assign to this
element an atomic weight of closely

[1] It seems very difficult at present to suggest
an end product
for thorium, unless we assume that, by loss of
electrons, thorium
E, or thorium-lead, reverts to a substance
chemically identical
with thorium itself. Such a change—whether
considered from the
point of view of the periodic law or of the
radioactive theory
would involve many interesting consequences. It
is, of course,
quite possible that the nature of the conditions
attending the
deposition of the uranium ores, many of which are
comparatively
recent, are responsible for the difficulties
observed. The
thorium and uranium ores are, again, specially
prone to
alteration.

207. By a somewhat similar calculation it is deduced that thorium-derived lead would possess the atomic weight of 208. Thus normal lead might be an admixture of uranium- and thorium-derived lead. However, as we have seen, the view that thorium gives rise to stable lead is beset with some difficulties.

If we are going upon reliable facts and figures, we must, then, assume: (a) That some other element than uranium, and genetically connected with it (probably as parent substance), gives rise, or formerly gave rise, to lead of heavier atomic weight than normal lead. It may be observed respecting this theory that there is some support for the view that a parent substance both to uranium and thorium has existed or possibly exists. The evidence is found in the proportionality frequently observed between the amounts of thorium and uranium in the primary rocks.[1] Or: (b) We may meet the difficulties in a simpler way, which may be stated as follows: If we assume that all stable lead is derived from uranium, and at the same time recognise that lead is not perfectly homogeneous in atomic weight, we must, of necessity, ascribe to uranium a similar want of homogeneity;

heavy atoms of
uranium giving rise to heavy

[1] Compare results for the thorium content of
such rocks

(appearing in a paper by the author Cong. Int. _de
Radiologie et

d'Electricité_, vol. i., 1910, p. 373), and those
for the radium

content, as collected in _Phil. Mag._, October,
1912, p. 697.

Also A. L. Fletcher, _Phil. Mag._, July, 1910;
January, 1911, and

June, 1911. J. H. J. Poole, _Phil. Mag._, April,
1915

26

atoms of lead and light atoms of uranium
generating light atoms

of lead. This assumption seems to be involved in
the figures

upon, which we are going. Still relying on these
figures, we

find, however, that existing uranium cannot give
rise to lead of

normal atomic weight. We can only conclude that
the heavier atoms

of uranium have decayed more rapidly than the
lighter ones. In

this connection it is of interest to note the
complexity of

uranium as recently established by Geiger,
although in this case

it is assumed that the shorter-lived isotope bears
the relation

of offspring to the longer-lived and largely
preponderating

constituent. However, there does not seem to be any direct proof of this as yet.

From these considerations it would seem that unless the atomic weight of lead in uraninites, etc., is 206, the former complexity and more accelerated decay of uranium are indicated in the data respecting the atomic weights of radium and lead[1]. As an alternative view, we may assume, as in our first hypothesis, that some elementally different but genetically connected substance, decaying along branching lines of descent at a rate sufficient to practically remove the whole of it during geological time, formerly existed. Whichever hypothesis we adopt

[1] Later investigation has shown that the atomic weight of lead in uranium-bearing ores is about 206.6 (see Richards and Lemberg, Journ. of Am. Chem. Soc., July, 1914). This result gives support to the view expressed above.

27

we are confronted by probabilities which invalidate time-measurements based on the lead and helium ratio in minerals. We have, in short, grave reason to question the measure of

uniformitarianism postulated in finding the age by any of the known radioactive methods.

That we have much to learn respecting our assumptions, whether we pursue the geological or the radioactive methods of approaching the age of our era, is, indeed, probable. Whatever the issue it is certain that the reconciling facts will leave us with much more light than we at present possess either as respects the Earth's history or the history of the radioactive elements. With this necessary admission we leave our study of the Birth-Time of the World.

It has led us a long way from Lucretius. We do not ask if other Iliads have perished; or if poets before Homer have vainly sung, becoming a prey to all-consuming time. We move in a greater history, the landmarks of which are not the birth and death of kings and poets, but of species, genera, orders. And we set out these organic events not according to the passing generations of man, but over scores or hundreds of millions of years.

How much Lucretius has lost, and how much we have gained, is bound up with the question of the intrinsic value

of knowledge
and great ideas. Let us appraise knowledge as we
would the
Homeric poems, as some-

28

thing which ennobles life and makes it happier.
Well, then, we
are, as I think, in possession today of some of
those lost Iliads
and Odysseys for which Lucretius looked in vain.
[1]

[1] The duration in the past of Solar heat is
necessarily bound
up with the geological age. There is no known
means (outside
speculative science) of accounting for more than
about 30 million
years of the existing solar temperature in the
past. In this
direction the age seems certainly limited to 100
million years.
See a review of the question by Dr. Lindemann in
Nature, April
5th, 1915.

29

DENUATION

THE subject of denudation is at once one of the
most interesting
and one of the most complicated with which the
geologist has to
deal. While its great results are apparent even to
the most
casual observer, the factors which have led to

these results are
in many cases so indeterminate, and in some cases
apparently so
variable in influence, that thoughtful writers
have even claimed
precisely opposite effects as originating from,
the same cause.
Indeed, it is almost impossible to deal with the
subject without
entering upon controversial matters. In the
following pages I
shall endeavour to keep to broad issues which are,
at the present
day, either conceded by the greater number of
authorities on the
subject, or are, from their strictly quantitative
character, not
open to controversy.

It is evident, in the first place, that denudation
—or the wearing
away of the land surfaces of the earth—is mainly a
result of the
circulation of water from the ocean to the land,
and back again
to the ocean. An action entirely conditioned by
solar heat, and
without which it would completely cease and
further change upon
the land come to an end.

To what actions, then, is so great a potency of
the

30

circulating water to be traced? Broadly speaking,
we may classify

them as mechanical and chemical. The first involves the separation of rock masses into smaller fragments of all sizes, down to the finest dust. The second involves the actual solution in the water of the rock constituents, which may be regarded as the final act of disintegration. The rivers bear the burden both of the comminuted and the dissolved materials to the sea. The mud and sand carried by their currents, or gradually pushed along their beds, represent the former; the invisible dissolved matter, only to be demonstrated to the eye by evaporation of the water or by chemical precipitation, represents the latter.

The results of these actions, integrated over geological time, are enormous. The entire bulk of the sedimentary rocks, such as sandstones, slates, shales, conglomerates, limestones, etc., and the salt content of the ocean, are due to the combined activity of mechanical and solvent denudation. We shall, later on, make an estimate of the magnitude of the quantities actually involved.

In the Swiss valleys we see torrents of muddy water hurrying along, and if we follow them up, we trace them to glaciers high among the mountains. From beneath the foot of the

glacier, we
find, the torrent has birth. The first debris
given to the river
is derived from the wearing of the rocky bed along
which the
glacier moves. The river of ice bequeaths to the
river of
water—of which it is the parent—the spoils which
it has won from
the rocks

31

The work of mechanical disintegration is, however,
not restricted
to the glacier's bed. It proceeds everywhere over
the surface of
the rocks. It is aided by the most diverse
actions. For instance,
the freezing and expansion of water in the chinks
and cracks in
those alpine heights where between sunrise and
sunset the heat of
summer reigns, and between sunset and sunrise the
cold of winter.
Again, under these conditions the mere change of
surface
temperature from night to day severely stresses
the surface
layers of the rocks, and, on the same principles
as we explain
the fracture of an unequally heated glass vessel,
the rocks
cleave off in slabs which slip down the steeps of
the mountain
and collect as screes in the valley. At lower
levels the
expansive force of vegetable growth is not

unimportant, as all
will admit who have seen the strong roots of the
pines
penetrating the crannies of the rocks. Nor does
the river which
flows in the bed of the valley act as a carrier
only. Listening
carefully we may detect beneath the roar of the
alpine torrent
the crunching and knocking of descending boulders.
And in the
potholes scooped by its whirling waters we
recognise the abrasive
action of the suspended sand upon the river bed.

A view from an Alpine summit reveals a scene of
remarkable
desolation (Pl. V, p. 40). Screes lie piled
against the steep
slopes. Cliffs stand shattered and ready to fall
in ruins. And
here the forces at work readily reveal themselves.
An occasional
wreath of white smoke among

32

the far-off peaks, followed by a rumbling
reverberation, marks
the fall of an avalanche. Water everywhere
trickles through the
shaly débris scattered around. In the full
sunshine the rocks are
almost too hot to bear touching. A few hours later
the cold is
deadly, and all becomes a frozen silence. In such
scenes of
desolation and destruction, detrital sediments are

actively being
generated. As we descend into the valley we hear
the deep voice
of the torrents which are continually hurrying the
disintegrated
rocks to the ocean.

A remarkable demonstration of the activity of
mechanical
denudation is shown by the phenomenon of "earth
pillars." The
photograph (Pl. IV.) of the earth pillars of the
Val d'Hérens
(Switzerland) shows the peculiar appearance these
objects
present. They arise under conditions where large
stones or
boulders are scattered in a deep deposit of clay,
and where much
of the denudation is due to water scour. The large
boulders not
only act as shelter against rain, but they bind
and consolidate
by their mere weight the clay upon which they
rest. Hence the
materials underlying the boulders become more
resistant, and as
the surrounding clays are gradually washed away
and carried to
the streams, these compacted parts persist, and,
finally, stand
like walls or pillars above the general level.
After a time the
great boulders fall off and the underlying clay
becomes worn by
the rainwash to fantastic spikes and ridges. In

the Val d'Hérens
the earth pillars are formed

33

of the deep moraine stuff which thickly overlies the slopes of the valley. The wall of pillars runs across the axis of the valley, down the slope of the hill, and crosses the road, so that it has to be tunnelled to permit the passage of traffic. It is not improbable that some additional influence—possibly the presence of lime—has hardened the material forming the pillars, and tended to their preservation.

Denudation has, however, other methods of work than purely mechanical; methods more noiseless and gentle, but not less effective, as the victories of peace are no less than those of war.

Over the immense tracts of the continents chemical work proceeds relentlessly. The rock in general, more especially the primary igneous rock, is not stable in presence of the atmosphere and of water. Some of the minerals, such as certain silicates and carbonates, dissolve relatively fast, others with extreme slowness. In the process of solution chemical

actions are involved; oxidation in presence of the free oxygen of the atmosphere; attack by the feeble acid arising from the solution of carbon dioxide in water; or, again, by the activity of certain acids—humous acids—which originate in the decomposition of vegetable remains. These chemical agents may in some instances, _e.g._ in the case of carbonates such as limestone or dolomite—bring practically the whole rock into solution. In other instances—_e.g._ granites, basalts, etc.—they may remove some of the

34

constituent minerals completely or partially, such as felspar, olivine, augite, and leave more resistant substances to be ultimately washed down as fine sand or mud into the river.

It is often difficult or impossible to appraise the relative efficiency of mechanical and chemical denudation in removing the materials from a certain area. There can be, indeed, little doubt that in mountainous regions the mechanical effects are largely predominant. The silts of glacial rivers are little different

from freshly-powdered rock. The water which carries them but little different from the pure rain or snow which falls from the sky. There has not been time for the chemical or solvent actions to take place. Now while gravitational forces favour sudden shock and violent motions in the hills, the effect of these on solvent and chemical denudation is but small. Nor is good drainage favourable to chemical actions, for water is the primary factor in every case. Water takes up and removes soluble combinations of molecules, and penetrates beneath residual insoluble substances. It carries the oxygen and acids downwards through the soils, and finally conveys the results of its own work to the rivers and streams. The lower mean temperature of the mountains as well as the perfect drainage diminishes chemical activities.

Hence we conclude that the heights are not generally favourable to the purely solvent and chemical actions. It is on the lower-lying land that soils tend to accumulate,

35

and in these the chief solvent and the chief chemical denudation of the Earth are effected.

The solvent and chemical effects which go on in the finely-divided materials of the soils may be observed in the laboratory. They proceed faster than would be anticipated. The observation is made by passing a measured quantity of water backwards and forwards for some months through a tube containing a few grammes of powdered rock. Finally the water is analysed, and in this manner the amount of dissolved matter it has taken up is estimated. The rock powder is examined under the microscope in order to determine the size of the grains, and so to calculate the total surface exposed to the action of the water. We must be careful in such experiments to permit free oxidation by the atmosphere. Results obtained in this way of course take no account of the chemical effects of organic acids such as exist in the soils. The quantities obtained in the laboratory will, therefore, be deficient as compared with the natural results.

In this manner it has been found that fresh basalt exposed to continually moving water will lose about 0.20 gramme per square metre of surface per year. The mineral orthoclase, which enters

largely into the constitution of many granites, was found to lose under the same conditions 0.025 gramme. A glassy lava (obsidian) rich in silica and in the chemical constituents of an average granite, was more resistant still; losing but 0.013 gramme per square metre per year. Hornblende, a mineral

36

abundant in many rocks, lost 0.075 gramme. The mean of the results showed that 0.08 gramme was washed in a year from each square metre. Such results give us some indication of the rate at which the work of solution goes on in the finely divided soils.[1]

It might be urged that, as the mechanical break up of rocks, and the production in this way of large surfaces, must be at the basis of solvent and chemical denudation, these latter activities should be predominant in the mountains. The answer to this is that the soils rarely owe their existence to mechanical actions. The alluvium of the valleys constitutes only narrow margins to the rivers; the finer débris from the mountains is rapidly brought into the ocean. The soils which cover the greater part of

continental areas have had a very different origin.

In any quarry where a section of the soil and of the underlying rock is visible, we may study the mode of formation of soils. Our observations are, we will suppose, pursued in a granite quarry.

We first note that the material of the soil nearest the surface is intermixed with the roots of grasses, trees, or shrubs.

Examining a handful of this soil, we see glistening flakes of mica which plainly are derived from the original granite. Washing off the finer particles, we find the largest remaining grains are composed of the all but indestructible quartz.

[1] Proc. Roy. Irish Acad., VIII., Ser. A, p. 21.

37

This also is from the granite. Some few of the grains are of chalky-looking felspar; again a granitic mineral. What is the finer silt we have washed off? It, too, is composed of mineral particles to a great extent; rock dust stained with iron oxide and intermixed with organic remains, both animal and vegetable.

But if we make a chemical analysis of the finer silt we find that the composition is by no means that of the granite

beneath. The chemist is able to say, from a study of his results, that there has been, in the first place, a large loss of material attending the conversion of the granite to the soil. He finds a concentration of certain of the more resistant substances of the granite arising from the loss of the less resistant. Thus the percentage amount of alumina is increased. The percentage of iron is also increased. But silica and most other substances show a diminished percentage. Notably lime has nearly disappeared. Soda is much reduced; so is magnesia. Potash is not so completely abstracted. Finally, owing to hydration, there is much more combined water in the soil than in the rock. This is a typical result for rocks of this kind.

Deeper in the soil we often observe a change of texture. It has become finer, and at the same time the clay is paler in colour. This subsoil represents the finer particles carried by rain from above. The change of colour is due to the state of the iron which is less oxidised low down in the soil. Beneath the subsoil the soil grows

again coarser. Finally, we recognise in it fragments of granite which ever grow larger as we descend, till the soil has become replaced by the loose and shattered rock. Beneath this the only sign of weathering apparent in the rock is the rusty hue imparted by the oxidised iron which the percolating rain has leached from iron-bearing minerals.

The soil we have examined has plainly been derived in situ from the underlying rock. It represents the more insoluble residue after water and acids have done their work. Each year there must be a very slow sinking of the surface, but the ablation is infinitesimal.

The depth of such a soil may be considerable. The total surface exposed by the countless grains of which it is composed is enormous. In a cubic foot of average soil the surface area of the grains may be 50,000 square feet or more. Hence a soil only two feet deep may expose 100,000 square feet for each square foot of surface area.

It is true that soils formed in this manner by atmospheric and organic actions take a very long time to grow. It must be

remembered, however, that the process is throughout attended by the removal in solution: of chemically altered materials.

Considerations such as the foregoing must convince us that while the accumulation of the detrital sediments around the continents is largely the result of activities progressing on the steeper slopes of the land, that is,

39

among the mountainous regions, the feeding of the salts to the ocean arises from the slower work of meteorological and organic agencies attacking the molecular constitution of the rocks; processes which best proceed where the drainage is sluggish and the quiescent conditions permit of the development of abundant organic growth and decay.

Statistics of the solvent denudation of the continents support this view. Within recent years a very large amount of work has been expended on the chemical investigation of river waters of America and of Europe. F. W. Clarke has, at the expense of much labour, collected and compared these results. They are expressed as so many tonnes removed in solution per square

mile per annum.

For North America the result shows 79 tonnes so removed; for

Europe 100 tonnes. Now there is a notable difference between the

mean elevations of these two continents. North America has a mean

elevation of 700 metres over sea level, whereas the mean

elevation of Europe is but 300 metres. We see in these figures

that the more mountainous land supplies less dissolved matter to

the ocean than the land of lower elevation, as our study has led

us to expect.

We have now considered the source of the detrital sediments, as

well as of the dissolved matter which has given to the ocean, in

the course of geological time, its present gigantic load of

salts. It is true there are further solvent and chemical effects

exerted by the sea water

40

upon the sediments discharged into it; but we are justified in

concluding that, relatively to the similar actions taking place

in the soils, the solvent and chemical work of the ocean is

small. The fact is, the deposited detrital sediments around the

continents occupy an area small when contrasted

with the vast stretches of the land. The area of deposition is much less than that of denudation; probably hardly as much as one twentieth. And, again, the conditions of aeration and circulation which largely promote chemical and solvent denudation in the soils are relatively limited and ineffective in the detrital oceanic deposits.

The summation of the amounts of dissolved and detrital materials which denudation has brought into the ocean during the long denudative history of the Earth, as we might anticipate, reveals quantities of almost unrealisable greatness. The facts are among the most impressive which geological science has brought to light. Elsewhere in this volume they have been mentioned when discussing the age of the Earth. In the present connection, however, they are deserving of separate consideration.

The basis of our reasoning is that the ocean owes its saltiness mainly if not entirely to the denudative activities we have been considering. We must establish this.

We may, in the first place, say that any other view at once

raises the greatest difficulties. The chemical composition of the detrital sediments which are spread over

41

the continents and which build up the mountains, differs on the average very considerably from that of the igneous rocks. We know the former have been derived from the latter, and we know that the difference in the composition of the two classes of materials is due to the removal in solution of certain of the constituents of the igneous rocks. But the ocean alone can have received this dissolved matter. We know of no other place in which to look for it. It is true that some part of this dissolved matter has been again rejected by the ocean; thus the formation of limestone is largely due to the abstraction of lime from sea water by organic and other agencies. This, however, in no way relieves us of the necessity of tracing to the ocean the substances dissolved from the igneous rocks. It follows that we have here a very *causa* for the saltiness of the ocean. The view that the ocean "was salt from the first" is without one known fact to support it, and leaves us with the burden of the entire dissolved salts of

geological time
to dispose of—Where and how?

The argument we have outlined above becomes convincingly strong when examined more closely. For this purpose we first compare the average chemical composition of the sedimentary and the igneous rocks. The following table gives the percentages of the chief chemical constituents: [1]

[1] F. W. Clarke: A Preliminary Study of Chemical Denudation,
p. 13

42

		Igneous.
Sedimentary.		
Silica (SiO_2) -	59.99	58.51
Alumina (Al_2O_3) -	15.04	13.07
Ferric oxide (Fe_2O_3) -	2.59	3.40
Ferrous oxide (FeO) -	3.34	2.00
Magnesia (MgO) -	3.89	2.52
Lime (CaO) -	4.81	5.42
Soda (Na_2O) -	3.41	1.12
Potash (K_2O) -	2.95	2.80
Water (H_2O) -	1.92	4.28
Carbon dioxide (CO_2) -	--	4.93
Minor constituents -	<u>2.06</u>	<u>1.95</u>
	100.00	100.00

In the derivation of the sediments from the igneous rocks there is a loss by solution of about 33 per cent; i.e. 100 tons of

igneous rock yields rather less than 70 tons of sedimentary rock.

This involves a concentration in the sediments of the more

insoluble constituents. To this rule the lime-content appears to

be an exception. It is not so in reality. Its high value in the

sediments is due to its restoration from the ocean to the land.

The magnesia and potash are, also, largely restored from the

ocean; the former in dolomites and magnesian limestones; the

latter in glauconite sands. The iron of the sediments shows

increased oxidation. The most notable difference in the two

analyses appears, however, in the soda percentages. This falls

from 3.41 in the igneous rock to 1.12 in the average sediment.

Indeed, this

43

deficiency of soda in sedimentary rocks is so characteristic of

secondary rocks that it may with some safety be applied to

discriminate between the two classes of substances in cases where

petrological distinctions of other kinds break down.

To what is this so marked deficiency of soda to be ascribed? It

is a result of the extreme solubility of the salts

of sodium in water. This has not only rendered its deposition by evaporation a relatively rare and unimportant incident of geological history, but also has protected it from abstraction from the ocean by organic agencies. The element sodium has, in fact, accumulated in the ocean during the whole of geological time.

We can use the facts associated with the accumulation of sodium salts in the ocean as a means of obtaining additional support to the view, that the processes of solvent denudation are responsible for the saltiness of the ocean. The new evidence may be stated as follows: Estimates of the amounts of sedimentary rock on the continents have repeatedly been made. It is true that these estimates are no more than approximations. But they undoubtedly are approximations, and as such may legitimately be used in our argument; more especially as final agreement tends to check and to support the several estimates which enter into them.

The most recent and probable estimates of the sediments on the land assign an average thickness of one mile of

secondary rocks over the land area of the world. To this some increase must be made to allow for similar materials concealed in the ocean, principally around the continental margins. If we add 10 per cent. and assign a specific gravity of 2.5 we get as the mass of the sediments 64×10^{16} tonnes. But as this is about 67 per cent. of the parent igneous rock—i.e.—the average igneous rock from which the sediments are derived—we conclude that the primary denuded rock amounted to a mass of about 95×10^{16} tonnes.

Now from the mean chemical composition of the secondary rocks we calculate that the mass of sediments as above determined contains 0.72×10^{16} tonnes of the sodium oxide, Na_2O . If to this amount we add the quantity of sodium oxide which must have been given to the ocean in order to account for the sodium salts contained therein, we arrive at a total quantity of oxide of sodium which must be that possessed by the primary rock before denudation began its work upon it. The mass of the ocean being well ascertained, we easily calculate that the sodium in the ocean converted to sodium oxide amounts to 2.1×10^{16}

tonnes. Hence
between the estimated sediments and the waters of
the ocean we
can account for 2.82×10^{16} tonnes of soda. When
now we put this
quantity back into the estimated mass of primary
rock we find
that it assigns to the primary rock a soda
percentage of 3.0. On
the average analysis given above this should be
3.41 per cent.
The agreement,

45

all things considered, more especially the
uncertainty in the
estimate of the sediments, is plainly in support
of the view that
oceanic salts are derived from the rocks; if,
indeed, it does not
render it a certainty.

A leading and fundamental inference in the
denudative history of
the Earth thus finds support: indeed, we may say,
verification.

In the light of this fact the whole work of
denudation stands
revealed. That the ocean began its history as a
vast fresh-water
envelope of the Globe is a view which accords with
the evidence
for the primitive high temperature of the Earth.
Geological
history opened with the condensation of an
atmosphere of immense
extent, which, after long fluctuations between the

states of
steam and water, finally settled upon the surface,
almost free of
matter in solution: an ocean of distilled water.
The epoch of
denudation then began. It will, probably, continue
till the
waters, undergoing further loss of thermal energy,
suffer yet
another change of state, when their circulation
will cease and
their attack upon the rocks come to an end.

From what has been reviewed above it is evident
that the sodium
in the ocean is an index of the total activity of
denudation
integrated over geological time. From this the
broad facts of the
results of denudation admit of determination with
considerable
accuracy. We can estimate the amount of rock which
has been
degraded by solvent and chemical actions, and the
amount of
sediments which has been derived from it. We are,

46

thus, able to amend our estimate of the sediments
which, as
determined by direct observation, served to
support the basis of
our argument.

We now go straight to the ocean for the amount of
sodium of
denudative origin. There may, indeed, have been

some primitive sodium dissolved by a more rapid denudation while the Earth's surface was still falling in temperature. It can be shown, however, that this amount was relatively small. Neglecting it we may say with safety that the quantity of sodium carried into the ocean by the rivers must be between 14,000 and 15,000 million million tonnes: i.e. $14,500 \times 10^{12}$ tonnes, say.

Keeping the figures to round numbers we find that this amount of sodium involves the denudation of about 80×10^{16} tonnes of average igneous rock to 53×10^{16} tonnes of average sediment. From these vast quantities we know that the parent rock denuded during geological time amounted to some 300 million cubic kilometres or about seventy million cubic miles. The sediments derived therefrom possessed a bulk of 220 million cubic kilometres or fifty million cubic miles. The area of the land surface of the Globe is 144 million square kilometres. The parent rock would have covered this to a uniform depth of rather more than two kilometres, and the derived sediment to more than 1.5 kilometres, or about one mile deep.

The slow accomplishment of results so vast conveys some idea of the great duration of geological time.

47

The foregoing method of investigating the statistics of solvent denudation is capable of affording information not only as to the amount of sediments upon the land, but also as to the quantity which is spread over the floor of the ocean.

We see this when we follow the fate of the 33 per cent. of dissolved salts which has been leached from the parent igneous rock, and the mass of which we calculate from the ascertained mass of the latter, to be 27×10^{16} tonnes. This quantity was at one time or another all in the ocean. But, as we saw above, a certain part of it has been again abstracted from solution, chiefly by organic agencies. Now the abstracted solids have not been altogether retained beneath the ocean. Movements of the land during geological time have resulted in some portion being uplifted along with other sediments. These substances constitute, mainly, the limestones.

We see, then, that the 27×10^{16} tonnes of substances leached

from the parent igneous rocks have had a threefold destination.

One part is still in solution; a second part has been

precipitated to the bottom of the ocean; a third part exists on the land in the form of calcareous rocks.

Observation on the land sediments shows that the calcareous rocks

amount to about 5 per cent. of the whole. From this we find that

3×10^{16} tonnes, approximately, of such rocks have been taken

from the ocean. This accounts for one of the three classes of material

48

into which the original dissolved matter has been divided.

Another of the three quantities is easily estimated: the amount

of matter still in solution in the ocean. The volume of the ocean

is 1,414 million cubic kilometres and its mass is 145×10^{16}

tonnes. The dissolved salts in it constitute 3.4 per cent. of its

mass; or, rather more than 5×10^{16} tonnes. The limestones on the

land and the salts in the sea water together make up about $8 \times$

10^{16} tonnes. If we, now, deduct this from the total of 27×10^{16}

tonnes, we find that about 19×10^{16} tonnes must

exist as
precipitated matter on the floor of the ocean.

The area of the ocean is 367×10^{12} square metres, so that if the precipitated sediment possesses an average specific gravity of 2.5, it would cover the entire floor to a uniform depth of 218 metres; that is 715 feet. This assumes that there was uniform deposition of the abstracted matter over the floor of the ocean.

Of course, this assumption is not justifiable. It is certain that the rate of deposition on the floor of the sea has varied

enormously with various conditions—principally with the depth.

Again, it must be remembered that this estimate takes no account

of solid materials otherwise brought into the oceanic deposits;

e.g., by wind-transported dust from the land or volcanic

ejectamenta in the ocean depths. It is not probable, however,

that any considerable addition to the estimated mean depth of

deposit from such sources would be allowable.

49

The greatness of the quantities involved in these determinations

is almost awe inspiring. Take the case of the dissolved salts in

the ocean. They are but a fraction, as we have

seen, of the total results of solvent denudation and represent the integration of the minute traces contributed by the river water. Yet the common salt (chloride of sodium) alone, contained in the ocean, would, if abstracted and spread over the dry land as a layer of rock salt having a specific gravity of 2.2, cover the whole to a depth of 107 metres or 354 feet. The total salts in solution in the ocean similarly spread over the land would increase the depth of the layer to 460 feet. After considering what this means we have to remember that this amount of matter now in solution in the seas is, in point of fact, less than a fifth part of the total dissolved from the rocks during geological time.

The transport by denudation of detrital and dissolved matter from the land to the ocean has had a most important influence on the events of geological history. The existing surface features of the earth must have been largely conditioned by the dynamical effects arising therefrom. In dealing with the subject of mountain genesis we will, elsewhere, see that all the great mountain ranges have originated in the accumulation of the

detrital sediments near the shore in areas which, in consequence of the load, gradually became depressed and developed into synclines of many thousands of feet in depth. The most impressive surface features of the Globe originated

50

in this manner. We will see too that these events were of a rhythmic character; the upraising of the mountains involving intensified mechanical denudation over the elevated area and in this way an accelerated transport of detritus to the sea; the formation of fresh deposits; renewed synclinal sinking of the sea floor, and, finally, the upheaval of a younger mountain range.

This extraordinary sequence of events has been determined by the events of detrital denudation acting along with certain general conditions which have all along involved the growth of compressive stresses in the surface crust of the Earth.

The effects of purely solvent denudation are less easily traced, but, very probably, they have been of not less importance. I refer here to the transport from the land to the sea of matter in solution.

Solvent denudation, as observed above, takes place mainly in the soils and in this way over the more level continental areas. It has resulted in the removal from the land and transfer to the ocean of an amount of matter which represents a uniform layer of one half a kilometre; that is of more than 1,600 feet of rock. The continents have, during geological time, been lightened to this extent. On the other hand all this matter has for the greater part escaped the geosynclines and become uniformly diffused throughout the ocean or precipitated over its floor principally on the continental slopes before the great depths are reached. Of this material the ocean

51

waters contain in solution an amount sufficient to increase their specific gravity by 2.7 per cent.

Taking the last point first, it is interesting to note the effects upon the bulk of the ocean which has resulted from the matter dissolved in it. From the known density of average sea water we find that 100 ccs. of it weigh just 102.7 grammes. Of this 3.5 per cent. by weight are solids in solution. That is to

say, 3.594 grammes. Hence the weight of water present is 99.1 grammes, or a volume of 99.1 ccs. From this we see that the salts present have increased the volume by 0.9 ccs. or 0.9 per cent.

The average depth of the ocean is 2,000 fathoms or 3,700 metres.

The increase of depth due to salts dissolved in the ocean has been, therefore, 108 feet or 33.24 metres. This result assumes that there has been no increased elastic compression due to the increased pressure, and no change of compressional elastic properties. We may be sure that the rise on the shore line of the land has not been less than 100 feet.

We see then that as the result of solvent denudation we have to do with a heavier and a deeper ocean, expanded in volume by nearly one per cent. and the floor of which has become raised, on an average, about 700 feet by precipitated sediment.

One of the first conceptions, which the student of geology has to dismiss from his mind, is that of the immobility or rigidity of the Earth's crust. The lane, we live on sways even to the gentle rise and fall of ocean tides

around the coasts. It suffers its own tidal oscillations due to the moon's attractions. Large tracts of semi-liquid matter underlie it. There is every evidence that the raised features of the Globe are sustained by such pressures acting over other and adjacent areas as serve to keep them in equilibrium against the force of gravity. This state of equilibrium, which was first recognised by Pratt, as part of the dynamics of the Earth's crust, has been named isostasy. The state of the crust is that of "mobile equilibrium."

The transfer of matter from the exposed land surfaces to the sub-oceanic slopes of the continents and the increase in the density of the ocean, must all along have been attended by isostatic readjustment. We cannot take any other view. On the one hand the land was being lightened; on the other the sea was increasing in mass and depth and the flanks of the continents were being loaded with the matter removed from the land and borne in solution to the ocean. How important the resulting movements must have been may be gathered from the fact that the existing

land of the Globe stands at a mean elevation of no more than 2,000 feet above sea level. We have seen that solvent denudation removed over 1,600 feet of rock. But we have no evidence that on the whole the elevation of land in the past was ever very different from what it now is.

We have, then, presented to our view the remarkable fact that throughout the past, and acting with extreme

53

slowness, the land has steadily been melted down into the sea and as steadily been upraised from the waters. It is possible that the increased bulk of the ocean has led to a certain diminution of the exposed land area. The point is a difficult one. One thing we may without much risk assume. The sub-aereal current of dissolved matter from the land to the ocean was accompanied by a sub-crustal flux from the ocean areas to the land areas; the heated viscous materials creeping from depths far beneath the ocean floor to depths beneath the roots of the mountains which arose around the oceans. Such movements took ages for their accomplishment. Indeed, they have been, probably, continuous all

along and are still proceeding. A low degree of viscosity will suffice to permit of movements so slow. Superimposed upon these movements the rhythmic alternations of depression and elevation of the geosynclines probably resulted in releasing the crust from local accumulation of strains arising in the more rigid surface materials. The whole sequence of movements presents an extraordinary picture of pseudo-vitality—reminding us of the circulatory and respiratory systems of a vast organism.

All great results in our universe are founded in motions and forces the most minute. In contemplating the Cause or the Effect we stand equally impressed with the spectacle presented to us. We shall now turn from the great effects of denudation upon the history and evolution of a world and consider for a moment activities

54

so minute in detail that their operations will probably for ever elude our bodily senses, but which nevertheless have necessarily affected and modified the great results we have been considering.

The ocean a little way from the land is generally so free from suspended sediments that it has a blackness as of ink. This blackness is due to its absolute freedom from particles reflecting the sun's light. The beautiful blue of the Swiss and Italian lakes is due to the presence of very fine particles carried into them by the rivers; the finest flour of the glaciers, which remain almost indefinitely suspended in the water. But in the ocean it is only in those places where rapid currents running over shallows stir continually the sediments or where the fresh water of a great river is carried far from the land, that the presence of silt is to be observed. The beautiful phenomenon of the coal-black sea is familiar to every yachtsman who has sailed to the west of our Islands.[1]

There is, in fact, a very remarkable difference in the manner of settlement of fine sediments in salt and in fresh water. We are here brought into contact with one of those subtle yet influential natural actions the explanation of which involves scientific advance along many apparently unconnected lines of investigation.

[1] See Tyndall's Voyage to Algeria in _Fragments of Science._ The cause of the blue colour of the lakes has been discussed by various observers, not always with agreement.

55

It is easy to observe in the laboratory the fact of the different behaviour of salt and fresh water towards finely divided substances. The nature of the insoluble substance is not important.

We place, in a good light, two glass vessels of equal dimensions; the one filled with sea water, the other with fresh water. Into each we stir the same weight of very finely powdered slate: just so much as will produce a cloudiness. In a few hours we find the sea water limpid. The fresh water is still cloudy, however; and, indeed, may be hardly different in appearance from what it was at starting. In itself this is a most extraordinary experiment. We would have anticipated quite the opposite result owing to the greater density of the sea water.

But a still more interesting experiment remains to be carried out. In the sea water we have many different salts in solution.

Let us see if these salts are equally responsible for the result we have obtained. For this purpose we measure out quantities of sodium chloride and magnesium chloride in the proportion in which they exist in sea water: that is about as seven to one. We add such an equal amount of water to each as represents the dilution of these salts in sea water. Then finally we stir a little of the finely powdered slate into each. It will be found that the magnesium chloride, although so much more dilute than the sodium chloride, is considerably more active in clearing out the suspension. We may now try such marine salts as magnesium sulphate,

56

or calcium sulphate against sodium chloride; keeping the marine proportions. Again we find that the magnesium and calcium salts are the most effective, although so much more dilute than the sodium salt.

There is no visible clue to the explanation of these results. But we must conclude as most probable that some action is at work in the sea water and in the salt solutions which clumps or

flocculates the sediment. For only by the gathering of the particles together in little aggregates can we explain their rapid fall to the bottom. It is not a question of viscosity (_i.e._ of resistance to the motion of the particles), for the salt solutions are rather more viscous than the fresh water. Still more remarkable is the fact that every dissolved substance will not bring about the result. Thus if we dissolve sugar in water we find that, if anything, the silt settles more slowly in the sugar solution than in fresh water.

Now there is one effect produced by the solution of such salts as we have dealt with which is not produced by such bodies as sugar. The water is rendered a conductor of electricity. Long ago Faraday explained this as due to the presence of free atoms of the dissolved salt in the solution, carrying electric charges. We now speak of the salt as "ionised." That is it is partly split up into ions or free electrified atoms of chlorine, sodium, magnesium, etc., according to the particular salt in solution. This fact leads us to think that these electrified

atoms moving about in the solution may be the cause of the clumping or flocculation. Such electrified atoms are absent from the sugar solution: sugar does not become "ionised" when it is dissolved.

The suspicion that the free electrified atoms play a part in the phenomenon is strengthened when we recall the remarkable difference in the action of sodium chloride and magnesium chloride. In each of the solutions of these substances there are free chlorine atoms each of which carries a single charge of negative electricity. As these atoms are alike in both solutions the different behaviour of the solutions cannot be due to the chlorine. But the metallic atom is very different in the two cases. The ionised sodium atom is known to be _monad_ or carries but _one_ positive charge; whereas the magnesium atom is _diad_ and carries _two_ positive charges. If, then, we assume that the metallic, positively electrified atom is in each case responsible, we have something to go on. It may be now stated that it has been found by experiment and supported by theory that the clumping power of an ion rises very rapidly

with its valency;
that is with the number of unit charges associated
with it. Thus
diads such as magnesium, calcium, barium, etc.,
are very much
more efficient than monads such as sodium,
potassium, etc., and
again, triads such as aluminium are, similarly,
very much more
powerful than diad atoms. Here, in short, we have
arrived at the
active cause of the phenomenon. Its inner
mechanism

58

is, however, harder to fathom. A plausible
explanation can be
offered, but a study of it would take us too far.
Sufficient has
been said to show the very subtle nature of the
forces at work.

We have here an effect due to the sea salts
derived by denudation
from the land which has been slowly augmenting
during geological
time. It is certain that the ocean was practically
fresh water in
remote ages. During those times the silt from the
great rivers
would have been carried very far from the land. A
Mississippi of
those ages would have sent its finer suspensions
far abroad on a
contemporary Gulf stream: not improbably right
across the
Atlantic. The earlier sediments of argillaceous

type were not collected in the geosynclines and the genesis of the mountains was delayed proportionately. But it was, probably, not for very long that such conditions prevailed. For the accumulation of calcium salts must have been rapid, and although the great salinity due to sodium salts was of slow growth the salts of the diad element calcium must have soon introduced the cooperation of the ion in the work of building the mountain.

59

THE ABUNDANCE OF LIFE [1]

WE had reached the Pass of Tre Croci[2] and from a point a little below the summit, looked eastward over the glorious Val Buona. The pines which clothed the floor and lower slopes of the valley, extended their multitudes into the furthest distance, among the many recesses of the mountains, and into the confluent Val di Misurina. In the sunshine the Alpine butterflies flitted from stone to stone. The ground at our feet and everywhere throughout the forests teemed with the countless millions of the small black ants.

It was a magnificent display of vitality; of the aggressiveness of vitality, assailing the barren heights of the limestone, wringing a subsistence from dead things. And the question suggested itself with new force: why the abundance of life and its unending activity?

In trying to answer this question, the present sketch originated.

I propose to refer for an answer to dynamic considerations. It is apparent that natural selection can only be concerned in a secondary way. Natural selection defines

[1] Proc. Roy. Dublin Soc., vol. vii., 1890.

[2] In the Dolomites of Southeast Tyrol; during the summer of 1890. Much of what follows was evolved in discussion with my fellow-traveller, Henry H. Dixon. Much of it is his.

60

a certain course of development for the organism; but very evidently some property of inherent progressiveness in the organism must be involved. The mineral is not affected by natural selection to enter on a course of continual variation and

multiplication. The dynamic relations of the organism with the environment are evidently very different from those of inanimate nature.

GENERAL DYNAMIC CONDITIONS ATTENDING INANIMATE ACTIONS

It is necessary, in the first place, to refer briefly to the phenomena attending the transfer of energy within and into inanimate material systems. It is not assumed here that these phenomena are restricted in their sphere of action to inanimate nature. It is, in fact, very certain that they are not; but while they confer on dead nature its own dynamic tendencies, it will appear that their effects are by various means evaded in living nature. We, therefore, treat of them as characteristic of inanimate actions. We accept as fundamental to all the considerations which follow the truth of the principle of the Conservation of Energy.[1]

[1] "The principle of the Conservation of Energy has acquired so much scientific weight during the last twenty years that no physiologist would feel any confidence in an experiment which showed a considerable difference between the work

done by the
animal and the balance of the account of Energy
received and
spent."—Clerk Maxwell, *Nature*, vol. xix., p.
142. See also
Helmholtz *On the Conservation of Force.*

61

Whatever speculations may be made as to the course
of events very
distant from us in space, it appears certain that
dissipation of
energy is at present actively progressing
throughout our sphere
of observation in inanimate nature. It follows, in
fact, from the
second law of thermodynamics, that whenever work
is derived from
heat, a certain quantity of heat falls in
potential without doing
work or, in short, is dissipated. On the other
hand, work may be
entirely converted into heat. The result is the
heat-tendency of
the universe. Heat, being an undirected form of
energy, seeks, as
it were, its own level, so that the result of this
heat-tendency
is continual approach to uniformity of potential.

The heat-tendency of the universe is also revealed
in the
far-reaching "law of maximum work," which defines
that chemical
change, accomplished without the intervention of
external energy,
tends to the production of the body, or system of

bodies, which
disengage the greatest quantity of heat.[1] And,
again, vast
numbers of actions going on throughout nature are
attended by
dissipatory thermal effects, as those arising from
the motions of
proximate molecules (friction, viscosity), and
from the fall of
electrical potential.

Thus, on all sides, the energy which was once most
probably
existent in the form of gravitational potential,
is being
dissipated into unavailable forms. We must

[1] Berthelot, *Essai de Mécanique Chimique.*

62

recognize dissipation as an inevitable attendant
on inanimate
transfer of energy.

But when we come to consider inanimate actions in
relation to
time, or time-rate of change, we find a new
feature in the
phenomena attending transfer of energy; a feature
which is really
involved in general statements as to the laws of
physical
interactions.[1] It is seen, that the attitude of
inanimate
material systems is very generally, if not in all
cases,
retardative of change—opposing it by effects
generated by the

primary action, which may be called "secondary" for convenience.

Further, it will be seen that these secondary effects are those concerned in bringing about the inevitable dissipation.

As example, let us endeavour to transfer gravitational potential energy contained in a mass raised above the surface of the Earth into an elastic body, which we can put into compression by resting the weight upon it. In this way work is done against elastic force and stored as elastic potential energy. We may deal with a metal spring, or with a mass of gas contained in a cylinder fitted with a piston upon which the weight may be placed. In either case we find the effect of compression is to raise the temperature of the substance, thus causing its

[1] Helmholtz, Ice and Glaciers. Atkinson's collection of his Popular Lectures. First Series, p.120. Quoted by Tate, Heat, p. 311.

63

expansion or increased resistance to the descent of the weight.

And this resistance continues, with diminishing intensity, till

all the heat generated is dissipated into the surrounding medium.

The secondary effect thus delays the final transfer of energy.

Again, if we suppose the gas in the cylinder replaced by a vapour

in a state of saturation, the effect of increased pressure, as of

a weight placed upon the piston, is to reduce the vapour to a

liquid, thereby bringing about a great diminution of volume and

proportional loss of gravitational potential by the weight. But

this change will by no means be brought about instantaneously.

When a little of the vapour is condensed, this portion parts with

latent heat of vaporisation, increasing the tension of the

remainder, or raising its point of saturation, so that before the

weight descends any further, this heat has to escape from the

cylinder.

Many more such cases might be cited. The heating of india-rubber

when expanded, its cooling when compressed, is a remarkable one;

for at first sight it appears as if this must render it

exceptional to the general law, most substances exhibiting the

opposite thermal effects when stressed. However, here, too, the

action of the stress is opposed by the secondary

effects

developed in the substance; for it is found that this substance contracts when heated, expands when cooled. Again, ice being a substance which contracts in melting, the effect of pressure is to facilitate melting, lowering its freezing point. But

64

so soon as a little melting occurs, the resulting liquid calls on the residual ice for an amount of heat equivalent to the latent heat of liquefaction, and so by cooling the whole, retards the change.

Such particular cases illustrate a principle controlling the interaction of matter and energy which seems universal in application save when evaded, as we shall see, by the ingenuity of life. This principle is not only revealed in the researches of the laboratory; it is manifest in the history of worlds and solar systems. Thus, consider the effects arising from the aggregation of matter in space under the influence of the mutual attraction of the particles. The tendency here is loss of gravitational potential. The final approach is however retarded by the

temperature, or vis viva of the parts attending collision and compression. From this cause the great suns of space radiate for ages before the final loss of potential is attained.

Clerk Maxwell[1] observes on the general principle that less force is required to produce a change in a body when the change is unopposed by constraints than when it is subjected to such. From this if we assume the external forces acting upon a system not to rise above a certain potential (which is the order of nature), the constraints of secondary actions may, under certain circumstances, lead to final rejection of some of the energy, or, in any

[1] _Theory of Heat_, p. 131.

65

case, to retardation of change in the system—dissipation of energy being the result.[1]

As such constraints seem inherently present in the properties of matter, we may summarise as follows:

_The transfer of energy into any inanimate material system is attended by effects retardative to the transfer

and conducive to
dissipation._

Was this the only possible dynamic order ruling in
material
systems it is quite certain the myriads of ants
and pines never
could have been, except all generated by creative
act at vast
primary expenditure of energy. Growth and
reproduction would have
been impossible in systems which retarded change
at every step
and never proceeded in any direction but in that
of dissipation.
Once created, indeed, it is conceivable that, as
heat engines,
they might have dragged out an existence of
alternate life and
death; life in the hours of sunshine, death in
hours of darkness:
no final death, however, their lot, till their
parts were simply
worn out by long use, never made good by repair.
But the
sustained and increasing activity of organized
nature is a fact;
therefore some other order of events must be
possible.

[1] The law of Least Action, which has been
applied, not alone in
optics, but in many mechanical systems, appears
physically based
upon the restraint and retardation opposing the
transfer of
energy in material systems.

GENERAL DYNAMIC CONDITIONS ATTENDING ANIMATE ACTIONS

What is the actual dynamic attitude of the primary organic engine—the vegetable organism? We consider, here, in the first place, not intervening, but resulting phenomena.

The young leaf exposed to solar radiation is small at first, and the quantity of radiant energy it receives in unit of time cannot exceed that which falls upon its surface. But what is the effect of this energy? Not to produce a retardative reaction, but an accelerative response: for, in the enlarging of the leaf by growth, the plant opens for itself new channels of supply.

If we refer to "the living protoplasm which, with its unknown molecular arrangement, is the only absolute test of the cell and of the organism in general,[1] we find a similar attitude towards external sources of available energy. In the act of growth increased rate of assimilation is involved, so that there is an acceleration of change till a bulk of maximum activity is attained. The surface, finally, becomes too small for the

absorption of energy adequate to sustain further increase of mass (Spencer[2]), and the acceleration ceases. The waste going on in the central parts is then just balanced by the renewal at the surface. By division, by spreading of the mass, by

[1] Claus, *_Zoology_*, p. 13.

[2] Geddes and Thomson, *_The Evolution of Sex_*, p. 220.

67

out-flowing processes, the normal activity of growth may be restored. Till this moment nothing would be gained by any of these changes. One or other of them is now conducive to progressive absorption of energy by the organism, and one or other occurs, most generally the best of them, subdivision. Two units now exist; the total mass immediately on division is unaltered, but paths for the more abundant absorption of energy are laid open.

The encystment of the protoplasm (occurring under conditions upon which naturalists do not seem agreed[1]) is to all appearance protective from an unfavourable environment, but it is often a period of internal change as well, resulting in a segregation

within the mass of numerous small units, followed by a breakup of the whole into these units. It is thus an extension of the basis of supply, and in an impoverished medium, where unit of surface is less active, is evidently the best means of preserving a condition of progress.

Thus, in the organism which forms the basis of all modes of life, a definite law of action is obeyed under various circumstances of reaction with the available energy of its environment.

Similarly, in the case of the more complex leaf, we see, not only in the phenomenon of growth, but in its extension in a flattened form, and in the orientation of greatest surface towards the source of energy, an attitude towards

[1] However, "In no way comparable with death."
Weismann,
Biological Memoirs, p. 158.

68

available energy causative of accelerated transfer. There is seemingly a principle at work, leading to the increase of organic activity.

Many other examples might be adduced. The gastrula stage in the

development of embryos, where by invagination such an arrangement of the multiplying cells is secured as to offer the greatest possible surface consistent with a first division of labour; the provision of cilia for drawing upon the energy supplies of the medium; and more generally the specialisation of organs in the higher developments of life, may alike be regarded as efforts of the organism directed to the absorption of energy. When any particular organ becomes unavailing in the obtainment of supplies, the organ in the course of time becomes aborted or disappears.[1] On the other hand, when a too ready and liberal supply renders exertion and specialisation unnecessary, a similar abortion of functionless organs takes place. This is seen in the degraded members of certain parasites.

During certain epochs of geological history, the vegetable world developed enormously; in response probably to liberal supplies of carbon dioxide. A structural adaptation to the rich atmosphere occurred, such as was calculated to cooperate in rapidly consuming the supplies, and to this obedience to a law of progressive transfer of energy we owe the vast

stores of energy
now accumulated

[1] Claus, Zoology, p. 157

69

in our coal fields. And when, further, we reflect that this store of energy had long since been dissipated into space but for the intervention of the organism, we see definitely another factor in organic transfer of energy—a factor acting conservatively of energy, or antagonistically to dissipation.

The tendency of organized nature in the presence of unlimited supplies is to "run riot." This seems so universal a relation, that we are safe in seeing here cause and effect, and in drawing our conclusions as to the attitude of the organism towards available energy. New species, when they come on the field of geological history, armed with fresh adaptations, irresistible till the slow defences of the subjected organisms are completed, attain enormous sizes under the stimulus of abundant supply, till finally, the environment, living and dead, reacts upon them with restraining influence. The exuberance of the organism in presence of energy is often so abundant as to lead by

deprivation to its self-destruction. Thus the growth of bacteria is often controlled by their own waste products. A moment's consideration shows that such progressive activity denotes an accelerative attitude on the part of the organism towards the transfer of energy into the organic material system. Finally, we are conscious in ourselves how, by use, our faculties are developed; and it is apparent that all such progressive developments must rest on actions which respond to supplies with fresh demands. Possibly in the present and ever-

70

increasing consumption of inanimate power by civilised races, we see revealed the dynamic attitude of the organism working through thought-processes.

Whether this be so or not, we find generally in organised nature causes at work which in some way lead to a progressive transfer of energy into the organic system. And we notice, too, that all is not spent, but both immediately in the growth of the individual, and ultimately in the multiplication of the species, there are actions associated with vitality which

retard the
dissipation of energy. We proceed to state the
dynamical
principles involved in these manifestations, which
appear
characteristic of the organism, as follows:—

—The transfer of energy into any animate material
system is
attended by effects conducive to the transfer, and
retardative of
dissipation.—

This statement is, I think, perfectly general. It
has been in
part advanced before, but from the organic more
than the physical
point of view. Thus, "hunger is an essential
characteristic of
living matter"; and again, "hunger is a dominant
characteristic
of living matter,"[1] are, in part, expressions of
the statement.

If it be objected against the generality of the
statement, that
there are periods in the life of individuals when
stagnation and
decay make their appearance, we may answer, that

[1] —Evolution of Sex.— Geddes and Thomson, chap.
xvi. See also a
reference to Cope's theory of "Growth Force," in
Wallace's
—Darwinism—, p. 425.

71

such phenomena arise in phases of life developed
under conditions

of external constraint, as will be urged more fully further on, and that in fact the special conditions of old age do not and cannot express the true law and tendency of the dynamic relations of life in the face of its evident advance upon the Earth. The law of the unconstrained cell is growth on an ever increasing scale; and although we assume the organic configuration, whether somatic or reproductive, to be essentially unstable, so that continual inflow of energy is required merely to keep it in existence, this does not vitiate the fact that, when free of all external constraint, growth gains on waste. Indeed, even in the case of old age, the statement remains essentially true, for the phenomena then displayed point to a breakdown of the functioning power of the cell, an approximation to configurations incapable of assimilation. It is not as if life showed in these phenomena that its conditions could obtain in the midst of abundance, and yet its law be suspended; but as if they represented a degradation of the very conditions of life, a break up, under the laws of the inanimate, of the animate contrivance; so that energy is no longer available to it, or the primary

condition, "the transfer of energy into the animate system," is imperfectly obeyed. It is to the perfect contrivance of life our statement refers.

That the final end of all will be general non-availability there seems little reason to doubt, and the organism, itself dependent upon differences of potential, cannot

72

hope to carry on aggregation of energy beyond the period when differences of potential are not. The organism is not accountable for this. It is being affected by events external to it, by the actions going on through inanimate agents. And although there be only a part of the received energy preserved, there is a part preserved, and this amount is continually on the increase. To see this it is only necessary to reflect that the sum of animate energy—capability of doing work in any way through animate means—at present upon the Earth, is the result, although a small one, of energy reaching the Earth since a remote period, and which otherwise had been dissipated in space. In inanimate actions throughout nature, as we know it, the

availability is continually diminishing. The change is all the one way. As, however, the supply of available energy in the universe is (probably) limited in amount, we must look upon the two as simply effecting the final dissipation of potential in very different ways. The animate system is aggressive on the energy available to it, spends with economy, and invests at interest till death finally deprives it of all. It has heirs, indeed, who inherit some of its gains, but they, too, must die, and ultimately there will be no successors, and the greater part must melt away as if it had never been. The inanimate system responds to the forces imposed upon it by sluggish changes; of that which is thrust upon it, it squanders uselessly. The path of the energy is very different in the two cases.

73

While it is true generally that both systems ultimately result in the dissipation of energy to uniform potential, the organism can, as we have seen, under particular circumstances evade the final doom altogether. It can lay up a store of potential energy which may be permanent. Thus, so long as there is free

oxygen in the universe, our coalfields might, at any time in the remote future, generate light and heat in the universal grave.

It is necessary to observe on the fundamental distinction between the growth of the protoplasm and the growth of the crystal. It is common to draw comparison between the two, and to point to metabolism as the chief distinction. But while this is the most obvious distinction the more fundamental one remains in the energy relations of the two with the environment.

[1] The growth of the crystal is the result of loss of energy; that of the organism the result of gain of energy. The crystal represents a last position of stable equilibrium assumed by molecules upon a certain loss of kinetic energy, and the formation of the crystal by evaporation and concentration of a liquid does not, in its dynamic aspect, differ much from the precipitation of an amorphous sediment. The organism, on the other hand, represents a more or less unstable condition formed and maintained by inflow of energy; its formation, indeed, often attended with a loss of kinetic energy (fixation of carbon in plants),

but, if so,
accompanied by

[1] It appears exceptional for the crystal line configuration to stand higher in the scale of energy than the amorphous.

74

a more than compensatory increase of potential molecular energy.

Thus, between growth in the living world and growth in the dead world, the energy relations with the environment reveal a marked contrast. Again, in the phenomena of combustion, there are certain superficial resemblances which have led to comparison between the two. Here again, however, the attitudes towards the energy of the environment stand very much as + and -. The life absorbs, stores, and spends with economy. The flame only recklessly spends. The property of storage by the organism calls out a further distinction between the course of the two processes. It secures that the chemical activity of the organism can be propagated in a medium in which the supply of energy is discontinuous or localised. The chemical activity of the combustion can, strictly speaking, only be

propagated among
contiguous particles. I need not dwell on the
latter fact; an
example of the former is seen in the action of the
roots of
plants, which will often traverse a barren place
or circumvent an
obstacle in their search for energy. In this
manner roots will
find out spots of rich nutriment.

Thus there is a dynamic distinction between the
progress of the
organism and the progress of the combustion, or of
the chemical
reaction generally. And although there be unstable
chemical
systems which absorb energy during reaction, these
are
(dynamically) no more than the expansion of the
compressed gas.
There is a certain

75

initial capacity in the system for a given
quantity of energy;
this satisfied, progress ceases. The progress of
the organism in
time is continual, and goes on from less to
greater so long as
its development is unconstrained and the supply of
energy is
unlimited.

We must regard the organism as a configuration
which is so
contrived as to evade the tendency of the

universal laws of nature. Except we are prepared to believe that a violation of the second law of thermodynamics occurs in the organism, that a "sorting demon" is at work within it, we must, I think, assume that the interactions going on among its molecules are accompanied by retardation and dissipation like the rest of nature. That such conditions are not incompatible with the definition of the dynamic attitude of the organism, can be shown by analogy with our inanimate machines which, by aid of hypotheses in keeping with the second law of thermodynamics, may be supposed to fulfil the energy-functions of the plant or animal, and, in fact, in all apparent respects conform to the definition of the organism.

We may assume this accomplished by a contrivance of the nature of a steam-engine, driven by solar energy. It has a boiler, which we may suppose fed by the action of the engine. It has piston, cranks, and other movable parts, all subject to resistance from friction, etc. Now there is no reason why this engine should not expend its surplus energy in shaping, fitting, and starting into

action other engines:—in fact, in reproductive sacrifice. All

76

these other engines represent a multiplied absorption of energy as the effects of the energy received by the parent engine, and may in time be supposed to reproduce themselves. Further, we may suppose the parent engine to be small and capable of developing very little power, but the whole series as increasing in power at each generation. Thus the primary energy relations of the vegetable organism are represented in these engines, and no violation of the second law of thermodynamics involved.

We might extend the analogy, and assuming these engines to spend a portion of their surplus energy in doing work against chemical forces—as, for example, by decomposing water through the intervention of a dynamo—suppose them to lay up in this way a store of potential energy capable of heating the boilers of a second order of engines, representing the graminivorous animal. It is obvious without proceeding to a tertiary or carnivorous order, that the condition of energy in the animal world may be

supposed fulfilled in these successive series of engines, and no violation of the principles governing the actions going on in our machines assumed. Organisms evolving on similar principles would experience loss at every transfer. Thus only a portion of the radiant energy absorbed by the leaf would be expended in actual work, chemical and gravitational, etc. It is very certain that this is, in fact, what takes place.

It is, perhaps, worth passing observation that, from the nutritive dependence of the animal upon the vegetable,

77

and the fact that a conversion of the energy of the one to the purposes of the other cannot occur without loss, the mean energy absorbed daily by the vegetable for the purpose of growth must greatly exceed that used in animal growth; so that the chemical potential energy of vegetation upon the earth is much greater than the energy of all kinds represented in the animal configurations.[1] It appears, too, that in the power possessed by the vegetable of remaining comparatively inactive, of surviving hard times by the expenditure and

absorption of but little, the vegetable constitutes a veritable reservoir for the uniform supply of the more unstable and active animal.

Finally, on the question of the manner of origin of organic systems, it is to be observed that, while the life of the present is very surely the survival of the fittest of the tendencies and chances of the past, yet, in the initiation of the organised world, a single chance may have decided a whole course of events: for, once originated, its own law secures its increase, although within the new order of actions, the law of the fittest must assert itself. That such a progressive material system as an organism was possible, and at some remote period was initiated, is matter of knowledge; whether or not the initiatory living configuration was rare and fortuitous, or the probable result of the general action of physical laws acting among innumerable chances, must remain matter of

[1] I find a similar conclusion arrived at in Semper's *_Animal Life_*, p. 52.

speculation. In the event of the former being the truth, it is evidently possible, in spite of a large finite number of habitable worlds, that life is non-existent elsewhere. If the latter is the truth, it is almost certain that there is life in all, or many of those worlds.

EVOLUTION AND ACCELERATION OF ACTIVITY

The primary factor in evolution is the "struggle for existence."

This involves a "natural selection" among the many variations of the organism. If we seek the underlying causes of the struggle, we find that the necessity of food and (in a lesser degree) the desire for a mate are the principal causes of contention. The former is much the more important factor, and, accordingly, we find the greater degree of specialisation based upon it.

The present view assumes a dynamic necessity for its demands involved in the nature of the organism as such. This assumption is based on observation of the outcome of its unconstrained growth, reproduction, and life-acts. We have the same right to assert this of the organism as we have to assert that retardation and degradation attend the actions of inanimate

machines, which
assertion, also, is based on observation of
results. Thus we pass
from the superficial statements that organisms
require food in
order to live, or that organisms desire food, to
the more
fundamental one that:

_The organism is a configuration of matter which
absorbs energy
acceleratively, without limit, when
unconstrained._

79

This is the dynamic basis for a "struggle for
existence." The
organism being a material system responding to
accession of
energy with fresh demands, and energy being
limited in amount,
the struggle follows as a necessity. Thus,
evolution guiding' the
steps of the energy-seeking organism, must
presuppose and find
its origin in that inherent property of the
organism which
determines its attitude in presence of available
energy.

Turning to the factor, "adaptation," we find that
this also must
presuppose, in order to be explicable, some
quality of
aggressiveness on the part of the organism. For
adaptation in
this or that direction is the result of repulse or

victory, and,
therefore, we must presuppose an attack. The
attack is made by
the organism in obedience to its law of demand; we
see in the
adaptation of the organism but the accumulated
wisdom derived
from past defeats and victories.

Where the environment is active, that is living,
adaptation
occurs on both sides. Improved means of defence or
improved means
of attack, both presuppose activity. Thus the
reactions to the
environment, animate and inanimate, are at once
the outcome of
the eternal aggressiveness of the organism, and
the source of
fresh aggressiveness upon the resources of the
medium.

As concerns the "survival of the fittest" (or
"natural
selection"), we can, I think, at once conclude
that the organism
which best fulfils the organic law under the
circumstances of
supply is the "fittest," *ipso facto*. In many

80

cases this is contained in the commonsense
consideration, that to
be strong, consistent with concealment from
enemies which are
stronger, is best, as giving the organism mastery
over foes which

are weaker, and generally renders it better able to secure supplies. Weismann points out that natural selection favours early and abundant reproduction. But whether the qualifications of the "fittest" be strength, fertility, cunning, fleetness, imitation, or concealment, we are safe in concluding that growth and reproduction must be the primary qualities which at once determine selection and are fostered by it. Inherent in the nature of the organism is accelerated absorption of energy, but the qualifications of the "fittest" are various, for the supply of energy is limited, and there are many competitors for it. To secure that none be wasted is ultimately the object of natural selection, deciding among the eager competitors what is best for each.

In short, the facts and generalisations concerning evolution must presuppose an organism endowed with the quality of progressive absorption of energy, and retentive of it. The continuity of organic activity in a world where supplies are intermittent is evidently only possible upon the latter condition. Thus it appears that the dynamic attitude of the organism,

considered in
these pages, occupies a fundamental position
regarding its
evolution.

We turn to the consideration of old age and death,
endeavouring
to discover in what relation they stand to the
innate
progressiveness of the organism.

81

THE PERIODICITY OF THE ORGANISM AND THE LAW OF PROGRESSIVE ACTIVITY

The organic system is essentially unstable. Its
aggressive
attitude is involved in the phenomenon of growth,
and in
reproduction which is a form of growth. But the
energy absorbed
is not only spent in growth. It partly goes, also,
to make good
the decay which arises from the instability of the
organic unit.

The cell is molecularly perishable. It possesses
its entity much
as a top keeps erect, by the continual inflow of
energy.

Metabolism is always taking place within it. Any
other condition
would, probably, involve the difficulties of
perpetual motion.

The phenomenon of old age is not evident in the
case of the
unicellular organism reproducing by fission. At

any stage of its
history all the individuals are of the same age:
all contain a
like portion of the original cell, so far as this
can be regarded
as persisting where there is continual flux of
matter and energy.
In the higher organisms death is universally
evident. Why is
this?

The question is one of great complexity.
Considered from the more
fundamental molecular point of view we should
perhaps look to
failure of the power of cell division as the
condition of
mortality. For it is to this phenomenon—that of
cell
division—that the continued life of the protozoon
is to be
ascribed, as we have already seen. Reproduction
is, in fact, the
saving factor here.

As we do not know the source or nature of the
stimulus

82

responsible for cell division we cannot give a
molecular account
of death in the higher organisms. However we shall
now see that,
philosophically, we are entitled to consider
reproduction as a
saving factor in this case also; and to regard the
death of the

individual much as we regard the fall of the leaf from the tree:

i.e. as the cessation of an outgrowth from a development

extending from the past into the future. The phenomena of old age

and natural death are, in short, not at variance with the

progressive activity of the organism. We perceive this when we

come to consider death from the evolutionary point of view.

Professor Weismann, in his two essays, "The Duration of Life,"

and "Life and Death,"[1] adopts and defends the view that "death

is not a primary necessity but that it has been secondarily

acquired by adaptation." The cell was not inherently limited in

its number of cell-generations. The low unicellular organisms are

potentially immortal, the higher multicellular forms with

well-differentiated organs contain the germs of death within

themselves.

He finds the necessity of death in its utility to the species.

Long life is a useless luxury. Early and abundant reproduction is

best for the species. An immortal individual would gradually

become injured and would be valueless or even harmful to the

species by taking the place of those that are

sound. Hence
natural selection will shorten life.

[1] See his *Biological Memoirs.* Oxford, 1888.

83

Weismann contends against the transmission of acquired characters as being unproved.[1] He bases the appearance of death on variations in the reproductive cells, encouraged by the ceaseless action of natural selection, which led to a differentiation into perishable somatic cells and immortal reproductive cells. The time-limit of any particular organism ultimately depends upon the number of somatic cell-generations and the duration of each generation. These quantities are "predestined in the germ itself" which gives rise to each individual. "The existence of immortal metazoan organisms is conceivable," but their capacity for existence is influenced by conditions of the external world; this renders necessary the process of adaptation. In fact, in the differentiation of somatic from reproductive cells, material was provided upon which natural selection could operate to shorten or to lengthen the life of the individual in accordance with the needs of the species. The soma is in a sense "a

secondary
appendage of the real bearer of life—the
reproductive cells." The
somatic cells probably lost their immortal
qualities, on this
immortality becoming useless to the species. Their
mortality may
have been a mere consequence of their
differentiation (loc. cit.,
p. 140), itself due to natural selection. "Natural
death was
not," in fact, "introduced from absolute intrinsic
necessity
inherent in the nature of living matter, but on
grounds of
utility,

[1] Biological Memoirs, p. 142.

84

that is from necessities which sprang up, not from
the general
conditions of life, but from those special
conditions which
dominate the life of multicellular organisms."

On the inherent immortality of life, Weismann
finally states:

"Reproduction is, in truth, an essential attribute
of living
matter, just as the growth which gives rise to
it.... Life is
continuous, and not periodically interrupted: ever
since its
first appearance upon the Earth in the lowest
organism, it has
continued without break; the forms in which it is

manifest have
alone undergone change. Every individual alive
today—even the
highest—is to be derived in an unbroken line from
the first and
lowest forms." [1]

At the present day the view is very prevalent that
the soma of
higher organisms is, in a sense, but the carrier
for a period of
the immortal reproductive cells (Ray Lankester)[2]
—an appendage
due to adaptation, concerned in their supply,
protection, and
transmission. And whether we regard the time-limit
of its
functions as due to external constraints,
recurrently acting till
their effects become hereditary, or to variations
more directly
of internal origin, encouraged by natural
selection, we see in
old age and death phenomena ultimately brought
about in obedience
to the action of an environment. These are not
inherent in the
properties of living matter. But, in spite

[1] Loc. cit., p. 159

[2] Geddes and Thomson, *The Evolution of Sex*,
chap. xviii.

85

of its mortality, the body remains a striking
manifestation of
the progressiveness of the organism, for to this

it must be ascribed. To it energy is available which is denied to the protozoon. Ingenious adaptations to environment are more especially its privilege. A higher manifestation, however, was possible, and was found in the development of mind. This, too, is a servant of the cell, as the genii of the lamp. Through it energy is available which is denied to the body. This is the masterpiece of the cell. Its activity dates, as it were, but from yesterday, and today it inherits the most diverse energies of the Earth.

Taking this view of organic succession, we may liken the individual to a particle vibrating for a moment and then coming to rest, but sweeping out in its motion one wave in the continuous organic vibration travelling from the past into the future. But as this vibration is one spreading with increased energy from each vibrating particle, its propagation involves a continual accelerated inflow of energy from the surrounding medium, a dynamic condition unknown in periodic effects transmitted by inanimate actions, and, indeed, marking the

fundamental difference between the dynamic attitudes of the animate and inanimate.

We can trace the periodic succession of individuals on a diagram of activity with some advantage. Considering, first, the case of the unicellular organism reproducing by subdivision and recalling that conditions, definite and inevitable, oppose a limit to the rate of growth, or, for our

86

present purpose, rate of consumption of energy, we proceed as follows:

{Fig. 1}

Along a horizontal axis units of time are measured; along a vertical axis units of energy. Then the life-history of the amoeba, for example, appears as a line such as A in Fig. 1.

During the earlier stages of its growth the rate of absorption of energy is small; so that in the unit interval of time, t , the small quantity of energy, e_1 , is absorbed. As life advances, the activity of the organism augments, till finally this rate attains a maximum, when e_2 units of energy are consumed in the unit of time.[1]

[1] Reference to p. 76, where the organic system is treated as purely mechanical, may help readers to understand what is involved in this curve. The solar engine may, unquestionably, have its activity defined by such a curve. The organism is, indeed, more complex; but neither this fact nor our ignorance of its mechanism, affects the principles which justify the diagram.

87

On this diagram reproduction, on the part of the organism, is represented by a line which repeats the curvature of the parent organism originating at such a point as P in the path of the latter, when the rate of consumption of energy has become constant. The organism A has now ceased to act as a unit. The products of fission each carry on the vital development of

{Fig. 2}

the species along the curve B, which may be numbered (2), to signify that it represents the activity of two individuals, and so on, the numbering advancing in geometrical progression. The particular curvature adopted in the diagram is, of course,

imaginary; but it is not of an indeterminate nature. Its course for any species is a characteristic of fundamental physical importance, regarding the part played in nature by the particular organism.

88

In Fig. 2 is represented the path of a primitive multicellular organism before the effects of competition produced or fostered its mortality. The lettering of Fig. 1 applies; the successive reproductive acts are marked P1, P2; Q1, Q2, etc., in the paths of the successive individuals.

{Fig. 3}

The next figure (Fig. 3) diagrammatically illustrates death in organic history. The path ever turns more and more from the axis of energy, till at length the point is reached when no more energy is available; a tangent to the curve at this point is at right angles to the axis of energy and parallel to the time axis. The death point is reached, and however great a length we measure along the axis of time, no further consumption of energy is

89

indicated by the path of the organism. Drawing the line beyond the death point is meaningless for our present purpose.

It is observable that while the progress of animate nature finds its representation on this diagram by lines sloping upwards from left to right, the course of events in inanimate nature—for example, the history of the organic configuration after death, or

{Fig. 4}

the changes progressing—let us say, in the solar system, or in the process of a crystallisation, would appear as lines sloping downwards from left to right.

Whatever our views on the origin of death may be, we have to recognise a periodicity of functions in the life-history of the successive individuals of the present day; and whether or not we trace this directly or indirectly to

90

a sort of interference with the rising wave of life, imposed by the activity of a series of derived units, each seeking energy, and in virtue of its adaptation each being more fitted to obtain it than its predecessor, or even leave the idea of

interference
out of account altogether in the origination or
perpetuation of
death, the truth of the diagram (Fig. 4) holds in
so far as it
may be supposed to graphically represent the
dynamic history of
the individual. The point chosen on the curve for
the origination
of a derived unit is only applicable to certain
organisms, many
reproducing at the very close of life. A chain of
units are
supposed here represented.[1]

THE LENGTH OF LIFE

If we lay out waves as above to a common scale of
time for
different species, the difference of longevity is
shown in the
greater or less number of vibrations executed in a
given time,
i.e. in greater or less "frequency." We cannot
indeed draw the
curvature correctly, for this would necessitate a
knowledge which
we have not of the activity of the organism at
different periods
of its life-history, and so neither can we plot
the direction of
the organic line of propagation with respect to
the

[1] Projecting upon the axes of time and energy
any one complete
vibration, as in Fig. 4, the total energy consumed
by the

organism during life is the length E on the axis of energy, and its period of life is the length T on the time-axis. The mean activity is the quotient E/T .

91

axes of reference as this involves a knowledge of the mean activity.[1]

The group of curves which follow, relating to typical animals possessing very different activities (Fig. 5), are therefore entirely diagrammatic, except in respect to the approximate

{Fig. 5}

longevity of the organisms. (1) might represent an animal of the length of life and of the activity of Man; (2), on the same scale of longevity,

[1] In the relative food-supply at various periods of life the curvature is approximately determinable.

92

one of the smaller mammals; and (3), the life-history of a cold blooded animal living to a great age; _e.g._ certain of the reptilia.

It is probable, that to conditions of structural development, under the influence of natural selection, the question of longer or shorter life is in a great degree referable. Thus, development along lines of large growth will tend to a slow rate of reproduction from the simple fact that unlimited energy to supply abundant reproduction is not procurable, whatever we may assume as to the strength or cunning exerted by the individual in its efforts to obtain its supplies. On the other hand, development along lines of small growth, in that reproduction is less costly, will probably lead to increased rate of reproduction. It is, in fact, matter of general observation that in the case of larger animals the rate of reproduction is generally slower than in the case of smaller animals. But the rate of reproduction might be expected to have an important influence in determining the particular periodicity of the organism. Were we to depict in the last diagram, on the same time-scale as Man, the vibrations of the smaller and shorter-lived living things, we would see but a straight line, save for secular variations in activity, representing the progress of the species in time:

the tiny
thrills of its units lost in comparison with the
yet brief period
of Man.

The interdependence of the rate of reproduction
and

93

the duration of the individual is, indeed, very
probably revealed
in the fact that short-lived animals most
generally reproduce
themselves rapidly and in great abundance, and
vice versa. In
many cases where this appears contradicted, it
will be found that
the young are exposed to such dangers that but few
survive (_e.g._
many of the reptilia, etc.), and so the rate of
reproduction is
actually slow.

Death through the periodic rigour of the inanimate
environment
calls forth phenomena very different from death
introduced or
favoured by competition. A multiplicity of effects
simulative of
death occur. Organisms will, for example, learn to
meet very
rigorous conditions if slowly introduced, and not
permanent. A
transitory period of want can be tided over by
contrivance. The
lily withdrawing its vital forces into the bulb,
protected from

the greatest extremity of rigour by seclusion in the Earth; the trance of the hibernating animal; are instances of such contrivances.

But there are organisms whose life-wave truly takes up the periodicity of the Earth in its orbit. Thus the smaller animals and plants, possessing less resources in themselves, die at the approach of winter, propagating themselves by units which, whether egg or seed, undergo a period of quiescence during the season of want. In these quiescent units the energy of the organism is potential, and the time-energy function is in abeyance. This condition is, perhaps, foreshadowed in the encyst-

94

ment of the amoeba in resistance to drought. In most cases of hibernation the time-energy function seems maintained at a loss of potential by the organism, a diminished vital consumption of energy being carried on at the expense of the stored energy of the tissues. So, too, even among the largest organisms there will be a diminution of activity periodically inspired by

climatological conditions. Thus, wholly or in part, the activity of organisms is recurrently affected by the great energy-tides set up by the Earth's orbital motion.

{Fig. 6}

Similarly in the phenomenon of sleep the organism responds to the Earth's axial periodicity, for in the interval of night a period of impoverishment has to be endured. Thus the diurnal waves of energy also meet a response in the organism. These tides and waves of activity would appear as larger and smaller ripples

95

on the life-curve of the organism. But in some, in which life and death are encompassed in a day, this would not be so; and for the annual among plants, the seed rest divides the waves with lines of no activity (Fig. 6).

Thus, finally, we regard the organism as a dynamic phenomenon passing through periodic variations of intensity. The material systems concerned in the transfer of the energy rise, flourish, and fall in endless succession, like cities of ancient dynasties. At points of similar phase upon the waves the rate of consumption

of energy is approximately the same; the functions, too, which demand and expend the energy are of similar nature.

That the rhythm of these events is ultimately based on harmony in the configuration and motion of the molecules within the germ seems an unavoidable conclusion. In the life of the individual rhythmic dynamic phenomena reappear which in some cases have no longer a parallel in the external world, or under conditions when the individual is no longer influenced by these external conditions.,, In many cases the periodic phenomena ultimately die out under new influences, like the oscillations of a body in a viscous medium; in others when they seem to be more deeply rooted in physiological conditions they persist.

The "length of life is dependent upon the number

[1] The _Descent of Man._

96

of generations of somatic cells which can succeed one another in the course of a single life, and furthermore the number as well as the duration of each single cell-generation is predestined in the germ itself." [1]

Only in the vague conception of a harmonising or formative structural influence derived from the germ, perishing in each cell from internal causes, but handed from cell to cell till the formative influence itself degrades into molecular discords, does it seem possible to form any physical representation of the successive events of life. The degradation of the molecular formative influence might be supposed involved in its frequent transference according to some such dynamic actions as occur in inanimate nature. Thus, ultimately, to the waste within the cell, to the presence of a force retardative of its perpetual harmonic motions, the death of the individual is to be ascribed. Perhaps in protoplasmic waste the existence of a universal death should be recognised. It is here we seem to touch inanimate nature; and we are led back to a former conclusion that the organism in its unconstrained state is to be regarded as a contrivance for evading the dynamic tendencies of actions in which lifeless matter participates.[2]

[1] Weismann, _Life and Death; Biological Memoirs_, p. 146.

[2] In connection with the predestinating power and possible complexity of the germ, it is instructive to reflect on the very great molecular population of even the smallest spores—giving rise to very simple forms. Thus, the spores of the unicellular Schizomycetes are estimated to dimensions as low as 1/10,000 of a millimetre in diameter (Cornil et Babes, *Les Batteries*, 1. 37). From Lord Kelvin's estimate of the number of molecules in water, comprised within the length of a wave-length of yellow light (*The Size of Atoms*, Proc. R. I., vol. x., p. 185) it is probable that such spores contain some 500,000 molecules, while one hundred molecules range along a diameter.

97

THE NUMERICAL ABUNDANCE OF LIFE

We began by seeking in various manifestations of life a dynamic principle sufficiently comprehensive to embrace its very various phenomena. This, to all appearance, found, we have been led to regard life, to a great extent, as a periodic dynamic phenomenon. Fundamentally, in that characteristic of the contrivance, which leads it to respond favourably to transfer of energy, its

enormous extension is due. It is probable that to its instability its numerical abundance is to be traced—for this, necessitating the continual supply of all the parts already formed, renders large, undifferentiated growth, incompatible with the limited supplies of the environment. These are fundamental conditions of abundant life upon the Earth.

Although we recognise in the instability of living systems the underlying reason for their numerical abundance, secondary evolutionary causes are at work. The most important of these is the self-favouring nature of the phenomenon of reproduction. Thus there is a tendency not only to favour reproductiveness, but early reproductiveness, in the form of one prolific reproductive.

98

act, after which the individual dies.[1] Hence the wavelength of the species diminishes, reproduction is more frequent, and correspondingly greater numbers come and go in an interval of time.

Another cause of the numerical abundance of life exists, as

already stated, in the conditions of nourishment. Energy is more readily conveyed to the various parts of the smaller mass, and hence the lesser organisms will more actively functionate; and this, as being the urging dynamic attitude, as well as that most generally favourable in the struggle, will multiply and favour such forms of life. On the other hand, however, these forms will have less resource within themselves, and less power of endurance, so that they are only suitable to fairly uniform conditions of supply; they cannot survive the long continued want of winter, and so we have the seasonal abundance of summer. Only the larger and more resistant organisms, whether animal or vegetable, will, in general, populate the Earth from year to year. From this we may conclude that, but for the seasonal energy-tides, the development of life upon the globe had gone along very different lines from those actually followed. It is, indeed, possible that the evolution of the larger organisms would not have occurred; there would have been no vacant place for their development, and a being so endowed as Man could hardly

[1] Weismann, _The Duration of Life._

99

have been evolved. We may, too, apply this reasoning elsewhere, and regard as highly probable, that in worlds which are without seasonal influences, the higher developments of life have not appeared; except they have been evolved under other conditions, when they might for a period persist. We have, indeed, only to picture to ourselves what the consequence of a continuance of summer would be on insect life to arrive at an idea of the antagonistic influences obtaining in such worlds to the survival of larger organisms.

It appears that to the dynamic attitude of life in the first place, and secondarily to the environmental conditions limiting undifferentiated growth, as well as to the action of heredity in transmitting the reproductive qualities of the parent to the offspring, the multitudes of the pines, and the hosts of ants, are to be ascribed. Other causes are very certainly at work, but these, I think, must remain primary causes.

We well know that the abundance of the ants and pines is not a

tithe of the abundance around us visible and
invisible. It is a
vain endeavour to realise the countless numbers of
our
fellow-citizens upon the Earth; but, for our
purpose, the
restless ants, and the pines solemnly quiet in the
sunshine, have
served as types of animate things. In the pine the
gates of the
organic have been thrown open that the vivifying
river of energy
may flow in. The ants and the butterflies sip for
a brief moment
of its waters, and again vanish into the

100

inorganic: life, love and death encompassed in a
day.

Whether the organism stands at rest and life comes
to it on the
material currents of the winds and waters, or in
the vibratory
energy of the æther; or, again, whether with
restless craving it
hurries hither and thither in search of it,
matters nothing. The
one principle—the accelerative law which is the
law of the
organic—urges all alike onward to development,
reproduction and
death. But although the individual dies death is
not the end; for
life is a rhythmic phenomenon. Through the passing
ages the waves
of life persist: waves which change in their form

and in the frequency to which they are attuned from one geologic period to the next, but which still ever persist and still ever increase. And in the end the organism outlasts the generations of the hills.

101

THE BRIGHT COLOURS OF ALPINE FLOWERS [1]

IT is admitted by all observers that many species of flowering plants growing on the higher alps of mountainous regions display a more vivid and richer colour in their bloom than is displayed in the same species growing in the valleys. That this is actually the case, and not merely an effect produced upon the observer by the scant foliage rendering the bloom more conspicuous, has been shown by comparative microscopic examination of the petals of species growing on the heights and in the valleys. Such examination has revealed that in many cases pigment granules are more numerous in the individuals growing at the higher altitudes. The difference is specially marked in *Myosotis sylvatica*, *Campanula rotundifolia*, *Ranunculus sylvaticus*, *Galium cruciatum*, and others. It is less marked in the case of

Thymus serpyllum and
Geranium sylvaticum; while in Rosa alpina and
Erigeron alpinus no
difference is observable.[2]

In the following cases a difference of intensity
of colour is,
according to Kerner ("Pflanzenleben," 11. 504),
especially
noticeable:— _Agrostemma githago, Campanula

[1] _Proc. Royal Dublin Society_, 1893.

[2] G. Bonnier, quoted by De Varigny,
Experimental Evolution,
p. 55.

102

pusilla, Dianthus inodorus (silvestris),
Gypsophila repens, Lotus
corniculatus, Saponaria ocymoides, Satureja
hortensis, Taraxacumm
officinale, Vicia cracca, and Vicia sepium._

To my own observation this beautiful phenomenon
has always
appeared most obvious and impressive. It appears
to have struck
many unprofessional observers. Helmholtz offers
the explanation
that the vivid colours are the result of the
brighter sunlight of
the heights. It has been said, too, that they are
the direct
chemical effects of a more highly ozonized
atmosphere. The latter
explanation I am unable to refer to its author.
The following

pages contain a suggestion on the matter, which occurred to me while touring, along with Henry H. Dixon, in the Linthal district of Switzerland last summer.[1]

If the bloom of these higher alpine flowers is especially pleasing to our own æsthetic instincts, and markedly conspicuous to us as observers, why not also especially attractive and conspicuous to the insect whose mission it is to wander from flower to flower over the pastures? The answer to this question involves the hypothesis I would advance as accounting for the bright colours of high-growing individuals. In short, I believe a satisfactory explanation is to be found in the conditions of insect life in the higher alps.

In the higher pastures the summer begins late and

[1] The summer of 1892.

103

closes early, and even in the middle of summer the day closes in with extreme cold, and the cold of night is only dispelled when the sun is well up. Again, clouds cover the heights when all is clear below, and cold winds sweep over them when there is warmth and shelter in the valleys. With these rigorous

conditions the
pollinating insects have to contend in their
search for food, and
that when the rival attractions of the valleys
below are so many.
I believe it is these rigorous conditions which
are indirectly
responsible for the bright colours of alpine
flowers. For such
conditions will bring about a comparative scarcity
of insect
activity on the heights; and a scarcity or
uncertainty in the
action of insect agency in effecting fertilization
will intensify
the competition to attract attention, and only the
brightest
blooms will be fertilized.[1]

This will be a natural selection of the brightest,
or the

[1] Grant Allen, I have recently learned, advances
in Science in
Arcady the theory that there is a natural
selective cause
fostering the bright blooms of alpine. The
selective cause is,
however, by him referred to the greater abundance
of butterfly
relatively to bee fertilizers. The former, he
says, display more
æsthetic instinct than bees. In the valley the
bees secure the
fertilization of all. I may observe that upon the
Fridolins Alp
all the fertilizers we observed were bees. I have
always found

butterflies very scarce at altitudes of 7,000 to 8,000 feet. The alpine bees are very light in body, like our hive bee, and I do not think rarefaction of the atmosphere can operate to hinder its ascent to the heights, as Grant Allen suggests. The observations on the death-rate of bees and butterflies on the glacier, to be referred to presently, seem to negative such a hypothesis, and to show that a large preponderance of bees over butterflies make their way to the heights.

104

brightest will be the fittest, and this condition, along with the influence of heredity, will encourage a race of vivid flowers. On the other hand, the more scant and uncertain root supply, and the severe atmospheric conditions, will not encourage the grosser struggle for existence which in the valleys is carried on so eagerly between leaves and branches—the normal offensive and defensive weapons of the plant—and so the struggle becomes refined into the more æsthetic one of colour and brightness between flower and flower. Hence the scant foliage and vivid bloom would be at once the result of a necessary economy, and a

resort to the best method of securing reproduction under the circumstances of insect fertilizing agency. Or, in other words, while the luxuriant growth is forbidden by the conditions, and thus methods of offence and defence, based upon vigorous development, reduced in importance, it would appear that the struggle is mainly referred to rivalry for insect preference. It is probable that this is the more economical manner of carrying on the contest.

In the valleys we see on every side the struggle between the vegetative organs of the plant; the soundless battle among the leaves and branches. The blossom here is carried aloft on a slender stem, or else, taking but a secondary part in the contest, it is relegated to obscurity (Pl. XII.). Further up on the mountains, where the conditions are more severe and the supplies less abundant, the leaf and branch assume lesser dimensions, for they are costly weapons to provide and the elements are unfriendly

105

to their existence (Pl. XIII.). Still higher, approaching the

climatic limit of vegetable life, the struggle for existence is mainly carried on by the æsthetic rivalry of lowly but conspicuous blossoms.

As regards the conditions of insect life in the higher alps, it came to my notice in a very striking manner that vast numbers of such bees and butterflies as venture up perish in the cold of night time. It appears as if at the approach of dusk these are attracted by the gleam of the snow, and quitting the pastures, lose themselves upon the glaciers and firns, there to die in hundreds. Thus in an ascent of the Tödi from the Fridolinshüte we counted in the early dawn sixty-seven frozen bees, twenty-nine dead butterflies, and some half-dozen moths on the Biferten Glacier and Firn. These numbers, it is to be remembered, only included those lying to either side of our way over the snow, so that the number must have mounted up to thousands when integrated over the entire glacier and firn. Approaching the summit none were found. The bees resembled our hive bee in appearance, the butterflies resembled the small white variety common in our gardens, which has yellow and black upon its

wings. One large moth, striped across the abdomen, and measuring nearly two inches in length of body, was found. Upon our return, long after the sun's rays had grown strong, we observed some of the butterflies showed signs of reanimation. We descended so quickly to avoid the inconvenience of the soft snow that we had time for no

106

close observation on the frozen bees. But dead bees are common objects upon the snows of the alps.

These remarks I noted down roughly while at Linthal last summer, but quite recently I read in Natural Science[1] the following note:

"Late Flowering Plants.—While we write, the ivy is in flower, and bees, wasps, and flies are jostling each other and struggling to find standing-room on the sweet-smelling plant. How great must be the advantage obtained by this plant through its exceptional habit of flowering in the late autumn, and ripening its fruit in the spring. To anyone who has watched the struggle to approach the ivy-blossom at a time when nearly all other plants are bare,

it is evident that, as far as transport of pollen and cross-fertilization go, the plant could not flower at a more suitable time. The season is so late that most other plants are out of flower, but yet it is not too late for many insects to be brought out by each sunny day, and each insect, judging by its behaviour, must be exceptionally hungry.

"Not only has the ivy the world to itself during its flowering season, but it delays to ripen its seed till the spring, a time when most other plants have shed their seed, and most edible fruits have been picked by the birds. Thus birds wanting fruit in the spring can obtain little but ivy, and how they appreciate the ivy berry is evident

[1] For December, 1892, vol. i., p. 730.

107

by the purple stains everywhere visible within a short distance of the bush."

These remarks suggest that the ivy adopts the converse attitude towards its visitors to that forced upon the alpine flower. The ivy bloom is small and inconspicuous, but then it has the season to itself, and its inconspicuousness is no

disadvantage, _i.e._
if one plant was more conspicuous than its
neighbours, it would
not have any decided advantage where the
pollinating insect is
abundant and otherwise unprovided for. Its dark-
green berries in
spring, which I would describe as very
inconspicuous, have a
similar advantage in relation to the necessities
of bird life.

The experiments of M. C. Flahault must be noticed.
This
naturalist grew seeds of coloured flowers which
had ripened in
Paris, part in Upsala, and part in Paris; and seed
which had
ripened in Upsala, part at Paris, and part at
Upsala. The flowers
opening in the more northern city were in most
cases the
brighter.[1] If this observation may be considered
indisputable,
as appears to be the case, the question arises,
Are we to regard
this as a direct effect of the more rigorous
climate upon the
development of colouring matter on the blooms
opening at Upsala?
If we suppose an affirmative answer, the theory of
direct effect
by sun brightness must I think be abandoned. But I
venture to
think that the explanation of the Upsala

[1] Quoted by De Varigny, _Experimental
Evolution_, p. 56.

experiment is not to be found in direct climatic influence upon the colour, but in causes which lie deeper, and involve some factors deducible from biological theory.

The organism, as a result of the great facts of heredity and of the survival of the fittest, is necessarily a system which gathers experience with successive generations; and the principal lesson ever being impressed upon it by external events is economy. Its success depends upon the use it makes of its opportunities for the reception of energy and the economy attained in disposing of what is gained.

With regard to using the passing opportunity the entire seasonal development of life is a manifestation of this attitude, and the fleetness, agility, etc., of higher organisms are developments in this direction. The higher vegetable organism is not locomotory, save in the transferences of pollen and seed, for its food comes to it, and the necessary relative motion between food and organism is preserved in the quick motion of radiated energy from the sun and the slower motion of the winds on the surface of the

earth. But, even so, the vegetable organism must stand ever ready and waiting for its supplies. Its molecular parts must be ready to seize the prey offered to it, somewhat as the waiting spider the fly. Hence, the plant stands ready; and every cloud with moving shadow crossing the fields handicaps the shaded to the benefit of the unshaded plant in the adjoining field. The open bloom

109

is a manifestation of the generally expectant attitude of the plant, but in relation to reproduction.

As regards economy, any principle of maximum economy, where many functions have to be fulfilled, will, we may very safely predict, involve as far as possible mutual helpfulness in the processes going on. Thus the process of the development towards meeting any particular external conditions, A, suppose, will, if possible, tend to forward the development towards meeting conditions B; so that, in short, where circumstances of morphology and physiology are favourable, the ideally economical system will be attained when in place of two separate processes, α , β , the one process γ ,

cheaper than $\alpha + \beta$, suffices to advance development simultaneously in both the directions A and B. The economy is as obvious as that involved in "killing two birds with the one stone"—if so crude a simile is permissible—and it is to be expected that to foster such economy will be the tendency of evolution in all organic systems subjected to restraints as those we are acquainted with invariably are.

Such economy might be simply illustrated by considering the case of a reservoir of water elevated above two hydraulic motors, so that the elevated mass of water possessed gravitational potential. The available energy here represents the stored-up energy in the organism. How best may the water be conveyed to the two motors [the organic systems reacting towards conditions A and B] so

110

that as little energy as possible is lost in transit? If the motors are near together it is most economical to use the one conduit, which will distribute the requisite supply of water to both. If the motors are located far asunder it will be most

economical to lay separate conduits. There is greatest economy in meeting a plurality of functions by the same train of physiological processes where this is consistent with meeting other demands necessitated by external or internal conditions.

But an important and obvious consequence arises in the supply of the two motors from the one conduit. We cannot work one motor without working the other. If we open a valve in the conduit both motors start into motion and begin consuming the energy stored in the tank. And although they may both under one set of conditions be doing useful and necessary work, in some other set of conditions it may be needless for both to be driven.

This last fact is an illustration of a consideration which must enter into the phenomenon which an eminent biologist speaks of as physiological or unconscious "memory,"[1] For the development of the organism from the ovum is but the starting of a train of interdependent events of a complexity depending upon the experience of the past.

[1] Ewald Hering, quoted by Ray Lankaster, _The Advancement of

Science_, p. 283.

111

In short, we may suppose the entire development of the plant, towards meeting certain groups of external conditions, physiologically knit together according as Nature tends to associate certain groups of conditions. Thus, in the case in point, climatic rigour and scarcity of pollinating agency will ever be associated; and in the long experience of the past the most economical physiological attitude towards both is, we may suppose, adopted; so that the presence of one condition excites the apparent unconscious memory of the other. In reality the process of meeting the one condition involves the process and development for meeting the other.

And this consideration may be extended very generally to such organisms as can survive under the same associated natural conditions, for the history of evolution is so long, and the power of locomotion so essential to the organism at some period in its life history, that we cannot philosophically assume a local history for members of a species even if widely severed

geographically at the present day. At some period in the past then, it is very possible that the individuals today thriving at Paris, acquired the experience called out at Upsala. The perfection of physiological memory inspires no limit to the date at which this may have occurred—possibly the result of a succession of severe seasons at Paris; possibly the result of migrations—and the seed of many flowering plants possess means of migration only inferior to those possessed by the flying and swimming animals. But, again, possibly the experi-

112

ence was acquired far back in the evolutionary history of the flower.[1]

But a further consideration arises. Not only at each moment in the life of the individual must maximum income and most judicious expenditure be considered, but in its whole life history, and even over the history of its race, the efficiency must tend to be a maximum. This principle is even carried so far that when necessary it leads to the death of the individual, as in the case of those organisms which, having accomplished the reproductive

act, almost immediately expire. This view of nature may be repellent, but it is, nevertheless, evident that we are parts of a system which ruthlessly sacrifices the individual on general grounds of economy. Thus, if the curve which defines the mean rate of reception of energy of all kinds at different periods in the life of the organism be opposed by a second curve, drawn below the axis along which time is measured, representing the mean rate of expenditure of energy on development, reproduction, etc. (Fig. 7), this latter curve, which is, of course,

[1] The blooms of self-fertilising, and especially of cleistogamic plants (_e.g._ Viola), are examples of unconscious memory, or unconscious "association of ideas" leading to the development of organs now functionless. The Pontederia crassipes of the Amazon, which develops its floating bladders when grown in water, but aborts them rapidly when grown on land, and seems to retain this power of adaptation to the environment for an indefinite period of time, must act in each case upon an unconscious memory based upon past experience.

Many other cases
might be cited.

113

physiologically dependent on the former, must be of such a nature from its origin to its completion in death, that the condition is realized of the most economical rate of expenditure at each period of life.[1] The rate of expenditure of energy at any period of life is, of course, in such a curve defined by the slope of the curve towards the axis of time at the period in question; but this particular slope _must be led to by a previous part of the curve, and involves its past and future course to a very great extent_.

{Fig. 7}

There will, therefore, be impressed upon the organism by the factors of evolution a unified course of economical expenditure completed only by its death, and which will give to the developmental progress of the individual its prophetic character.

In this way we look to the unified career of each organic unit, from its commencement in the ovum to the day

[1] See _The Abundance of Life_.

when it is done with vitality, for that preparation for momentous organic events which is in progress throughout the entire course of development; and to the economy involved in the welding of physiological processes for the phenomenon of physiological memory, wherein we see reflected, as it were, in the development of the organism, the association of inorganic restraints occurring in nature which at some previous period impressed itself upon the plastic organism. We may picture the seedling at Upsala, swayed by organic memory and the inherited tendency to an economical preparation for future events, gradually developing towards the æsthetic climax of its career. In some such manner only does it appear possible to account for the prophetic development of organisms, not alone to be observed in the alpine flowers, but throughout nature.

And thus, finally, to the effects of natural selection and to actions defined by general principles involved in biology, I would refer for explanation of the manner in which flowers on the Alps develop their peculiar beauty.

MOUNTAIN GENESIS

OUR ancestors regarded mountainous regions with feelings of horror, mingled with commiseration for those whom an unkindly destiny had condemned to dwell therein. We, on the other hand, find in the contemplation of the great alps of the Earth such peaceful and elevated thoughts, and such rest to our souls, that it is to those very solitudes we turn to heal the wounds of life. It is difficult to explain the cause of this very different point of view. It is probably, in part, to be referred to that cloud of superstitious horror which, throughout the Middle Ages, peopled the solitudes with unknown terrors; and, in part, to the asceticism which led the pious to regard the beauty and joy of life as snares to the soul's well-being. In those eternal solitudes where the overwhelming forces of Nature are most in evidence, an evil principle must dwell or a dragon's dreadful brood must find a home.

But while in our time the aesthetic aspect of the hills appeals to all, there remains in the physical history of the mountains

much that is lost to those who have not shared in the scientific studies of alpine structure and genesis. They lose a past history which for interest com-

116

petes with anything science has to tell of the changes of the Earth.

Great as are the physical features of the mountains compared with the works of Man, and great as are the forces involved compared with those we can originate or control, the loftiest ranges are small contrasted with the dimensions of the Earth. It is well to bear this in mind. I give here (Pl. XV.) a measured drawing showing a sector cut from a sphere of 50 cms. radius; so much of it as to exhibit the convergence of its radial boundaries which if prolonged will meet at the centre. On the same scale as the radius the diagram shows the highest mountains and the deepest ocean. The average height of the land and the average depth of the ocean are also exhibited. We see how small a movement of the crust the loftiest elevation of the Himalaya represents and what a little depression holds the ocean.

Nevertheless, it is not by any means easy to explain the genesis of those small elevations and depressions. It would lead us far from our immediate subject to discuss the various theoretical views which have been advanced to account for the facts. The idea that mountain folds, and the lesser rugosities of the Earth's surface, arose in a wrinkling of the crust under the influence of cooling and shrinkage of the subcrustal materials, is held by many eminent geologists, but not without dissent from others.

The most striking observational fact connected with mountain structure is that, without exception, the ranges

117

of the Earth are built essentially of sedimentary rocks: that is of rocks which have been accumulated at some remote past time beneath the surface of the ocean. A volcanic core there may sometimes be—probably an attendant or consequence of the uplifting—or a core of plutonic igneous rocks which has arisen under the same compressive forces which have bowed and arched the strata from their original horizontal position. It is not uncommon to meet among unobservant people those

who regard all mountain ranges as volcanic in origin. Volcanoes, however, do not build mountain ranges. They break out as more or less isolated cones or hills. Compare the map of the Auvergne with that of Switzerland; the volcanoes of South Italy with the Apennines.

Such great ranges as those which border with triple walls the west coast of North America are in no sense volcanic: nor are the Pyrenees, the Caucasus, or the Himalaya. Volcanic materials are poured out from the summits of the Andes, but the range itself is built up of folded sediments on the same architecture as the other great ranges of the Earth.

Before attempting an explanation of the origin of the mountains we must first become more closely acquainted with the phenomena attending mountain elevation.

At the present day great accumulations of sediment are taking place along the margins of the continents where the rivers reach the ocean. Thus, the Gulf of Mexico receiving the sediment of the Mississippi and Rio Grande;

the northeast coast of South America receiving the sediments of the Amazons; the east coast of Asia receiving the detritus of the Chinese rivers; are instances of such areas of deposition. Year by year, century by century, the accumulation progresses, and as it grows the floor of the sea sinks under the load. Of the yielding of the crust under the burthen of the sediments we are assured; for otherwise the many miles of vertically piled strata which are uplifted to our view in the mountains, never could have been deposited in the coastal seas of the past. The flexure and sinking of the crust are undeniable realities.

Such vast subsiding areas are known as geosynclines. From the accumulated sediments of the geosynclines the mountain ranges of the past have in every case originated; and the mountains of the future will assuredly arise and lofty ranges will stand where now the ocean waters close over the collecting sediments. Every mountain range upon the Earth enforces the certainty of this prediction.

The mountain-forming movement takes place after a certain great depth of sediment is collected. It is most intense where the

thickness of deposit is greatest. We see this when we examine the structure of our existing mountain ranges. At either side where the sediments thin out, the disturbance dies away, till we find the comparatively shallow and undisturbed level sediments which clothe the continental surface.

Whatever be the connection between the deposition and

119

the subsequent upheaval, _the element of great depth of accumulation seems a necessary condition and must evidently enter as a factor into the Physical Processes involved_. The mountain range can only arise where the geosyncline is deeply filled by long ages of sedimentation.

Dana's description of the events attending mountain building is impressive:

"A mountain range of the common type, like that to which the Appalachians belong, is made out of the sedimentary formations of a long preceding era; beds that were laid down conformably, and in succession, until they had reached the needed thickness; beds spreading over a region tens of thousands of square miles in

area. The region over which sedimentary formations were in progress in order to make, finally, the Appalachian range, reached from New York to Alabama, and had a breadth of 100 to 200 miles, and the pile of horizontal beds along the middle was 40,000 feet in depth. The pile for the Wahsatch Mountains was 60,000 feet thick, according to King. The beds for the Appalachians were not laid down in a deep ocean, but in shallow waters, where a gradual subsidence was in progress; and they at last, when ready for the genesis, lay in a trough 40,000 feet deep, filling the trough to the brim. It thus appears that epochs of mountain-making have occurred only after long intervals of quiet in the history of a continent." [1]

[1] Dana, Manual of Geology, third edition, p. 794

120

On the western side of North America the work of mountain-building was, indeed, on the grandest scale. For long ages and through a succession of geological epochs, sedimentation had proceeded so that the accumulations of Palaeozoic and Mesozoic times had collected in the geosyncline formed by their

own ever increasing weight. The site of the future Laramide range was in late Cretaceous times occupied by some 50,000 feet of sedimentary deposits; but the limit had apparently been attained, and at this time the Laramide range, as well as its southerly continuation into the United States, the Rockies, had their beginning. Chamberlin and Salisbury[1] estimate that the height of the mountains developed in the Laramide range at this time was 20,000 feet, and that, owing to the further elevation which has since taken place, from 32,000 to 35,000 feet would be their present height if erosion had not reduced them. Thus on either side of the American continent we have the same forces at work, throwing up mountain ridges where the sediments had formerly been shed into the ocean.

These great events are of a rhythmic character; the crust, as it were, pulsating under the combined influences of sedimentation and denudation. The first involves downward movements under a gathering load, and ultimately a reversal of the movement to one of upheaval; the second factor, which throughout has been in

[1] Chamberlin and Salisbury, *_Geology_*, 1906, iii., 163.

121

operation as creator of the sediments, then intervenes as an assailant of the newly-raised mountains, transporting their materials again to the ocean, when the rhythmic action is restored to its first phase, and the age-long sequence of events must begin all over again.

It has long been inferred that compressive stress in the crust must be a primary condition of these movements. The work required to effect the upheavals must be derived from some preexisting source of energy. The phenomenon—intrinsically one of folding of the crust—suggests the adjustment of the earth-crust to a lessening radius; the fact that great mountain-building movements have simultaneously affected the entire earth is certainly in favour of the view that a generally prevailing cause is at the basis of the phenomenon.

The compressive stresses must be confined to the upper few miles of the crust, for, in fact, the downward increase of temperature and pressure soon confers fluid properties on the

medium, and
slow tangential compression results in hydrostatic
pressure
rather than directed stresses. Thus the folding
visible in the
mountain range, and the lateral compression
arising therefrom,
are effects confined to the upper parts of the
crust.

The energy which uplifts the mountain is probably
a surviving
part of the original gravitational potential
energy of the crust
itself. It must be assumed that the crust in
following downwards
the shrinking subcrustal magma, develops immense
compressive
stresses in

122

its materials, vast geographical areas being
involved. When
folding at length takes place along the axis of
the elongated
syncline of deposition, the stresses find relief
probably for
some hundreds of miles, and the region of folding
now becomes
compressed in a transverse direction. As an
illustration, the
Laramide range, according to Dawson, represents
the reduction of
a surface-belt 50 miles wide to one of 25 miles.
The marvellous
translatory movements of crustal folds from south
to north

arising in the genesis of the Swiss Alps, which recent research has brought to light, is another example of these movements of relief, which continue to take place perhaps for many millions of years after they are initiated.

The result of this yielding of the crust is a buckling of the surface which on the whole is directed upwards; but depression also is an attendant, in many cases at least, on mountain upheaval. Thus we find that the ocean floor is depressed into a syncline along the western coast of South America; a trough always parallel to the ranges of the Andes. The downward deflection of the crust is of course an outcome of the same compressive stresses which elevate the mountain.

The fact that the yielding of the crust is always situated where the sediments have accumulated to the greatest depth, has led to attempts from time to time of establishing a physical connexion between the one and the other. The best-known of these theories is that of Babbage and Herschel. This seeks the connexion in the rise of the

geotherms into the sinking mass of sediment and the consequent increase of temperature of the earth-crust beneath. It will be understood that as these isogeotherms, or levels at which the temperature is the same, lie at a uniform distance from the surface all over the Earth, unless where special variations of conductivity may disturb them, the introduction of material pressed downwards from above must result in these materials partaking of the temperature proper to the depth to which they are depressed. In other words the geotherms rise into the sinking sediments, always, however, preserving their former average distance from the surface. The argument is that as this process undoubtedly involves the heating up of that portion of the crust which the sediments have displaced downwards, the result must be a local enfeeblement of the crust, and hence these areas become those of least resistance to the stresses in the crust.

When this theory is examined closely, we see that it only amounts to saying that the bedded rocks, which have taken the place of the igneous materials beneath, as a part of the rigid crust of

the Earth, must be less able to withstand compressive stress than the average crust. For there has been no absolute rise of the geotherms, the thermal conductivities of both classes of materials differing but little. Sedimentary rock has merely taken the place of average crust-rock, and is subjected to the same average temperature and pressure prevailing in the surrounding crust. But are there any grounds for the

124

assumption that the compressive resistance of a complex of sedimentary rocks is inferior to one of igneous materials? The metamorphosed siliceous sediments are among the strongest rocks known as regards resistance to compressive stress; and if limestones have indeed plastic qualities, it must be remembered that their average amount is only some 5 per cent. of the whole. Again, so far as rise of temperature in the upper crust may affect the question, a temperature which will soften an average igneous rock will not soften a sedimentary rock, for the reason that the effect of solvent denudation has been to remove those alkaline silicates which confer fusibility.

When, however, we take into account the radioactive content of the sediments the matter assumes a different aspect.

The facts as to the general distribution of radioactive substances at the surface, and in rocks which have come from considerable depths in the crust, lead us to regard as certain the widespread existence of heat-producing radioactive elements in the exterior crust of the Earth. We find, indeed, in this fact an explanation—at least in part—of the outflow of heat continually taking place at the surface as revealed by the rising temperature inwards. And we conclude that there must be a thickness of crust amounting to some miles, containing the radioactive elements.

Some of the most recent measurements of the quantities of radium and thorium in the rocks of igneous origin—_e.g._ granites, syenites, diorites, basalts, etc., show that the

125

radioactive heat continually given out by such rocks amounts to about one millionth part of 0.6 calories per second per cubic metre of average igneous rock. As we have to

account for the escape of about 0.0014 calorie[1] per square metre of the Earth's surface per second (assuming the rise of temperature downwards, _i.e._ the "gradient" of temperature, to be one degree centigrade in 35 metres) the downward extension of such rocks might, _prima facie_, be as much as 19 kilometres.

About this calculation we have to observe that we assume the average radioactivity of the materials with which we have dealt at the surface to extend uniformly all the way down, _i.e._ that our experiments reveal the average radioactivity of a radioactive crust. There is much to be said for this assumption. The rocks which enter into the measurements come from all depths of the crust. It is highly probable that the less silicious, _i.e._ the more basic, rocks, mainly come from considerable depths; the more acid or silica-rich rocks, from higher levels in the crust. The radioactivity determined as the mean of the values for these two classes of rock closely agrees with that found for intermediate rocks, or rocks containing an intermediate amount of silica. Clarke contends that this last class of material probably

represents the average composition of the Earth's crust so far as it has been explored by us.

[1] The calorie referred to is the quantity of heat required to heat one gram of water, _i.e._ one cubic centimetre of water—through one degree centigrade.

126

It is therefore highly probable that the value found for the mean radioactivity of acid and basic rocks, or that found for intermediate rocks, truly represents the radioactive state of the crust to a considerable depth. But it is easy to show that we cannot with confidence speak of the thickness of this crust as determinable by equating the heat outflow at the surface with the heat production of this average rock.

This appears in the failure of a radioactive layer, taken at a thickness of about 19-kilometres, to account for the deep-seated high temperatures which we find to be indicated by volcanic phenomena at many places on the surface. It is not hard to show that the 19-kilometre layer would account for a temperature no higher than about $270^{\circ} >C.$ at its base.

It is true that this will be augmented beneath the sedimentary deposits as we shall presently see; and that it is just in association with these deposits that deep-seated temperatures are most in evidence at the surface; but still the result that the maximum temperature beneath the crust in general attains a value no higher than 270° C. is hardly tenable. We conclude, then, that some other source of heat exists beneath. This may be radioactive in origin and may be easily accounted for if the radioactive materials are more sparsely distributed at the base of the upper crust. Or, again, the heat may be primeval or original heat, still escaping from a cooling world. For our present purpose it does not much matter which view

127

we adopt. But we must recognise that the calculated depth of 19 kilometres of crust, possessing the average radioactivity of the surface, is excessive; for, in fact, we are compelled by the facts to recognise that some other source of heat exists beneath.

If the observed surface gradient of temperature persisted

uniformly downwards, at some 35 kilometres beneath the surface there would exist temperatures (of about 1000° C.) adequate to soften basic rocks. It is probable, however, that the gradient diminishes downwards, and that the level at which such temperatures exist lies rather deeper than this. It is, doubtless, somewhat variable according to local conditions; nor can we at all approximate closely to an estimate of the depth at which the fusion temperatures will be reached, for, in fact, the existence of the radioactive layer very much complicates our estimates. In what follows we assume the depth of softening to lie at about 40 kilometres beneath the surface of the normal crust; that is 25 miles down. It is to be observed that Prestwich and other eminent geologists, from a study of the facts of crust-folding, etc., have arrived at similar estimates.[1] As a further assumption we are probably not far wrong if we assign to the radioactive part of this crust a thickness of about 10 or 12 kilometres; _i.e._ six or seven miles. This is necessarily a rough approximation only; but the conclusions at which

[1] Prestwich, _Proc. Royal Soc._, xii., p. 158
et seq.

128

we shall arrive are reached in their essential features allowing a wide latitude in our choice of data. We shall speak of this part of the crust as the normal radioactive layer.

An important fact is evolved from the mathematical investigation of the temperature conditions arising from the presence of such a radioactive layer. It is found that the greatest temperature, due to the radioactive heat everywhere evolved in the layer—i.e.—the temperature at its base—is proportional to the square of the thickness of the layer. This fact has a direct bearing on the influence of radioactivity upon mountain elevation; as we shall now find.

The normal radioactive layer of the Earth is composed of rocks extending—as we assume—approximately to a depth of 12 kilometres (7.5 miles). The temperature at the base of this layer due to the heat being continually evolved in it, is, say, t_1° . Now, let us suppose, in the trough of the geosyncline, and upon the top of the normal layer, a deposit of, say, 10 kilometres

(6.2 miles) of sediments is formed during a long period of continental denudation. What is the effect of this on the temperature at the base of the normal layer depressed beneath this load? The total thickness of radioactive rocks is now 22 kilometres. Accordingly we find the new temperature t_2° , by the proportion $t_1^\circ : t_2^\circ :: 12^\circ : 22^\circ$ That is, as 144 to 484. In fact, the temperature is more than trebled. It is true we here assume the radioactivity of the sediments

129

and of the normal crust to be the same. The sediments are, however, less radioactive in the proportion of 4 to 3. Nevertheless the effects of the increased thickness will be considerable.

Now this remarkable increase in the temperature arises entirely from the condition attending the radioactive heating; and involves something _additional_ to the temperature conditions determined by the mere depression and thickening of the crust as in the Babbage-Herschel theory. The latter theory only involves a _shifting_ of the temperature levels (or

geotherms) into the deposited materials. The radioactive theory involves an actual rise in the temperature at any distance from the surface; so that _the level in the crust at which the rocks are softened is nearer to the surface in the geosynclines than it is elsewhere in the normal crust_ (Pl. XV, p. 118).

In this manner the rigid part of the crust is reduced in thickness where the great sedimentary deposits have collected. A ten-kilometre layer of sediment might result in reducing the effective thickness of the crust by 30 per cent.; a fourteen-kilometre layer might reduce it by nearly 50 per cent. Even a four-kilometre deposit might reduce the effective resistance of the crust to compressive forces, by 10 per cent.

Such results are, of course, approximate only. They show that as the sediments grow in depth there is a rising of the geotherm of plasticity—whatever its true temperature may be—gradually reducing the thickness of that part

130

of the upper crust which is bearing the simultaneously increasing

compressive stresses. Below this geotherm long-continued stress resolves itself into hydrostatic pressure; above it (there is, of course, no sharp line of demarcation) the crust accumulates elastic energy. The final yielding and flexure occur when the resistant cross-section has been sufficiently diminished. It is probable that there is also some outward hydrostatic thrust over the area of rising temperature, which assists in determining the upward throw of the folds.

When yielding has begun in any geosyncline, and the materials are faulted and overthrust, there results a considerably increased thickness. As an instance, consider the piling up of sediments over the existing materials of the Alps, which resulted from the compressive force acting from south to north in the progress of Alpine upheaval. Schmidt of Basel has estimated that from 15 to 20 kilometres of rock covered the materials of the Simplon as now exposed, at the time when the orogenic forces were actively at work folding and shearing the beds, and injecting into their folds the plastic gneisses coming from beneath.[1] The lateral compression of the area of deposition of the

Laramide, already
referred to, resulted in a great thickening of the
deposits. Many
other cases might be cited; the effect is always
in some degree
necessarily produced.

[1] Schmidt, Ec. Geol. *_Helvelix_*, vol. ix., No. 4, p. 590

131

If time be given for the heat to accumulate in the lower depths of the crushed-up sediments, here is an additional source of increased temperature. The piled-up masses of the Simplon might have occasioned a rise due to radioactive heating of one or two hundred degrees, or even more; and if this be added to the interior heat, a total of from 800° to 1000° might have prevailed in the rocks now exposed at the surface of the mountain. Even a lesser temperature, accompanied by the intense pressure conditions, might well occasion the appearances of thermal metamorphism described by Weinschenk, and for which, otherwise, there is difficulty in accounting.[1]

This increase upon the primarily developed temperature conditions takes place concurrently with the progress of compression; and while it cannot be taken into account in estimating the conditions of initial yielding of the crust, it adds an element of instability, inasmuch as any progressive thickening by lateral

compression results in an accelerated rise of the
geotherms. It
is probable that time sufficient for these effects
to develop, if
not to their final, yet to a considerable extent,
is often
available. The viscous movements of siliceous
materials, and the
out-pouring of igneous rocks which often attend
mountain
elevation, would find an explanation in such
temperatures.

[1] Weinschenk, _Congrès Géol. Internat._, 1900,
i., p. 332.

132

There is no more striking feature of the part here
played by
radioactivity than the fact that the rhythmic
occurrence of
depression and upheaval succeeding each other
after great
intervals of time, and often shifting their
position but little
from the first scene of sedimentation, becomes
accounted for. The
source of thermal energy, as we have already
remarked, is in fact
transported with the sediments—that energy which
determines the
place of yielding and upheaval, and ordains that
the mountain
ranges shall stand around the continental borders.
Sedimentation
from this point of view is a convection of energy.

When the consolidated sediments are by these and by succeeding movements forced upwards into mountains, they are exposed to denudative effects greatly exceeding those which affect the plains. Witness the removal during late Tertiary times of the vast thickness of rock enveloping the Alps. Such great masses are hurried away by ice, rivers, and rain. The ocean receives them; and with infinite patience the world awaits the slow accumulation of the radioactive energy beginning afresh upon its work. The time for such events appears to us immense, for millions of years are required for the sediments to grow in thickness, and the geotherms to move upwards; but vast as it is, it is but a moment in the life of the parent radioactive substances, whose atoms, hardly diminished in numbers, pursue their changes while the mountains come and go, and the

133

rudiments of life develop into its highest consummations.

To those unacquainted with the results of geological investigation the history of the mountains as deciphered in the rocks seems almost incredible.

The recently published sections of the Himalaya, due to H. H.

Hayden and the many distinguished men who have contributed to the Geological Survey of India, show these great ranges to be essentially formed of folded sediments penetrated by vast masses of granite and other eruptives. Their geological history may be summarised as follows

The Himalayan area in pre-Cambrian times was, in its southwestern extension, part of the floor of a sea which covered much of what is now the Indian Peninsula. In the northern shallows of this sea were laid down beds of conglomerate, shale, sandstone and limestone, derived from the denudation of Archæan rocks, which, probably, rose as hills or mountains in parts of Peninsular India and along the Tibetan edge of the Himalayan region. These beds constitute the record of the long Purana Era[1] and are probably coeval with the Algonkian of North America. Even in these early times volcanic disturbances affected this area and the lower beds of the Purana deposits were penetrated by volcanic outflows, covered by sheets of lava, uplifted, denuded and again submerged

[1] See footnote, p. 139.

beneath the waters. Two such periods of instability have left their records in the sediments of the Purana sea.

The succeeding era—the Dravidian Era—opens with Haimanta

(Cambrian) times. A shallow sea now extended over Kumaun, Garwal, and Spiti, as well as Kashmir and ultimately over the Salt Range

region of the Punjab as is shown by deposits in these areas. This

sea was not, however, connected with the Cambrian sea of Europe.

The fossil faunas left by the two seas are distinct.

After an interval of disturbance during closing Haimanta times,

geographical changes attendant on further land movements

occurred. The central sea of Asia, the Tethys, extended westwards

and now joined with the European Paleozoic sea; and deposits of

Ordovician and Silurian age were laid down:—the Muth deposits.

The succeeding Devonian Period saw the whole Northern Himalayan

area under the waters of the Tethys which, eastward, extended to

Burma and China and, westward, covered Kashmir, the Hindu Kush

and part of Afghanistan. Deposits continued to be

formed in this
area till middle Carboniferous times.

Near the close of the Dravidian Era Kashmir became convulsed by volcanic disturbance and the Penjal traps were ejected. It was a time of worldwide disturbance and of redistribution of land and water. Carboniferous times had begun, and the geographical changes in

135

the southern limits of the Tethys are regarded as ushering in a new and last era in Indian geological history the Aryan Bra.

India was now part of Gondwanaland; that vanished continent which then reached westward to South Africa and eastward to Australia.

A boulder-bed of glacial origin, the Talchir Boulder-bed, occurs in many surviving parts of this great land. It enters largely into the Salt Range deposits. There is evidence that extensive sheets of ice, wearing down the rocks of Rajputana, shoved their moraines northward into the Salt Range Sea; then, probably, a southern extension of the Tethys.

Subsequent to this ice age the Indian coalfields of the Gondwana were laid down, with beds rich in the Glossopteris

and
Gangamopteris flora. This remarkable carboniferous
flora extends
to Southern Kashmir, so that it is to be inferred
that this
region was also part of the main Gondwanaland. But
its emergence
was but for a brief period. Upper Carboniferous
marine deposits
succeeded; and, in fact, there was no important
discontinuity in
the deposits in this area from Panjal times till
the early
Tertiaries. During the whole of which vast period
Kashmir was
covered with the waters of the Tethys.

The closing Dravidian disturbances of the Kashmir
region did not,
apparently, extend to the eastern Himalayan area.
But the
Carboniferous Period was, in this

136

eastern area, one of instability, culminating, at
the close of
the Period, in a steady rise of the land and a
northward retreat
of the Tethys. Nearly the entire Himalaya east of
Kashmir became
a land surface and remained exposed to denudative
forces for so
long a time that in places the whole of the
Carboniferous,
Devonian, and a large part of the Silurian and
Ordovician
deposits were removed—some thousands of feet in

thickness—before
resubmergence in the Tethys occurred.

Towards the end of the Palaeozoic Age the Aryan Tethys receded westwards, but still covered the Himalaya and was still connected with the European Palæozoic sea. The Himalayan area (as well as Kashmir) remained submerged in its waters throughout the entire Mesozoic Age.

During Cretaceous times the Tethys became greatly extended, indicating a considerable subsidence of northwestern India, Afghanistan, Western Asia, and, probably, much of Tibet. The shallow-water character of the deposits of the Tibetan Himalaya indicates, however, a coast line near this region. Volcanic materials, now poured out, foreshadow the incoming of the great mountain-building epoch of the Tertiary Era. The enormous mass of the Deccan traps, possessing a volume which has been estimated at as much as 6,000 cubic miles, was probably extruded over the Northern Peninsular region during late Cretaceous times. The sea now began to retreat, and by the close of

the Eocene, it had moved westward to Sind and Baluchistan. The movements of the Earth's crust were attended by intense volcanic activity, and great volumes of granite were injected into the sediments, followed by dykes and outflows of basic lavas.

The Tethys vanished to return no more. It survives in the Mediterranean of today. The mountain-building movements continued into Pliocene times. The Nummulite beds of the Eocene were, as the result, ultimately uplifted 18,500 feet over sea level, a total uplift of not less than 20,000 feet.

Thus with many vicissitudes, involving intervals of volcanic activity, local uplifting, and extensive local denudation, the Himalaya, which had originated in the sediments of the ancient Purana sea, far back in pre-Cambrian times, and which had developed potentially in a long sequence of deposits collecting almost continuously throughout the whole of geological time, finally took their place high in the heavens, where only the winds—faint at such altitudes—and the lights of heaven can visit their eternal snows.[1]

In this great history it is significant that the longest continuous series of sedimentary deposits which the world has known has become transfigured into the loftiest elevation upon its surface.

[1] See A Sketch of the _Geography and Geology of the Himalaya Mountains and Tibet_. By Colonel S. G. Burrard, R.E., F.R.S., and H. H. Hayden, F.G.S., Part IV. Calcutta, 1908.

138

The diagrammatic sections of the Himalaya accompanying this brief description are taken from the monograph of Burrard and Hayden (loc. cit.) on the Himalaya. Looking at the sections we see that some of the loftiest summits are sculptured in granite and other crystalline rocks. The appearance of these materials at the surface indicates the removal by denudation and the extreme metamorphism of much sedimentary deposit. The crystalline rocks, indeed, penetrate some of the oldest rocks in the world. They appear in contact with Archaean, Algonkian or early Palaeozoic rocks. A study of the sections reveals not only the severe earth movements, but also the immense amount of sedimentary deposits

involved in the genesis of these alps. It will be noted that the vertical scale is not exaggerated relatively to the horizontal.[1] Although there is no evidence of mountain building

[1] To those unacquainted with the terminology of Indian geology the following list of approximate equivalents in time will be of use

Ngari Khorsum Beds -	Pleistocene.
Siwalik Series -	Miocene and Pliocene.
Sirmur Series -	Oligocene.
Kampa System -	Eocene and Cretaceous.
Lilang System -	Triassic.
Kuling System -	Permian.
Gondwana System -	Carboniferous.
Kenawar System -	Carboniferous and Devonian
Muth System -	Silurian.
Haimanta System -	Mid. and Lower Cambrian.
Purana Group -	Algonkian.
Vaikrita System -	Archæan.
Daling Series -	Archæan.

139

on a large scale in the Himalayan area till the Tertiary upheaval, it is, in the majority of cases, literally correct to speak of the mountains as having their generations like organic beings, and passing through all the stages of birth, life, death

and reproduction. The Alps, the Jura, the Pyrenees, the Andes, have been remade more than once in the course of geological time, the _débris_ of a worn-out range being again uplifted in succeeding ages.

Thus to dwell for a moment on one case only: that of the Pyrenees. The Pyrenees arose as a range of older Palmozoic rocks in Devonian times. These early mountains, however, were sufficiently worn out and depressed by Carboniferous times to receive the deposits of that age laid down on the up-turned edges of the older rocks. And to Carboniferous succeeded Permian, Triassic, Jurassic and Lower Cretaceous sediments all laid down in conformable sequence. There was then fresh disturbance and upheaval followed by denudation, and these mountains, in turn, became worn out and depressed beneath the ocean so that Upper Greensand rocks were laid down unconformably on all beneath. To these now succeeded Upper Chalk, sediments of Danian age, and so on, till Eocene times, when the tale was completed and the existing ranges rose from the sea. Today we find the folded Nummulitic strata of Eocene age uplifted 11,000

feet, or within
200 feet of the greatest heights of the Pyrenees.
And so they
stand awaiting

140

the time when once again they shall "fall into the
portion of
outworn faces." [1]

Only mountains can beget mountains. Great
accumulations of
sediment are a necessary condition for the
localisation of
crust-flexure. The earliest mountains arose as
purely igneous or
volcanic elevations, but the generations of the
hills soon
originated in the collection of the *_débris_*,
under the law of
gravity, in the hollow places. And if a foundered
range is
exposed now to our view encumbered with thousands
of feet of
overlying sediments we know that while the one
range was sinking,
another, from which the sediments were derived,
surely existed.
Through the "windows" in the deep-cut rocks of the
Swiss valleys
we see the older Carboniferous Alps looking out,
revisiting the
sun light, after scores of millions of years of
imprisonment. We
know that just as surely as the Alps of today are
founding by
their muddy torrents ranges yet to arise, so other

primeval Alps
fed into the ocean the materials of these buried
pre-Permian
rocks.

This succession of events only can cease when the
rocks have been
sufficiently impoverished of the heat-producing
substances, or
the forces of compression shall have died out in
the surface
crust of the earth.

It seems impossible to escape the conclusion that
in the great
development of ocean-encircling areas of

[1] See Prestwich, _Chemical and Physical
Geology_, p. 302.

141

deposition and crustal folding, the heat of
radioactivity has
been a determining factor. We recognise in the
movements of the
sediments not only an influence localising and
accelerating
crustal movements, but one which, in subservience
to the primal
distribution of land and water, has determined
some of the
greatest geographical features of the globe.

It is no more than a step to show that bound up
with the
radioactive energy are most of the earthquake and
volcanic
phenomena of the earth. The association of

earthquakes with the great geosynclines is well known. The work of De Montessus showed that over 94 per cent. of all recorded shocks lie in the geosynclinal belts. There can be no doubt that these manifestations of instability are the results of the local weakness and flexure which originated in the accumulation of energy denuded from the continents. Similarly we may view in volcanoes phenomena referable to the same fundamental cause. The volcano was, in fact, long regarded as more intimately connected with earthquakes than it, probably, actually is; the association being regarded in a causative light, whereas the connexion is more that of possessing a common origin. The girdle of volcanoes around the Pacific and the earthquake belt coincide. Again, the ancient and modern volcanoes and earthquakes of Europe are associated with the geosyncline of the greater Mediterranean, the Tethys of Mesozoic times. There is no difficulty in understanding in a

142

general way the nature of the association. The earthquake is the manifestation of rupture and slip, and, as Suess

has shown, the
epicentres shift along that fault line where the
crust has
yielded.[1] The volcano marks the spot where the
zone of fusion
is brought so high in the fractured crust that the
melted
materials are poured out upon the surface.

In a recent work on the subject of earthquakes
Professor Hobbs
writes: "One of the most interesting of the
generalisations which
De Montessus has reached as a result of his
protracted studies,
is that the earthquake districts on the land
correspond almost
exactly to those belts upon the globe which were
the almost
continuous ocean basins of the long Secondary era
of geological
history. Within these belts the sedimentary
formations of the
crust were laid down in the greatest thickness,
and the
formations follow each other in relatively
complete succession.
For almost or quite the whole of this long era it
is therefore
clear that the ocean covered these zones. About
them the
formations are found interrupted, and the lacuna
indicate that
the sea invaded the area only to recede from it,
and again at
some later period to transgress upon it. For a
long time,

therefore, these earthquake belts were the sea basins—the geosynclines. They became later the rising mountains of the Tertiary period, and mountains they

[1] Suess, *The Face of the Earth*, vol. ii., chap. ii.

143

are today. The earthquake belts are hence those portions of the earth's crust which in recent times have suffered the greatest movements in a vertical direction—they are the most mobile portions of the earth's crust." [1] Whether the movements attending mountain elevation and denudation are a connected and integral part of those wide geographical changes which result in submergence and elevation of large continental areas, is an obscure and complex question. We seem, indeed, according to the views of some authorities, hardly in a position to affirm with certainty that such widespread movements of the land have actually occurred, and that the phenomena are not the outcome of fluctuations of oceanic level; that our observations go no further than the recognition of positive and negative movements of the strand. However this may be, the greater

part of
mechanical denudation during geological time has
been done on the
mountain ranges. It is, in short, indisputable
that the orogenic
movements which uplift the hills have been at the
basis of
geological history. To them the great
accumulations of sediments
which now form so large a part of continental land
are mainly
due. There can be no doubt of the fact that these
movements have
swayed the entire history, both inorganic and
organic, of the
world in which we live.

[1] Hobbs, Earthquakes, p. 58.

144

To sum the contents of this essay in the most
general terms, we
find that in the conception of denudation as
producing the
convection and accumulation of radiothermal energy
the surface
features of the globe receive a new significance.
The heat of the
earth is not internal only, but rather a heat-
source exists at
the surface, which, as we have seen, cannot
prevail to the same
degree within; and when the conditions become
favourable for the
aggregation of the energy, the crust, heated both
from beneath
and from above, assumes properties more akin to

those of its
earlier stages of development, the secular heat-
loss being
restored in the radioactive supplies. These causes
of local
mobility have been in operation, shifting somewhat
from place to
place, and defined geographically by the
continental masses
undergoing denudation, since the earliest times.

145

ALPINE STRUCTURE

AN intelligent observer of the geological changes
progressing in
southern Europe in Eocene times would have seen
little to inspire
him with a premonition of the events then
developing. The
Nummulitic limestones were being laid down in that
enlarged
Mediterranean which at this period, save for a few
islands,
covered most of south Europe. Of these stratified
remains, as
well as of the great beds of Cretaceous, Jurassic,
Triassic, and
Permian sediments beneath, our hypothetical
observer would
probably have been regardless; just as today we
observe, with an
indifference born of our transitoriness, the
deposits rapidly
gathering wherever river discharge is distributing
the sediments
over the sea-floor, or the lime-secreting

organisms are actively
at work. And yet it took but a few millions of
years to uplift
the deposits of the ancient Tethys; pile high its
sediments in
fold upon fold in the Alps, the Carpathians, and
the Himalayas;
and—exposing them to the rigours of denudation at
altitudes where
glaciation, landslip, and torrent prevail—
inaugurate a new epoch
of sedimentation and upheaval.

146

In the case of the Alps, to which we wish now
specially to refer,
the chief upheaval appears to have been in
Oligocene times,
although movement continued to the close of the
Pliocene. There
was thus a period of some millions of years within
which the
entire phenomena were comprised. Availing
ourselves of Sollas'
computations,[1] we may sum the maximum depths of
sedimentary
deposits of the geological periods concerned as
follows:—

Pliocene - - - - - 3,950 m.

Miocene - - - - - 4,250 m.

Oligocene - - - - - 3,660 m.

Eocene - - - - - 6,100 m.

and assuming that the orogenic forces began their
work in the

last quarter of the Eocene period, we have a total of 13,400 m.

as some measure of the time which elapsed. At the rate of 10

centimetres in a century these deposits could not have collected

in less than 13.4 millions of years. It would appear that not

less than some ten millions of years were consumed in the genesis

of the Alps before constructive movements finally ceased.

The progress of the earth-movements was attended by the usual

volcanic phenomena. The Oligocene and Miocene volcanoes extended

in a band marked by the Auvergne, the Eiffel, the Bohemian, and

the eastern Carpathian eruptions; and, later, towards the close

of the movements in Pliocene times, the south border

[1] Sollas, Anniversary Address, Geol. Soc., London, 1909.

147

regions of the Alps became the scene of eruptions such as those

of Etna, Santorin, Somma (Vesuvius), etc.

We have referred to these well-known episodes with two objects in

view: to recall to mind the time-interval involved, and the

evidence of intense crustal disturbance, both dynamic and

thermal. According to views explained in a previous essay, the energetic effects of radium in the sediments and upper crust were a principal factor in localising and bringing about these results. We propose now to inquire if, also, in the more intimate structure of the Alps, the radioactive energy may not have borne a part.

What we see today in the Alps is but a residue spared by denudation. It is certain that vast thicknesses of material have disappeared. Even while constructive effects were still in progress, denudative forces were not idle. Of this fact the shingle accumulations of the Molasse, where, on the northern borders of the Alps, they stand piled into mountains, bear eloquent testimony. In the sub-Apennine series of Italy, the great beds of clays, marls, and limestones afford evidence of these destructive processes continued into Pliocene times. We have already referred to Schmidt's estimate that the sedimentary covering must have in places amounted to from 15,000 to 20,000 metres. The evidence for this is mainly tectonic or structural; but is partly forthcoming in the changes which the

materials now
open to our inspection plainly reveal. Thus it is impos-

148

sible to suppose that gneissic rocks can become so far plastic as to flow in and around the calcareous sediments, or be penetrated by the latter—as we see in the Jungfrau and elsewhere—unless great pressures and high temperatures prevailed. And, according to some writers, the temperatures revealed by the intimate structural changes of rock-forming minerals must have amounted to those of fusion. The existence of such conditions is supported by the observation that where the crystallisation is now the most perfect, the phenomena of folding and injection are best developed.[1] These high temperatures would appear to be unaccountable without the intervention of radiothermal effects; and, indeed, have been regarded as enigmatic by observers of the phenomena in question. A covering of 20,000 metres in thickness would not occasion an earth-temperature exceeding 500° C. if the gradients were such as obtain in mountain regions generally; and 600° is about the limit we could ascribe to the purely passive

effects of such a layer in elevating the geotherms.

Those who are still unacquainted with the recently published observations on the structure of the Alps may find it difficult to enter into what has now to be stated; for the facts are, indeed, very different from the generally preconceived ideas of mountain formation. Nor can we wonder that many geologists for long held

[1] Weinschenk, C. R. _Congrès Géol._, 1900, p. 321, et seq.

149

back from admitting views which appeared so extreme. Receptivity is the first virtue of the scientific mind; but, with every desire to lay aside prejudice, many felt unequal to the acceptance of structural features involving a folding of the earth-crust in laps which lay for scores of miles from country to country, and the carriage of mountainous materials from the south of the Alps to the north, leaving them finally as Alpine ranges of ancient sediments reposing on foundations of more recent date. The historian of the subject will have to relate how some who

finally were most active in advancing the new views were at first opposed to them. In the change of conviction of these eminent geologists we have the strongest proof of the convincing nature of the observations and the reality of the tectonic features upon which the recent views are founded.

The lesser mountains which stand along the northern border of the great limestone Alps, those known as the Préalpes, present the strange characteristic of resting upon materials younger than themselves. Such mountains as the remarkable-looking Mythen, near Schwyz, for instance, are weathered from masses of Triassic and Jurassic rock, and repose on the much more recent Flysch. In sharp contrast to the Flysch scenery, they stand as abrupt and gigantic erratics, which have been transported from the central zone of the Alps lying far to the south. They are strangers petrologically,

150

stratigraphically, and geographically,[1] to the locality in which they now occur. The exotic materials may be dolomites, limestones, schists, sandstones, or rocks of igneous origin. They

show in every case traces of the severe dynamic actions to which they have been subjected in transit. The igneous, like the sedimentary, klippen, can be traced to distant sources; to the massif of Belladonne, to Mont Blanc, Lugano, and the Tyrol. The Préalpes are, in fact, mountains without local roots.

In this last-named essential feature, the Préalpes do not differ from the still greater limestone Alps which succeed them to the south. These giants, _e.g._ the Jungfrau, Wetterhorn, Eiger, etc., are also without local foundations. They have been formed from the overthrown and drawn-out anticlines of great crust-folds, whose synclines or roots are traceable to the south side of the Rhone Valley. The Bernese Oberland originated in the piling-up of four great sheets or recumbent folds, one of which is continued into the Préalpes. With Lugeon[2] we may see in the phenomenon of the formation of the Préalpes a detail; regarding it as a normal expression of that mechanism which has created the Swiss Alps. For these limestone masses of the Oberland are not indications of a merely local shift of the sedimentary covering

of the Alps.

Almost the whole covering has

[1] De Lapparent, *Traité de Géologie*, p. 1,785.

[2] Lugeon, *Bulletin Soc. Géol. de France*, 1901, p. 772.

151

been pushed over and piled up to the north. Lugeon[l] concludes that, before denudation had done its work and cut off the

Préalpes from their roots, there would have been found sheets, to

the number of eight, superimposed and extending between the Mont

Blanc massif and the massif of the Finsteraarhorn: these sheets

being the overthrown folds of the wrinkled sedimentary covering.

The general nature of the alpine structure

{Fig. 8}

will be understood from the presentation of it diagrammatically

after Schmidt of Basel (Fig. 8).[2] The section extends from

north to south, and brings out the relations of the several

recumbent folds. We must imagine almost the whole of these

superimposed folds now removed from the central regions of the

Alps by denudation,

[1] Lugeon, *loc. cit.*

[2] Schmidt, *_Ec. Geol. Helvetiae_*, vol. ix., No. 4.

152

and leaving the underlying gneisses rising through the remains of Permian, Triassic, and Jurassic sediments; while to the north the great limestone mountains and further north still, the Préalpes, carved from the remains of the recumbent folds, now stand with almost as little resemblance to the vanished mountains as the memories of the past have to its former intense reality.

These views as to the origin of the Alps, which are shared at the present day by so many distinguished geologists, had their origin in the labours of many now gone; dating back to Studer; finding their inspiration in the work of Heim, Suess, and Marcel Bertrand; and their consummation in that of Lugeon, Schardt, Rothpletz, Schmidt, and many others. Nor must it be forgotten that nearer home, somewhat similar phenomena, necessarily on a smaller scale, were recognised by Lapworth, twenty-six years ago, in his work on the structure of the Scottish Highlands.

An important tectonic principle underlies the development of the phenomena we have just been reviewing. The uppermost of the superimposed recumbent folds is more extended in its development than those which lie beneath. Passing downwards from the highest of the folds, they are found to be less and less extended both in the northerly and in the southerly direction, speaking of the special case—the Alps—now before us. This feature might be described somewhat differently. We might say that those folds which had their roots farther

153

to the south were the most drawn-out towards the north: or again we might say that the synclinal or deep-seated part of the fold has lagged behind the anticlinal or what was originally the highest part of the fold, in the advance of the latter to the north. The anticline has advanced relatively to the syncline. To this law one exception only is observed in the Swiss Alps; the sheet of the Brèche (_Byecciendecke_) falls short, in its northerly extension, of the underlying fold, which extends to form the Préalpes.

Contemplating such a generalised section as Professor Schmidt's, or, indeed, more particular sections, such as those in the Mont Blanc Massif by Marcel Bertrand,[1] of the Dent de Morcles, Diablerets, Wildhorn, and Massif de la Brèche by Lugeon,[2] or finally Termier's section of the Pelvoux Massif, [3] one is reminded of the breaking of waves on a sloping beach. The wave, retarded at its base, is carried forward above by its momentum, and finally spreads far up on the strand; and if it could there remain, the succeeding wave must necessarily find itself superimposed upon the first. But no effects of inertia, no kinetic effects, may be called to our aid in explaining the formation of mountains. Some geologists have accordingly supposed that in order to account for

[1] Marcel Bertrand, _Cong. Géol. Internat._, 1900, Guide Géol., xiii. a, p. 41.

[2] Lugeon, _loc. cit._, p. 773.

[3] De Lapparent, _Traite de Géol._, p. 1,773.

154

the recumbent folds and the peculiar phenomena of increasing overlap, or _déferlement_, an obstacle, fixed and

deep-seated, must have arrested the roots or synclines of the folds, and held them against translational motion, while a movement of the upper crust drew out and carried forward the anticlines. Others have contented themselves by recording the facts without advancing any explanatory hypothesis beyond that embodied in the incontestable statement that such phenomena must be referred to the effects of tangential forces acting in the Earth's crust.

It would appear that the explanation of the phenomena of recumbent folds and their déferlement is to be obtained directly from the temperature conditions prevailing throughout the stressed pile of rocks; and here the subject of mountain tectonics touches that with which we were elsewhere specially concerned—the geological influence of accumulated radioactive energy.

As already shown[1], a rise of temperature due to this source of several hundred degrees might be added to such temperatures as would arise from the mere blanketing of the Earth, and the consequent upward movement of the geotherms. The time element is here the most important consideration. The whole

sequence of
events from the first orogenic movements to the
final upheaval in
Pliocene times must probably have occupied not
less than ten
million years.

[1] Mountain Genesis, p. 129, et seq.

155

Unfortunately the full investigation of the
distribution of
temperature after any given time is beset with
difficulties; the
conditions being extremely complex. If the
radioactive heating
was strictly adiabatic—that is, if all the heat
was conserved and
none entered from without—the time required for
the attainment of
the equilibrium radioactive temperature would be
just about six
million years. The conditions are not, indeed,
adiabatic; but, on
the other hand, the rocks upraised by lateral
pressure were by no
means at 0° C. to start with. They must be assumed
to have
possessed such temperatures as the prior
radiothermal effects,
and the conducted heat from the Earth's interior,
may have
established.

It would from this appear probable that if a
duration of ten
million years was involved, the equilibrium

radioactive

temperatures must nearly have been attained. The effects of heat

conducted from the underlying earthcrust have to be added,

leading to a further rise in temperature of not less than 500° or

600°. In such considerations the observed indications of high

temperatures in materials now laid bare by denudation, probably

find their explanation (Pl. XIX).

The first fact that we infer from the former existence of such a

temperature distribution is the improbability, indeed the

impossibility, that anything resembling a rigid obstacle, or

deep-seated "horst," can have existed beneath the present

surface-level, and opposed the northerly movement of the

deep-lying synclines. For

156

such a horst can only have been constituted of some siliceous

rock-material such as we find everywhere rising through the

worn-down sediments of the Alps; and the idea that this could

retain rigidity under the prevailing temperature conditions, must

be dismissed. There is no need to labour this question; the horst

cannot have existed. To what, then, is the

retardation of the
lower parts of the folds, their overthrow, above,
to the north,
and their déferlement, to be ascribed?

A little consideration shows that the very
conditions of high
temperature and viscosity, which render untenable
the hypothesis
of a rigid obstacle, suffice to afford a full
explanation of the
retardation of the roots of the folds. For
directed translatory
movements cannot be transmitted through a fluid,
pressure in
which is necessarily hydrostatic, and must be
exerted equally in
every direction. And this applies, not only to a
fluid, but to a
body which will yield viscously to an impressed
force. There will
be a gradation, according as viscosity gives place
to rigidity,
between the states in which the applied force
resolves itself
into a purely hydrostatic pressure, and in which
it is
transmitted through the material as a directed
thrust. The nature
of the force, in the most general case, of course,
has to be
considered; whether it is suddenly applied and of
brief duration,
or steady and long-continued. The latter
conditions alone apply
to the present case.

It follows from this that, although a tangential force

157

or pressure be engendered by a crustal movement occurring to the south, and the resultant effects be transmitted northwards, these stresses can only mechanically affect the rigid parts of the crust into which they are carried. That is to say, they may result in folding and crushing, or horizontally transporting, the upper layers of the Earth's crust; but in the deeper-lying viscous materials they must be resolved into hydrostatic pressure which may act to upheave the overlying covering, but must refuse to transmit the horizontal translatory movements affecting the rigid materials above.

Between the regions in which these two opposing conditions prevail there will be no hard and fast line; but with the downward increase of fluidity there will be a gradual failure of the mechanical conditions and an increase of the hydrostatic.

Thus while the uppermost layers of the crust may be transported to the full amount of the crustal displacement acting from the south (speaking still of the Alps) deeper down

there will be a
lesser horizontal movement, and still deeper there
is no
influence to urge the viscous rock-materials in a
northerly
direction. The consequences of these conditions
must be the
recumbence of the folds formed under the crust-
stress, and their
déferlement towards the north. To see this, we
must follow the
several stages of development.

The earliest movements, we may suppose, result in
flexures of the
Jura-Mountain type—that is, in a

158

succession of undulations more or less
symmetrical. As the
orogenic force continues and develops, these
undulations give
place to folds, the limbs of which are
approximately vertical,
and the synclinal parts of which become ever more
and more
depressed into the deeper, and necessarily hotter,
underlying
materials; the anticlines being probably
correspondingly
elevated. These events are slowly developed, and
the temperature
beneath is steadily rising in consequence of the
conducted
interior heat, and the steady accumulation of
radioactive energy
in the sedimentary rocks and in the buried

radioactive layer of the Earth. The work expended on the crushed and sheared rock also contributes to the developing temperature. Thus the geotherms must move upwards, and the viscous conditions extend from below; continually diminishing the downward range of the translatory movements progressing in the higher parts. While above the folded sediments are being carried northward, beneath they are becoming anchored in the growing viscosity of the medium. The anticlines will bend over, and the most southerly of the folds will gradually become pushed or bent over those lying to the north. Finally, the whole upper part of the sheaf will become horizontally recumbent; and as the uppermost folds will be those experiencing the greatest effects of the continued displacement, the déferlement or overlap must necessarily arise.

We may follow these stages of mountain evolution

159

in a diagram (Fig. 9) in which we eliminate intermediate conditions, and regard the early and final stages of development only. In the upper sketch we suppose the lateral compression much

developed and the upward movement of the geotherms in progress.

The dotted line may be assumed to be a geotherm having a temperature of viscosity. If the conditions here shown persist

{Fig. 9}

indefinitely, there is no doubt that the only further

developments possible are the continued crushing of the sediments

and the bodily displacement of the whole mass to the north. The

second figure is intended to show in what manner these results

are evaded. The geotherm of viscosity has risen. All above it is

affected mechanically by the continuing stress, and borne

northwards in varying

160

degree depending upon the rigidity. The folds have been

overthrown and drawn out; those which lay originally most to the

south have become the uppermost; and, experiencing the maximum

amount of displacement, overlap those lying beneath. There has

also been a certain amount of upthrow owing to the hydrostatic

pressure. This last-mentioned element of the phenomena is of

highly indeterminate character, for we know not

the limits to which the hydrostatic pressure may be transmitted, and where it may most readily find relief. While, according to some of the published sections, the uplifting force would seem to have influenced the final results of the orogenic movements, a discussion of its effects would not be profitable.

161

OTHER MINDS THAN OURS?

IN the year 1610 Galileo, looking through his telescope then newly perfected by his own hands, discovered that the planet Jupiter was attended by a train of tiny stars which went round and round him just as the moon goes round the Earth.

It was a revelation too great to be credited by mankind. It was opposed to the doctrine of the centrality of the Earth, for it suggested that other worlds constituted like ours might exist in the heavens.

Some said it was a mere optic illusion; others that he who looked through such a tube did it at the peril of his soul—it was but a delusion of Satan. Galileo converted a few of the unbelievers who had the courage to look through his telescope. To

the others he
said, he hoped they would see those moons on their
way to heaven.
Old as this story is it has never lost its pathos
or its
teaching.

The spirit which assailed Galileo's discoveries
and which finally
was potent to overshadow his declining years,
closed in former
days the mouths of those who asked the question
written at the
head of this lecture: "Are we to believe that
there are other
minds than ours?"

162

Today we consider the question in a very different
spirit. Few
would regard it as either foolish or improper. Its
intense
interest would be admitted by all, and but for the
limitations
closing our way on every side it would, doubtless,
attract the
most earnest investigation. Even on the mere
balance of judgment
between the probable and the improbable, we have
little to go on.
We know nothing definitely as to the conditions
under which life
may originate: whether these are such as to be
rare almost to
impossibility, or common almost to certainty. Only
within narrow
limits of temperature and in presence of certain

of the elements,
can life like ours exist, and outside these
conditions life, if
such there be, must be different from ours. Once
originated it is
so constituted as to assail the energies around it
and to advance
from less to greater. Do we know more than these
vague facts?

Yes, we have in our experience one other fact and
one involving
much.

We know that our world is very old; that life has
been for many
millions of years upon it; and that Man as a
thinking being is
but of yesterday. Here is then a condition to be
fulfilled. To
every world is physically assigned a limit to the
period during
which it is habitable according to our knowledge
of life and its
necessities. This limit passed and rationality
missed, the chance
for that world is gone for ever, and other minds
than ours
assuredly will not from it contemplate the
universe. Looking at
our own world we see that the tree of life has,

163

indeed, branched, leaved and, possibly, budded
many times; it
never bloomed but once.

All difficulties dissolve and speculations become needless under one condition only: that in which rationality may be inferred directly or indirectly by our observations on some sister world in space, This is just the evidence which in recent years has been claimed as derived from a study of the surface of Mars. To that planet our hope of such evidence is restricted. Our survey in all other directions is barred by insurmountable difficulties. Unless some meteoric record reached our Earth, revelatory of intelligence on a perished world, our only hope of obtaining such evidence rests on the observation of Mars' surface features. To this subject we confine our attention in what follows.

The observations made during recent years upon the surface features of Mars have, excusably enough, given rise to sensational reports. We must consider under what circumstances these observations have been made.

Mars comes into particularly favourable conditions for observation every fifteen years. It is true that every two years and two months we overtake him in his orbit and he is then in "opposition." That is, the Earth is between him

and the sun: he
is therefore in the opposite part of the heavens
to the sun. Now
Mars' orbit is very excentric, sometimes he is 139
million miles
from the sun, and sometimes he as as much as 154
million miles
from the sun. The Earth's orbit is, by comparison,
almost

164

a circle. Evidently if we pass him when he is
nearest to the sun
we see him at his best; not only because he is
then nearest to
us, but because he is then also most brightly lit.
In such
favourable oppositions we are within 35 million
miles of him; if
Mars was in aphelion we would pass him at a
distance of 61
million miles. Opposition occurs under the most
favourable
circumstances every fifteen years. There was one
in 1862, another
in 1877, one in 1892, and so on.

When Mars is 35 million miles off and we apply a
telescope
magnifying 1,000 diameters, we see him as if
placed 35,000 miles
off. This would be seven times nearer than we see
the moon with
the naked eye. As Mars has a diameter about twice
as great as
that of the moon, at such a distance he would look
fourteen times

the diameter of the moon. Granting favourable conditions of atmosphere much should be seen.

But these are just the conditions of atmosphere of which most of the European observatories cannot boast. It is to the honour of Schiaparelli, of Milan, that under comparatively unfavourable conditions and with a small instrument, he so far outstripped his contemporaries in the observation of the features of Mars that those contemporaries received much of his early discoveries with scepticism. Light and dark outlines and patches on the planet's surface had indeed been mapped by others, and even a couple of the canals sighted; but at the opposition of 1877 Schiaparelli first mapped any considerable

165

number of the celebrated "canals" and showed that these constituted an extraordinary and characteristic feature of the planet's geography. He called them "canali," meaning thereby "channels." It is remarkable indeed that a mistranslation appears really responsible for the initiation of the idea that these features are canals.

In 1882 Schiaparelli startled the astronomical world by declaring that he saw some of the canals double—that is appearing as two parallel lines. As these lines span the planet's surface for distances of many thousands of miles the announcement naturally gave rise to much surprise and, as I have said, to much scepticism. But he resolutely stuck to his statement. Here is his map of 1882. It is sufficiently startling.

In 1892 he drew a new map. It adds a little to the former map, but the doubling was not so well seen. It is just the strangest feature about this doubling that at times it is conspicuous, at times invisible. A line which is distinctly seen as a single line at one time, a few weeks later will appear distinctly to consist of two parallel lines; like railway tracks, but tracks perhaps 200 miles apart and up to 3,000 or even 4,000 miles in length.

Many speculations were, of course, made to account for the origin of such features. No known surface peculiarity on the Earth or moon at all resembles these features. The moon's surface as you know is cracked and

streaked. But the cracks are what we generally find cracks to be—either aimless, wandering lines, or, if radiating from a centre, then lines which contract in width as they leave the point of rupture. Where will we find cracks accurately parallel to one another sweeping round a planet's face with steady curvature for, 4,000 miles, and crossing each other as if quite unhampered by one another's presence? If the phenomenon on Mars be due to cracks they imply a uniformity in thickness and strength of crust, a homogeneity, quite beyond all anticipation. We will afterwards see that the course of the lines is itself further opposed to the theory that haphazard cracking of the crust of the planet is responsible for the lines. It was also suggested that the surface of the planet was covered with ice and that these were cracks in the ice. This theory has even greater difficulties than the last to contend with. Rivers have been suggested. A glance at our own maps at once disposes of this hypothesis. Rivers wander just as cracks do and parallel rivers like parallel cracks are unknown.

In time the many suggestions were put aside. One only remained.

That the lines are actually the work of intelligence; actually are canals, artificially made, constructed for irrigation purposes on a scale of which we can hardly form any conception based on our own earthly engineering structures.

During the opposition of 1894, Percival Lowell, along with A. E. Douglass, and W. H. Pickering,

167

observed the planet from the summit of a mountain in Arizona, using an 18-inch refracting telescope and every resource of delicate measurement and spectroscopy. So superb a climate favoured them that for ten months the planet was kept under continual observation. Over 900 drawings were made and not only were Schiaparelli's channels confirmed, but they added 116 to his 79, on that portion of the planet visible at that opposition. They made the further important discovery that the lines do not stop short at the dark regions of the planet's surface, as hitherto believed, but go right on in many cases; the curvature of the lines being unaltered.

Lowell is an uncompromising advocate of the "canal" theory. If his arguments are correct we have at once an answer to our question, "Are there other minds than ours?"

We must consider a moment Lowell's arguments; not that it is my intention to combat them. You must form your own conclusions. I shall lay before you another and, as I venture to think, more adequate hypothesis in explanation of the channels of Schiaparelli. We learn, however, much from Lowell's book—it is full of interest.[1]

Lowell lays a deep foundation. He begins by showing that Mars has an atmosphere. This must be granted him till some counter observations are made.

[1] *_Mars_*, by Percival Lowell (Longmans, Green & Co.), 1896,

168

It is generally accepted. What that atmosphere is, is another matter. He certainly has made out a good case for the presence of water as one of its constituents,

It was long known that Mars possessed white regions at his poles, just as our Earth does. The waning of these polar snows—if indeed

they are such—with the advance of the Martian summer, had often been observed. Lowell plots day by day this waning. It is evident from his observations that the snowfall must be light indeed. We see in his map the south pole turned towards us. Mars in perihelion always turns his south pole towards the sun and therefore towards the Earth. We see that between the dates June 3rd to August 3rd—or in two months—the polar snow had almost completely vanished. This denotes a very scanty covering. It must be remembered that Mars even when nearest to the sun receives but half our supply of solar heat and light.

But other evidence exists to show that Mars probably possesses but little water upon his surface. The dark places are not water-covered, although they have been named as if they were, indeed, seas and lakes. Various phenomena show this. The canals show it. It would never do to imagine canals crossing the seas. No great rivers are visible. There is a striking absence of clouds. The atmosphere of Mars seems as serene as that of Venus appears to be cloudy. Mists and clouds, however, sometime appear to veil his face and add to the difficulty of

making observations near the limb of the planet. Lowell concludes it must be a calm and serene atmosphere; probably only one-seventh of our own in density. The normal height of the barometer in Mars would then be but four and a half inches. This is a pressure far less than exists on the top of the highest terrestrial mountain. A mountain here must have an altitude of about ten miles to possess so low a pressure on its summit. Drops of water big enough to form rain can hardly collect in such a rarefied atmosphere. Moisture will fall as dew or frost upon the ground. The days will be hot owing to the unimpeded solar radiation; the nights bitterly cold owing to the free radiation into space.

We may add that in such a climate the frost will descend principally upon the high ground at night time and in the advancing day it will melt. The freer radiation brings about this phenomenon among our own mountains in clear and calm weather.

With the progressive melting of the snow upon the pole Lowell connected many phenomena upon the planet's surface

of much
interest. The dark spaces appear to grow darker
and more
greenish. The canals begin to show themselves and
reveal their
double nature. All this suggests that the moisture
liberated by
the melting of the polar snow with the advancing
year, is
carrying vitality and springtime over the surface
of the planet.
But how is the water conveyed?

Lowell believes principally by the canals. These
are

170

constructed triangulating the surface of the
planet in all
directions. What we see, according to Lowell, is
not the canal
itself, but the broad band of vegetation which
springs up on the
arrival of the water. This band is perhaps thirty
or forty miles
wide, but perhaps much less, for Lowell reports
that the better
the conditions of observation the finer the lines
appeared, so
that they may be as narrow, possibly, as fifteen
miles. It is to
be remarked that a just visible dot on the surface
of Mars must
possess a diameter of 30 miles. But a chain of
much smaller dots
will be visible, just as we can see such fine
objects as spiders'

webs. The widening of the canals is then accounted for, according to Lowell, by the growth of a band of vegetation, similar to that which springs into existence when the floods of the Nile irrigate the plains of Egypt.

If no other explanation of the lines is forthcoming than that they are the work of intelligence, all this must be remembered.

If all other theories fail us, much must be granted Lowell. We must not reason like fishes—as Lowell puts it—and deny that intelligent beings can thrive in an atmospheric pressure of four and half inches of mercury. Zurbriggen has recently got to the top of Aconcagua, a height of 24,000 feet. On the summit of such a mountain the barometer must stand at about ten inches. Why should not beings be developed by evolution with a lung capacity capable of living at two and a half times this altitude. Those steadily

171

curved parallel lines are, indeed, very unlike anything we have experience of. It would be rather to be expected that another civilisation than our own would present many wide

differences in
its development.

What then is the picture we have before us
according to Lowell?

It is a sufficiently dramatic one.

Mars is a world whose water supply, never probably
very abundant,
has through countless years been drying up,
sinking into his
surface. But the inhabitants are making a brave
fight for it,
They have constructed canals right round their
world so that the
water, which otherwise would run to waste over the
vast deserts,
is led from oasis to oasis. Here the great centres
of
civilisation are placed: their Londons, Viennas,
New Yorks. These
gigantic works are the works of despair. A great
and civilised
world finds death staring it in the face. They
have had to triple
their canals so that when the central canal has
done its work the
water is turned into the side canals, in order to
utilise it as
far as possible. Through their splendid telescopes
they must view
our seas and ample rivers; and must die like
travellers in the
desert seeing in a mirage the cool waters of a
distant lake.

Perhaps that lonely signal reported to have been
seen in the

twilight limb of Mars was the outcome of pride in
their splendid
and perishing civilisation. They would leave some
memory of it:
they would have us witness how great was that
civilisation before
they perish!

I close this dramatic picture with the poor
comfort

172

that several philanthropic people have suggested
signalling to
them as a mark of sympathy. It is said that a
fortune was
bequeathed to the French Academy for the purpose
of communicating
with the Martians. It has been suggested that we
could flash
signals to them by means of gigantic mirrors
reflecting the light
of our Sun. Or, again, that we might light
bonfires on a
sufficiently large scale. They would have to be
about ten miles
in diameter! A writer in the Pall Mall Gazette
suggested that
there need really be no difficulty in the matter.
With the kind
cooperation of the London Gas Companies (this was
before the days
of electric lighting) a signal might be sent
without any
additional expense if the gas companies would
consent to
simultaneously turn off the gas at intervals of

five minutes over
the whole of London, a signal which would be
visible to the
astronomers in Mars would result. He adds,
naively: "If only
tried for an hour each night some results might be
obtained."

II

We have reviewed the theory of the artificial
construction of the
Martian lines. The amount of consideration we are
disposed to
give to the supposition that there are upon Mars
other minds than
ours will—as I have stated—necessarily depend upon
whether or not
we can assign a probable explanation of the lines
upon purely
physical grounds. If it is apparent that such

173

lines would be formed with great probability under
certain
conditions, which conditions are themselves
probable, then the
argument by exclusion for the existence of
civilisation on Mars,
at once breaks down.

{Fig. 10}

As a romance writer is sometimes under the
necessity of
transporting his readers to other scenes, so I
must now ask you
to consent to be transported some millions

of miles into the region of the heavens which lies outside Mars' orbit.

Between Mars and Jupiter is a chasm of 341 millions of miles.

This gap in the sequence of planets was long known to be quite

out of keeping with the orderly succession of worlds outward from

the Sun. A society was formed at the close of the last century

for the detection of the missing world. On the first day of the

last century, Piazzini—who, by the way, was not a member of the

society—discovered a tiny world in the vacant gap. Although

eagerly welcomed, as better than nothing, it was a disappointing

find. The new world was a mere rock. A speck of about 160 miles

in diameter. It was obviously never intended that such a body

should have all this space to itself. And, sure enough, shortly

after, another small world was discovered. Then another was

found, and another, and so on; and now more than 400 of these

strange little worlds are known.

But whence came such bodies? The generally accepted belief is

that these really represent a misbegotten world. When the Sun was

younger he shed off the several worlds of our system as so many rings. Each ring then coalesced into a world. Neptune being the first born; Mercury the youngest born.

After Jupiter was thrown off, and the Sun had shrunk away inwards some 200 million miles, he shed off another ring. Meaning that this offspring of his should grow up like the rest, develop into a stable world with the

175

potentiality even, it may be, of becoming the abode of rational beings. But something went wrong. It broke up into a ring of little bodies, circulating around him.

It is probable on this hypothesis that the number we are acquainted with does not nearly represent the actual number of past and present asteroids. It would take 125,000 of the biggest of them to make up a globe as big as our world. They, so far as they are known, vary in size from 10 miles to 160 miles in diameter. It is probable then—on the assumption that this failure of a world was intended to be about the mass of our Earth—that they numbered, and possibly number, many hundreds of thousands.

Some of these little bodies are very peculiar in respect to the orbits they move in. This peculiarity is sometimes in the eccentricity of their orbits, sometimes in the manner in which their orbits are tilted to the general plane of the ecliptic, in which all the other planets move.

The eccentricity, according to Proctor, in some cases may attain such extremes as to bring the little world inside Mars' mean distance from the sun. This, as you will remember, is very much less than his greatest distance from the sun. The entire belt of asteroids—as known—lie much nearer to Mars than to Jupiter.

As regards the tilt of their orbits, some are actually as much as 34 degrees inclined to the ecliptic, so that in fact they are seen from the Earth among our polar constellations.

176

From all this you see that Mars occupies a rather hot comer in the solar system. Is it not possible that more than once in the remote past Mars may have encountered one of these wanderers? If he came within a certain distance of the small body his great

mass would sway it from its orbit, and under certain conditions he would pick up a satellite in this manner. That his present satellites were actually so acquired is the suggestion of Newton, of Yale College.

Mars' satellites are indeed suspiciously and most abnormally small. I have not time to prove this to you by comparison with the other worlds of the solar system. In fact, they were not discovered till 1877—although they were predicted in a most curious manner, with the most uncannily accurate details, by Swift.

One of these bodies is about 36 miles in diameter. This is Phobos. Phobos is only 3,700 miles from the surface of Mars. The other is smaller and further off. He is named Deimos, and his diameter is only 10 miles. He is 12,500 miles from Mars' surface. With the exception of Phobos the next smallest satellite known in the solar system is one of Saturn's—Hyperion; almost 800 miles in diameter. The inner one goes all round Mars in 7½ hours. This is Phobos' month. Mars turns on his axis in 24 hours and 40 minutes, so that people in Mars would see the rise of Phobos twice in the

course of a day and night; lie would apparently cross the sky

177

going against the other satellite; that is, he would move apparently from west to east.

We may at least assume as probable that other satellites have been gathered by Mars in the past from the army of asteroids.

Some of the satellites so picked up would be direct: that is, would move round the planet in the direction of his axial rotation. Others, on the chances, would be retrograde: that is, would move against his axial rotation. They would describe orbits making the same various angles with the ecliptic as do the asteroids; and we may be sure they would be of the same varying dimensions.

We go on to inquire what would be the consequence to Mars of such captures.

A satellite captured in this manner is very likely to be pulled into the Planet. This is a probable end of a satellite in any case. It will probably be the end of our satellite too. The satellite Phobos is indeed believed to be about to

take this very
plunge into his planet. But in the case when the
satellite picked
up happens to be rotating round the planet in the
opposite
direction to the axial rotation of the planet, it
is pretty
certain that its career as a satellite will be a
brief one. The
reasons for this I cannot now give. If, then, Mars
picked up
satellites he is very sure to have absorbed them
sooner or later.
Sooner if they happened to be retrograde
satellites, later if
direct satellites. His present satellites are
recent additions.
They are direct.

178

The path of an expiring satellite will be a slow
spiral described
round the planet. The spiral will at last, after
many years,
bring the satellite down upon the surface of the
primary. Its
final approach will be accelerated if the planet
possesses an
atmosphere, as Mars probably does. A satellite of
the dimensions
of Phobos—that is 36 miles in diameter—would
hardly survive more
than 30 to 60 years within seventy miles of Mars'
surface. It
will then be rotating round Mars in an hour and
forty minutes,
moving, in fact, at the rate of 2.2 miles per

second. In the course of this 30 or 60 years it will, therefore, get round perhaps 200,000 times, before it finally crashes down upon the Martians. During this closing history of the satellite there is reason to believe, however, that it would by no means pursue continually the same path over the surface of the planet. There are many disturbing factors to be considered. Being so small any large surface features of Mars would probably act to perturb the orbit of the satellite.

The explanation of Mars' lines which I suggest, is that they were formed by the approach of such satellites in former times. I do not mean that they are lines cut into his surface by the actual infall of a satellite. The final end of the satellite would be too rapid for this, I think. But I hope to be able to show you that there is reason to believe that the mere passage of the satellite, say at 70 miles above the surface of the planet, will, in itself, give rise to effects on the crust of the planet capable

of accounting for just such single or parallel lines as we see.

In the first place we have to consider the stability of the satellite. Even in the case of a small satellite we cannot overlook the fact that the half of the satellite near the planet is pulled towards the planet by a gravitational force greater than that attracting the outer half, and that the centrifugal force is less on the inner than on the outer hemisphere. Hence there exists a force tending to tear the satellite asunder on the equatorial section tangential

{Fig. 11}

to the planet's surface. If in a fluid or plastic state, Phobos, for instance, could not possibly exist near the planet's surface.

The forces referred to would decide its fate. It may be shown by calculation, however, that if Phobos has the strength of basalt or glass there would remain a considerable coefficient of safety in favour of the satellite's stability; even when the surfaces of planet and satellite were separated by only five miles.

We have now to consider some things which we expect will happen

before the satellite takes its final plunge into the planet.

180

This diagram (Fig. 11) shows you the satellite travelling above the surface of the planet. The satellite is advancing towards, or away from, the spectator. The planet is supposed to show its solid crust in cross section, which may be a few miles in thickness. Below this is such a hot plastic magma as we have reason to believe underlies much of the solid crust of our own Earth. Now there is an attraction between the satellite and the crust of the planet; the same gravitational attraction which exists between every particle of matter in the universe. Let us consider how this attraction will affect the planet's crust. I have drawn little arrows to show how we may consider the attraction of the satellite pulling the crust of the planet not only upwards, but also pulling it inwards beneath the satellite. I have made these arrows longer where calculation shows the stress is greater. You see that the greatest lifting stress is just beneath the satellite, whereas the greatest stress pulling the crust in under the satellite is at a point

which lies out from under the satellite, at a considerable distance. At each side of the satellite there is a point where the stress pulling on the crust is the greatest. Of the two stresses the lifting stress will tend to raise the crust a little; the pulling stress may in certain cases actually tear the crust across; as at A and B.

It is possible to calculate the amount of the stress at the point at each side of the satellite where the stress is at its greatest. We must assume the satellite to be a certain size and density; we must also assume the crust of

181

Mars to be of some certain density. To fix our ideas on these points I take the case of the present satellite Phobos. What amount of stress will he exert upon the crust of Mars when he approaches within, say, 40 miles of the planet's surface? We know his size approximately—he is about 36 miles in diameter. We can guess his density to be between four times that of water and eight times that of water. We may assume the density of Mars' surface to be about the same as that of our

Earth's surface, that is three times as dense as water. We now find that the greatest stress tending to rend open the surface crust of Mars will be between 4,000 and 8,000 pounds to the square foot according to the density we assign to Phobos.

Will such a stress actually tear open the crust? We are not able to answer this question with any certainty. Much will depend upon the nature and condition of the crust. Thus, suppose that we are here (Fig. 12) looking down upon the satellite which is moving along slowly relatively to Mars' surface, in the direction of the arrow. The satellite has just passed over a weak and cracked part of the planet's crust. Here the stress has been sufficient to start two cracks. Now you know how easy it is to tear a piece of cloth when you go to the edge of it in order to make a beginning. Here the stress from the satellite has got to the edge of the crust. It is greatly concentrated just at the extremities of the cracks. It will, under such circumstances probably carry on the

182

tear. If it does not do so this time, remember the satellite will

some hours later be coming over the same place again, and then again for, at least, many hundreds of times. Then also we are not limited to the assumption that the

{Fig. 12}

satellite is as small as Phobos. Suppose we consider the case of a satellite approaching Mars which has a diameter double that of Phobos; a diameter still much less than that of the larger class of asteroids. Even at the distance

183

of 65 miles the stress will now amount to as much as from 15 to 30 tons per square foot. It is almost certain that such a stress repeated a comparatively few times over the same parts of the planet's surface would so rend the crust as to set up lines along which plutonic action would find a vent. That is, we might expect along these lines all the phenomena of upheaval and volcanic eruption which give rise to surface elevations.

The probable effect of a satellite of this dimension travelling slowly relatively to the surface of Mars is, then, to leave a very conspicuous memorial of his presence behind him. You see from the diagram that this memorial will consist

o: two parallel
lines of disturbance.

The linear character of the gravitational effects of the satellite is due entirely to the motion of the satellite relatively to the surface of the planet. If the satellite stood still above the surface the gravitational stress in the crust would, of course, be exerted radially outwards from the centre of the satellite. It would attain at the central point beneath the satellite its maximum vertical effect, and at some radial distance measured outwards from this point, which distance we can calculate, its maximum horizontal tearing effect. When the satellite moves relatively to the planet's crust, the horizontal tearing force acts differently according to whether it is directed in the line of motion or at right angles to this line.

In the direction of motion we see that the satellite

184

creates as it passes over the crust a wave of rarefaction or tension as at D, followed by compression just beneath the satellite and by a reversed direction of

gravitational pull as the satellite passes onwards. These stresses rapidly replace one another as the satellite travels along. They are resisted by the inertia of the crust, and are taken up by its elasticity. The nature of this succession of alternate compressions and rarefactions in the crust possess some resemblance to those arising in an earthquake shock.

If we consider the effects taking place laterally to the line of motion we see that there are no such changes in the nature of the forces in the crust. At each passage of the satellite the horizontal tearing stress increases to a maximum, when it is exerted laterally, along the line passing through the horizontal projection of the satellite and at right angles to the line of motion, and again dies away. It is always a tearing stress, renewed again and again.

This effect is at its maximum along two particular parallel lines which are tangents to the circle of maximum horizontal stress and which run parallel with the path of the satellite. The distance separating these lines depend upon the elevation of the satellite above the planet's surface. Such lines mark out

the theoretical
axes of the "double canals" which future crustal
movements will
more fully develop.

It is interesting to consider what the effect of
such

185

conditions would be if they arose at the surface
of our own
planet. We assume a horizontal force in the crust
adequate to set
up tensile stresses of the order, say, of fifteen
tons to the
square foot and these stresses to be repeated
every few hours;
our world being also subject to the dynamic
effects we recognise
in and beneath its crust.

It is easy to see that the areas over which the
satellite exerted
its gravitational stresses must become the foci –
foci of linear
form-of tectonic developments or crust movements.
The relief of
stresses, from whatever cause arising, in and
beneath the crust
must surely take place in these regions of
disturbance and along
these linear areas. Here must become concentrated
the folding
movements, which are under existing conditions
brought into the
geosynclines, along with their attendant volcanic
phenomena. In

the case of Mars such a concentration of tectonic events would not, owing to the absence of extensive subaerial denudation and great oceans, be complicated by the existence of such synclinal accumulations as have controlled terrestrial surface development. With the passage of time the linear features would probably develop; the energetic substratum continually asserting its influence along such lines of weakness. It is in the highest degree probable that radioactivity plays no less a part in Martian history than in terrestrial. The fact of radioactive heating allows us to assume the thin surface crust and continued sub-crustal energy throughout the entire period of the planet's history.

186

How far will these effects resemble the double canals of Mars? In this figure and in the calculations I have given you I have supposed the satellite engaged in marking the planet's surface with two lines separated by about the interval separating the wider double canals of Mars—that is about 220 miles apart. What the distance between the lines will be, as already stated, will

depend upon the height of the satellite above the surface when it comes upon a part of the crust in a condition to be affected by the stresses it sets up in it. If the satellite does its work at a point lower down above the surface the canal produced will be narrower. The stresses, too, will then be much greater. I must also observe that once the crust has yielded to the pulling stress, there is great probability that in future revolutions of the satellite a central fracture will result. For then all the pulling force adds itself to the lifting force and tends to crush the crust inwards on the central line beneath the satellite. It is thus quite possible that the passage of a satellite may give rise to triple lines. There is reason to believe that the canals on Mars are in some cases triple.

I have spoken all along of the satellite moving slowly over the surface of Mars. I have done so as I cannot at all pronounce so readily on what will happen when the satellite's velocity over the surface of Mars is very great. To account for all the lines mapped by Lowell some of them must have been produced by satellities moving relatively to the surface of

Mars at
velocities so great

187

as three miles a second or even rather more. The stresses set up are, in such cases, very difficult to estimate. It has not yet been done. Parallel lines of greatest stress or impulse ought to be formed as in the other case.

I now ask your attention to another kind of evidence that the lines are due in some way to the motion of satellites passing over the surface of Mars.

I may put the fresh evidence to which I refer, in this way: In Lowell's map (Pl. XXII, p. 192), and in a less degree in Schiaparelli's map (ante p. 166), we are given the course of the lines as fragments of incomplete curves. Now these curves might have been anything at all. We must take them as they are, however, when we apply them as a test of the theory that the motion of a satellite round Mars can strike such lines. If it can be shown that satellites revolving round Mars might strike just such curves then we assume this as an added confirmation of the hypothesis.

We must begin by realising what sort of curves a satellite which disturbs the surface of a planet would leave behind it after its demise. You might think that the satellite revolving round and round the planet must simply describe a circle upon the spherical surface of the planet: a "great circle" as it is called; that is the greatest circle which can be described upon a sphere. This great circle can, however, only be struck, as you will see, when the planet is not turning upon its axis: a condition not likely to be realised.

This diagram (PI. XXI) shows the surface of a globe

188

covered with the usual imaginary lines of latitude and longitude. The orbit of a supposed satellite is shown by a line crossing the sphere at some assumed angle with the equator. Along this line the satellite always moves at uniform velocity, passing across and round the back of the sphere and again across. If the sphere is not turning on its polar axis then this satellite, which we will suppose armed with a pencil which draws a line upon the sphere, will strike a great circle right round the

sphere. But
the sphere is rotating. And it is to be expected
that at
different times in a planet's history the rate of
rotation varies
very much indeed. There is reason to believe that
our own day was
once only $2\frac{1}{2}$ hours long, or thereabouts. After a
preliminary rise
in velocity of axial rotation, due to shrinkage
attending rapid
cooling, a planet as it advances in years rotates
slower and
slower. This phenomenon is due to tidal influences
of the sun or
of satellites. On the assumption that satellites
fell into Mars
there would in his case be a further action
tending to shorten
his day as time went on.

The effect of the rotation of the planet will be,
of course, that
as the satellite advances with its pencil it finds
the surface of
the sphere being displaced from under it. The line
struck ceases
to be the great circle but wanders off in another
curve—which is
in fact not a circle at all.

You will readily see how we find this curve.
Suppose the sphere
to be rotating at such a speed that while the
satellite is
advancing the distance $_{0a_}$, the point $_{b_}$ on the

sphere will be carried into the path of the satellite. The pencil will mark this point. Similarly we find that all the points along this full curved line are points which will just find themselves under the satellite as it passes with its pencil. This curve is then the track marked out by the revolving satellite. You see it dotted round the back of the sphere to where it cuts the equator at a certain point. The course of the curve and the point where it cuts the equator, before proceeding on its way, entirely depend upon the rate at which we suppose the sphere to be rotating and the satellite to be describing the orbit. We may call the distance measured round the planet's equator separating the starting point of the curve from the point at which it again meets the equator, the "span" of the curve. The span then depends entirely upon the rate of rotation of the planet on its axis and of the satellite in its orbit round the planet.

But the nature of events might have been somewhat different. The satellite is, in the figure, supposed to be rotating round the sphere in the same direction as that in which the sphere is turning. It might have been that Mars had picked

up a satellite
travelling in the opposite direction to that in
which he was
turning. With the velocity of planet on its axis
and of satellite
in its orbit the same as before, a different curve
would have
been described. The span of the curve due to a
retrograde
satellite will be greater than that due to a
direct satellite.
The retrograde satellite will have a span more
than half

190

way round the planet, the direct satellite will
describe a curve
which will be less than half way round the planet:
that is a span
due to a retrograde satellite will be more than
180 degrees,
while the span due to a direct satellite will be
less than 180
degrees upon the planet's equator.

I would draw your attention to the fact that what
the span will
be does not depend upon how much the orbit of the
satellite is
inclined to the equator. This only decides how far
the curve
marked out by the satellite will recede from the
equator.

We find then, so far, that it is easy to
distinguish between the
direct and the retrograde curves. The span of one

is less, of the other greater, than 180 degrees. The number of degrees which either sort of curve subtends upon the equator entirely depends upon the velocity of the satellite and the axial velocity of the planet.

But of these two velocities that of the satellite may be taken as sensibly invariable, when close enough to use his pencil. This depends upon the law of centrifugal force, which teaches us that the mass of the planet alone decides the velocity of a satellite in its orbit at any fixed distance from the planet's centre. The other velocity—that of the planet upon its axis—was, as we have seen, not in the past what it is now. If then Mars, at various times in his past history, picked up satellites, these satellites will describe curves round him having different spans which will depend upon the velocity of axial rotation of Mars at the time and upon this only.

191

In what way now can we apply this knowledge of the curves described by a satellite as a test of the lunar origin of the lines on Mars?

To do this we must apply to Lowell's map. We pick out preferably, of course, the most complete and definite curves. The chain of canals of which Acheron and Erebus are members mark out a fairly definite curve. We produce it by eye, preserving the curvature as far as possible, till it cuts the equator. Reading the span on the equator we find' it to be 255 degrees. In the first place we say then that this curve is due to a retrograde satellite. We also note on Lowell's map that the greatest rise of the curve is to a point about 32 degrees north of the equator. This gives the inclination of the satellite's orbit to the plane of Mars' equator.

With these data we calculate the velocity which the planet must have possessed at the time the canal was formed on the hypothesis that the curve was indeed the work of a satellite. The final question now remains If we determine the curve due to this velocity of Mars on its axis, will this curve fit that one which appears on Lowell's map, and of which we have really availed ourselves of only three points? To answer this question we plot upon a sphere, the curve of a satellite, in the

manner I have described, assigning to this sphere the velocity derived from the span of 255 degrees. Having plotted the curve on the sphere it only remains to transfer it to Lowell's map. This is easily done.

192

This map (Pl. XXII) shows you the result of treating this, as well as other curves, in the manner just described. You see that whether the fragmentary curves are steep and receding far from the equator; or whether they are flat and lying close along the equator; whether they span less or more than 180 degrees; the curves determined on the supposition that they are the work of satellites revolving round Mars agree with the mapped curves; following them with wonderful accuracy; possessing their properties, and, indeed, in some cases, actually coinciding with them.

I may add that the inadmissible span of 180 degrees and spans very near this value, which are not well admissible, are so far as I can find, absent. The curves are not great circles.

You will require of me that I should explain the centres of radiation so conspicuous here and there on Lowell's map. The meeting of more than two lines at the oases is a phenomenon possibly of the same nature and also requiring explanation.

In the first place the curves to which I have but briefly referred actually give rise in most cases to nodal, or crossing points; sometimes on the equator, sometimes off the equator; through which the path of the satellite returns again and again. These nodal points will not, however, afford a general explanation of the many-branched radiants.

It is probable that we should refer such an appearance

193

as is shown at the Sinus Titanum to the perturbations of the satellite's path due to the surface features on Mars. Observe that the principal radiants are situated upon the boundary of the dark regions or at the oases. Higher surface levels may be involved in both cases. Some marked difference in topography must characterise both these features. The latter may possibly

originate in the destruction of satellites. Or again, they may arise in crustal disturbance of a volcanic nature, primarily induced or localised by the crossing of two canals. Whatever the origin of these features it is only necessary to assume that they represent elevated features of some magnitude to explain the multiplication of crossing lines. We must here recall what observers say of the multiplicity of the canals. According to Lowell, "What their number maybe lies quite beyond the possibility of count at present; for the better our own air, the more of them are visible."

Such innumerable canals are just what the present theory requires. An in-falling satellite will, in the course of the last 60 or 80 years of its career, circulate some 100,000 times over Mars' surface. Now what will determine the more conspicuous development of a particular canal? The mass of the satellite; the state of the surface crust; the proximity of the satellite; and the amount of repetition over the same ground. The after effects may be taken as proportional to the primary disturbance.

It is probable that elevated surface features will influence two of these conditions: the number of repetitions and the proximity to the surface. A tract 100 miles in diameter and elevated 5,000 or 10,000 feet would seriously perturb the orbit of such a body as Phobos. It is to be expected that not only would it be effective in swaying the orbit of the satellite in the horizontal direction but also would draw it down closer to the surface. It is even to be considered if such a mass might not become nodal to the satellite's orbit, so that this passed through or above this point at various inclinations with its primary direction. If acting to bring down the orbit then this will quicken the speed and cause the satellite further on its path to attain a somewhat higher elevation above the surface. The lines most conspicuous in the telescope are, in short, those which have been favoured by a combination of circumstances as reviewed above, among which crustal features have, in some cases, played a part.

I must briefly refer to what is one of the most interesting features of the Martian lines: the manner in which

they appear to
come and go like visions.

Something going on in Mars determines the phenomenon. On a particular night a certain line looks single. A few nights later signs of doubling are perceived, and later still, when the seeing is particularly good, not one but two lines are seen. Thus, as an example, we may take the case of Phison and Euphrates. Faint glimpses of the dual state were detected in the summer

195

and autumn, but not till November did they appear as distinctly double. Observe that by this time the Antarctic snows had melted, and there was in addition, sufficient time for the moisture so liberated to become diffused in the planet's atmosphere.

This increase in the definition and conspicuousness of certain details on Mars' surface is further brought into connection with the liberation of the polar snows and the diffusion of this water through the atmosphere, by the fact that the definition appeared progressively better from the south pole upwards as the snow disappeared. Lowell thinks this points to

vegetation springing up
under the influence of moisture; he considers,
however, as we
have seen, that the canals convey the moisture. He
has to assume
the construction of triple canals to explain the
doubling of the
lines.

If we once admit the canals to be elevated ranges—
not necessarily
of great height—the difficulty of accounting for
increased
definition with increase of moisture vanishes. We
need not
necessarily even suppose vegetation concerned.
With respect to
this last possibility we may remark that the
colour observations,
upon which the idea of vegetation is based, are
likely to be
uncertain owing to possible fatigue effects where
a dark object
is seen against a reddish background.

However this may be we have to consider what the
effects of
moisture increasing in the atmosphere of Mars will
be with regard
to the visibility of elevated ranges,

196

We assume a serene and rare atmosphere: the nights
intensely
cold, the days hot with the unveiled solar
radiation. On the hill
tops the cold of night will be still more intense

and so, also,
will the solar radiation by day. The result of
this state of
things will be that the moisture will be
precipitated mainly on
the mountains during the cold of night—in the form
of frost—and
during the day this covering of frost will melt;
and, just as we
see a heavy dew-fall darken the ground in summer,
so the melting
ice will set off the elevated land against the
arid plains below.
Our valleys are more moist than our mountains only
because our
moisture is so abundant that it drains off the
mountains into the
valleys. If moisture was scarce it would distil
from the plains
to the colder elevations of the hills. On this
view the
accentuation of a canal is the result of
meteorological effects
such as would arise in the Martian climate;
effects which must be
influenced by conditions of mountain elevation,
atmospheric
currents, etc. We, thus, follow Lowell in
ascribing the
accentuation of the canals to the circulation of
water in Mars;
but we assume a simple and natural mode of
conveyance and do not
postulate artificial structures of all but
impossible magnitude.
That vegetation may take part in the darkening of
the elevated

tracts is not improbable. Indeed we would expect that in the Martian climate these tracts would be the only fertile parts of the surface.

Clouds also there certainly are. More recent observations

197

appear to have set this beyond doubt. Their presence obviously brings in other possible explanations of the coming and going of elevated surface features.

Finally, we may ask what about the reliability of the maps? About this it is to be said that the most recent map—that by Lowell—has been confirmed by numerous drawings by different observers, and that it is, itself the result of over 900 drawings. It has become a standard chart of Mars, and while it would be rash to contend for absence of errors it appears certain that the trend of the principal canals may be relied on, as, also, the general features of the planet's surface.

The question of the possibility of illusion has frequently been raised. What I have said above to a great extent answers such objections. The close agreement between the drawings of different

observers ought really to set the matter at rest. Recently, however, photography has left no further room for scepticism.

First photographed in 1905, the planet has since been photographed many thousands of times from various observatories.

A majority of the canals have been so mapped. The doubling of the canals is stated to have been also so recorded.[1]

The hypothesis which I have ventured to put before you involves no organic intervention to account for the

[1] E. C. Slipher's paper in Popular Astronomy for March, 1914, gives a good account of the recent work.

198

details on Mars' surface. They are physical surface features.

Mars presents his history written upon his face in the scars of

former encounters—like the shield of Sir Launcelot. Some of the

most interesting inferences of mathematical and physical

astronomy find a confirmation in his history. The slowing down in

the rate of axial rotation of the primary; the final inevitable

destruction of the satellite; the existence in the past of a far

larger number of asteroids than we at present are acquainted

with; all these great facts are involved in the theory now advanced. If justifiably, then is Mars' face a veritable Principia.

To fully answer the question which heads these lectures, we should go out into the populous solitudes (if the term be permitted) which lie beyond our system. It is well that there is now no time left to do so; for, in fact, there we can only dream dreams wherein the limits of the possible and the impossible become lost.

The marvel of the infinite number of stars is not so marvellous as the rationality that fain would comprehend them. In seeking other minds than ours we seek for what is almost infinitely complex and coordinated in a material universe relatively simple and heterogeneous. In our mental attitude towards the great question, this fact must be regarded as fundamental.

I can only fitly close a discourse which has throughout weighed the question of the living thought against the unthinking laws of matter, by a paraphrase of the words

of a great poet when he, in higher and, perhaps,
more philosophic
language, also sought to place the one in
comparison with the
other.[1]

Richter thought that he was—with his human heart
unstrengthened—taken by an angel among the
universe of stars.

Then, as they journeyed, our solar system was
sunken like a faint
star in the abyss, and they travelled yet further,
on the wings
of thought, through mightier systems: through all
the countless
numbers of our galaxy. But at length these also
were left behind,
and faded like a mist into the past. But this was
not all. The
dawn of other galaxies appeared in the void. Stars
more countless
still with insufferable light emerged. And these
also were
passed. And so they went through galaxies without
number till at
length they stood in the great Cathedral of the
Universe. Endless
were the starry aisles; endless the starry
columns; infinite the
arches and the architraves of stars. And the poet
saw the mighty
galaxies as steps descending to infinity, and as
steps going up
to infinity.

Then his human heart fainted and he longed for
some narrow cell;
longed to lie down in the grave that he might hide

from infinity.

And he said to the angel:

"Angel, I can go with thee no farther. Is there, then, no end to the universe of stars?"

[1] De Quincy in his System of the Heavens gives a fine paraphrase of "Richter's Dream."

200

Then the angel flung up his glorious hands to the heaven of heavens, saying "End is there none to the universe of God? Lo! also there is no beginning."

201

THE LATENT IMAGE [1]

My inclination has led me, in spite of a lively dread of incurring a charge of presumption, to address you principally on that profound and most subtle question, the nature and mode of formation of the photographic image. I am impelled to do so, not only because the subject is full of fascination and hopefulness, but because the wide topics of photographic methods or photographic applications would be quite unfittingly handled by the president you have chosen.

I would first direct your attention to Sir James Dewar's remarkable result that the photographic plate retains considerable power of forming the latent image at temperatures approaching the absolute zero—a result which, as I submit, compels us to regard the fundamental effects progressing in the film under the stimulus of light undulations as other than those of a purely chemical nature. But few, if any, instances of chemical combination or decomposition are known at so low a temperature. Purely chemical actions cease, indeed, at far higher temperatures, fluorine being among the few bodies which still show

[1] Presidential address to the Photographic Convention of the United Kingdom, July, 1905. *Nature*, Vol. 72, p. 308.

202

chemical activity at the comparatively elevated temperature of -180° C. In short, this result of Sir James Dewar's suggests that we must seek for the foundations of photographic action in some physical or intra-atomic effect which, as in the case of radioactivity or fluorescence, is not restricted

to intervals of temperature over which active molecular vis viva prevails. It compels us to regard with doubt the role of oxidation or other chemical action as essential, but rather points to the view that such effects must be secondary or subsidiary. We feel, in a word, that we must turn for guidance to some purely photo-physical effect.

Here, in the first place, we naturally recall the views of Bose.

This physicist would refer the formation of the image to a strain of the bromide of silver molecule under the electric force in the light wave, converting it into what might be regarded as an allotropic modification of the normal bromide which subsequently responds specially to the attack of the developer. The function of the sensitiser, according to this view, is to retard the recovery from strain. Bose obtained many suggestive parallels between the strain phenomena he was able to observe in silver and other substances under electromagnetic radiation and the behaviour of the photographic plate when subjected to long-continued exposure to light.

This theory, whatever it may have to recommend it, can hardly be regarded as offering a fundamental explanation. In the first place, we are left in the dark as to what

203

the strain may be. It may mean many and various things. We know nothing as to the inner mechanism of its effects upon subsequent chemical actions—or at least we cannot correlate it with what is known of the physics of chemical activity. Finally, as will be seen later, it is hardly adequate to account for the varying degrees of stability which may apparently characterise the latent image. Still, there is much in Bose's work deserving of careful consideration. He has by no means exhausted the line of investigation he has originated.

Another theory has doubtless been in the minds of many. I have said we must seek guidance in some photo-physical phenomenon. There is one such which preeminently connects light and chemical phenomena through the intermediary of the effects of the former upon a component part of the atom. I refer to the phenomena of photo-electricity.

It was ascertained by Hertz and his immediate successors that light has a remarkable power of discharging negative electrification from the surface of bodies—especially from certain substances. For long no explanation of the cause of this appeared. But the electron—the ubiquitous electron—is now known with considerable certainty to be responsible. The effect of the electric force in the light wave is to direct or assist the electrons contained in the substance to escape from the surface of the body. Each electron carries away a very small charge of negative electrification. If, then, a body is

204

originally charged negatively, it will be gradually discharged by this convective process. If it is not charged to start with, the electrons will still be liberated at the surface of the body, and this will acquire a positive charge. If the body is positively charged at first, we cannot discharge it by illumination.

It would be superfluous for me to speak here of the nature of electrons or of the various modes in which their presence may be detected. Suffice it to say, in further connection

with the Hertz effect, that when projected among gaseous molecules the electron soon attaches itself to one of these. In other words, it ionises a molecule of the gas or confers its electric charge upon it. The gaseous molecule may even be itself disrupted by impact of the electron, if this is moving fast enough, and left bereft of an electron.

We must note that such ionisation may be regarded as conferring potential chemical properties upon the molecules of the gas and upon the substance whence the electrons are derived. Similar ionisation under electric forces enters, as we now believe, into all the chemical effects progressing in the galvanic cell, and, indeed, generally in ionised solutes.

An experiment will best illustrate the principles I wish to remind you of. A clean aluminium plate, carefully insulated by a sulphur support, is faced by a sheet of copper-wire-gauze placed a couple of centimetres away from it. The gauze is maintained at a high positive

potential by this dry pile. A sensitive gold-leaf electroscope is attached to the aluminium plate, and its image thrown upon the screen. I now turn the light from this arc lamp upon the wire gauze, through which it in part passes and shines upon the aluminium plate. The electroscope at once charges up rapidly.

There is a liberation of negative electrons at the surface of the aluminium; these, under the attraction of the positive body, are rapidly removed as ions, and the electroscope charges up positively.

Again, if I simply electrify negatively this aluminium plate so that the leaves of the attached electroscope diverge widely, and now expose it to the rays from the arc lamp, the charge, as you see, is very rapidly dissipated. With positive electrification of the aluminium there is no effect attendant on the illumination.

Thus from the work of Hertz and his successors we know that light, and more particularly what we call actinic light, is an effective means of setting free electrons from certain substances. In short, our photographic agent, light, has the power of expelling from certain substances the

electron which is
so potent a factor in most, if not in all,
chemical effects. I
have not time here to refer to the work of Elster
and Geitel
whereby they have shown that this action is to be
traced to the
electric force in the light wave, but must turn to
the probable
bearing of this phenomenon on the familiar facts
of photography.
I assume that the experiment I have shown you is
the most

206

fundamental photographic experiment which it is
now in our power
to make.

We must first ask from what substances can light
liberate
electrons. There are many—metals as well as non-
metals and
liquids. It is a very general phenomenon and must
operate widely
throughout nature. But what chiefly concerns the
present
consideration is the fact that the haloid salts of
silver are
vigorously photo-electric, and, it is suggestive,
possess,
according to Schmidt, an activity in the
descending order
bromide, chloride, iodide. This is, in other
words, their order
of activity as ionisers (under the proper
conditions) when

exposed to ultra-violet light. Photographers will recognise that this is also the order of their photographic sensitiveness.

Another class of bodies also concerns our subject: the special sensitisers used by the photographer to modify the spectral distribution of sensibility of the haloid salts, _e.g._ eosine, fuchsine, cyanine. These again are electron-producers under light stimulus. Now it has been shown by Stoletow, Hallwachs, and Elster and Geitel that there is an intimate connection between photo-electric activity and the absorption of light by the substance, and, indeed, that the particular wavelengths absorbed by the substance are those which are effective in liberating the electrons. Thus we have strong reason for believing that the vigorous photo-electric activity displayed by the special sensitisers must be dependent upon their colour absorption. You will recognise that this is just

207

the connection between their photographic effects and their behaviour towards light.

There is yet another suggestive parallel. I referred to the observation of Sir James Dewar as to the continued sensitiveness of the photographic film at the lowest attained extreme of temperature, and drew the inference that the fundamental photographic action must be of intra-atomic nature, and not dependent upon the vis viva of the molecule or atom. In then seeking the origin of photographic action in photo-electric phenomena we naturally ask, Are these latter phenomena also traceable at low temperatures? If they are, we are entitled to look upon this fact as a qualifying characteristic or as another link in the chain of evidence connecting photographic with photo-electric activity.

I have quite recently, with the aid of liquid air supplied to me from the laboratory of the Royal Dublin Society, tested the photo-sensibility of aluminium and also of silver bromide down to temperatures approaching that of the liquid air. The mode of observation is essentially that of Schmidt—what he terms his static method. The substance undergoing observation is, however, contained at the bottom of a thin copper tube, 5

cm. in diameter,
which is immersed to a depth of about 10 cm in
liquid air. The
tube is closed above by a paraffin stopper which
carries a thin
quartz window as well as the sulphur tubes through
which the
connections pass. The air within is very carefully
dried by
phosphorus

208

pentoxide before the experiment. The arc light is
used as source
of illumination. It is found that a vigorous
photo-electric
effect continues in the case of the clean
aluminium. In the case
of the silver bromide a distinct photo-electric
effect is still
observed. I have not had leisure to make, as yet,
any trustworthy
estimate of the percentage effect at this
temperature in the case
of either substance. Nor have I determined the
temperature
accurately. The latter may be taken as roughly
about -150° C,

Sir James Dewar's actual measurements afforded
twenty per cent.
of the normal photographic effect at -180° C. and
ten per cent.
at the temperature of -252.5° C.

With this much to go upon, and the important
additional fact that

the electronic discharge—as from the X-ray tube or from radium—generates the latent image, I think we are fully entitled to suggest, as a legitimate lead to experiment, the hypothesis that the beginnings of photographic action involve an electronic discharge from the light-sensitive molecule; in other words that the latent image is built up of ionised atoms or molecules the result of the photo-electric effect on the illuminated silver haloid, and it is upon these ionised atoms that the chemical effects of the developer are subsequently directed. It may be that the liberated electrons ionise molecules not directly affected, or it may be that in their liberation they disrupt complex molecules built up in the ripening of the

209

emulsion. With the amount we have to go upon we cannot venture to particularise. It will be said that such an action must be in part of the nature of a chemical effect. This must be admitted, and, in so far as the rearrangement of molecular fabrics is involved, the result will doubtless be controlled by temperature conditions. The facts observed by Sir James Dewar support this.

But there is involved a fundamental process—the liberation of the electron by the electric force in the light wave, which is a physical effect, and which, upon the hypothesis of its reality as a factor in forming the latent image, appears to explain completely the outstanding photographic sensitiveness of the film at temperatures far below those at which chemical actions in general cease.

Again, we may assume that the electron-producing power of the special sensitiser or dye for the particular ray it absorbs is responsible, or responsible in part, for the special sensitiveness it confers upon the film. Sir Wm. Abney has shown that these sensitisers are active even if laid on as a varnish on the sensitive surface and removed before development. It must be remembered, however, that at temperatures of about -50° these sensitisers lose much of their influence on the film; as I have pointed out in a paper read before the Photographic Convention of 1894.

It. appears to me that on these views the curious phenomenon of recurrent reversals does not present a problem

hopeless of
explanation. The process of photo-

210

ionisation constituting the latent image, where the ion is probably not immediately neutralised by chemical combination, presents features akin to the charging of a capacity—say a Leyden jar. There may be a rising potential between the groups of ions until ultimately a point is attained when there is a spontaneous neutralisation. I may observe that the phenomena of reversal appear to indicate that the change in the silver bromide molecule, whatever be its nature, is one of gradually increasing intensity, and finally attains a maximum when a return to the original condition occurs. The maximum is the point of most intense developable image. It is probable that the sensitiser—in this case the gelatin in which the bromide of silver is immersed—plays a part in the conditions of stability which are involved.

Of great interest in all our considerations and theories is the recent work of Wood on photographic reversal. The result of this work is—as I take it—to show that the stability of

the latent image may be very various according to the mode of its formation. Thus it appears that the sort of latent effect which is produced by pressure or friction is the least stable of any. This may be reversed or wiped out by the application of any other known form of photographic stimulus. Thus an exposure to X-rays will obliterate it, or a very brief exposure to light. The latent image arising from X-rays is next in order of increasing stability. Light action will remove this. Third in order is a very brief light-shock or sudden flash. This

211

cannot be reversed by any of the foregoing modes of stimulation, but a long-continued undulatory stimulus, as from lamp-light, will reverse it. Last and most stable of all is the gradually built-up configuration due to long-continued light exposure. This can only be reversed by overdoing it according to the known facts of recurrent reversal. Wood takes occasion to remark that these phenomena are in bad agreement with the strain theory of Bose. We have, in fact, but the one resource—the allotropic modification of the haloid—whereby to explain all these orders

of stability.

It appears to me that the elasticity of the electronic theory is greater. The state of the ionised system may be very various according as it arises from continued rhythmic effects or from unorganised shocks. The ionisation due to X-rays or to friction will probably be quite unorganised, that due to light more or less stable according to the gradual and gentle nature of the forces at work. I think we are entitled to conclude that on the whole there is nothing in Wood's beautiful experiments opposed to the photo-electric origin of photographic effects, but that they rather fall in with what might be anticipated according to that theory.

When we look for further support to the views I have laid before you we are confronted with many difficulties. I have not as yet detected any electronic discharge from the film under light stimulus. This may be due to my defective experiments, or to a fact noted by Elster and Geitel concerning the photo-electric properties of gelatin.

They obtained a vigorous effect from Balmain's luminous paint, but when this was mixed in gelatin there was no external effect.

Schmidt's results as to the continuance of photo-electric

activity when bodies in general are dissolved in each other lead

us to believe that an actual conservative property of the medium

and not an effect of this on the luminous paint is here involved.

This conservative effect of the gelatin may be concerned with its efficacy as a sensitiser.

In the views I have laid before you I have endeavoured to show

that the recent addition to our knowledge of the electron as an

entity taking part in many physical and chemical effects should

be kept in sight in seeking an explanation of the mode of origin

of the latent image.[1]

[1] For a more detailed account of the subject, and some

ingenious extensions of the views expressed above, see

Photo-Electricity, by H. Stanley Allen: Longmans, Green & Co., 1913.

IT is now well established that a helium atom is expelled from certain of the radioactive elements at the moment of transformation. The helium atom or alpha ray leaves the transforming atom with a velocity which varies in the different radioactive elements, but which is always very great, attaining as much as 2×10^9 cms. per second; a velocity which, if unchecked, would carry the atom round the earth in less than two seconds. The alpha ray carries a positive charge of double the ionic amount.

When an alpha ray is discharged from the transforming element into a gaseous medium its velocity is rapidly checked and its energy absorbed. A certain amount of energy is thus transferred from the transforming atom to the gas. We recognise this energy in the gas by the altered properties of the latter; chiefly by the fact that it becomes a conductor of electricity. The mechanism by which this change is effected is in part known. The atoms of the gas, which appear to be freely penetrated by the alpha ray, are so far dismembered as to yield charged electrons or ions; the atoms remaining charged with an equal

and opposite
charge. Such a medium of

[1] Being the Huxley Lecture, delivered at the
University of
Birmingham on October 30th, 1912. Bedrock, Jan.,
1913.

214

free electric charges becomes a conductor of
electricity by
convection when an electromotive force is applied.
The gas also
acquires other properties in virtue of its
ionisation. Under
certain conditions it may acquire chemical
activity and new
combinations may be formed or existing ones broken
up. When its
initial velocity is expended the helium atom gives
up its
properties as an alpha ray and thenceforth remains
possessed of
the ordinary varying velocity of thermal
agitation. Bragg and
Kleeman and others have investigated the career of
the alpha ray
when its path or range lies in a gas at ordinary
or obtainable
conditions of pressure and temperature. We will
review some of
the facts ascertained.

The range or distance traversed in a gas at
ordinary pressures is
a few centimetres. The following table, compiled

by Geiger, gives
the range in air at the temperature of 15° C.:

	cms.		cms.
Uranium 1 -	2.50	Thorium -	2.72
Radioactinium	4.60		
Uranium 2 -	2.90	Radiothorium	3.87
Actinium X -	4.40		
Ionium -	3.00	Thorium X -	4.30
Emanation	5.70		
Radium -	3.30	Th Emanation	5.00
Actinium A -	6.50		
Ra Emanation	4.16	Thorium A -	5.70
Actinium C -	5.40		
Radium A -	4.75	Thorium C1 -	4.80
Radium C -	6.94	Thorium C2 -	8.60
Radium F -	3.77		

It will be seen that the ray of greatest range is that proceeding from thorium C2, which reaches a distance of 8.6 cms. In the uranium family the fastest ray is

215

that of radium C. It attains 6.94 cms. There is thus an appreciable difference between the ultimate distances traversed by the most energetic rays of the two families. The shortest ranges are those of uranium 1 and 2.

The ionisation effected by these rays is by no means uniform along the path of the ray. By examining the conductivity of the

gas at different points along the path of the ray, the ionisation at these points may be determined. At the limits of the range the ionisation

{Fig. 13}

ceases. In this manner the range is, in fact, determined. The dotted curve (Fig. 13) depicts the recent investigation of the ionisation effected by a sheaf of parallel rays of radium C in air, as determined by Geiger. The range is laid out horizontally in centimetres. The numbers of ions are laid out vertically. The remarkable nature of the results will be at once apparent. We should have expected that the ray at the beginning of its path, when its velocity and kinetic energy were greatest, would have been more effective than towards the end of its range

216

when its energy had almost run out. But the curve shows that it is just the other way. The lagging ray, about to resign its ionising properties, becomes a much more efficient ioniser than it was at first. The maximum efficiency is, however, in the case of a bundle of parallel rays, not quite at the end

of the range,
but about half a centimetre from it. The increase
to the maximum
is rapid, the fall from the maximum to nothing is
much more
rapid.

It can be shown that the ionisation effected
anywhere along the
path of the ray is inversely proportional to the
velocity of the
ray at that point. But this evidently does not
apply to the last
5 or 10 mms. of the range where the rate of
ionisation and of the
speed of the ray change most rapidly. To what are
the changing
properties of the rays near the end of their path
to be ascribed?
It is only recently that this matter has been
elucidated.

When the alpha ray has sufficiently slowed down,
its power of
passing right through atoms, without appreciably
experiencing any
effects from them, diminishes. The opposing atoms
begin to exert
an influence on the path of the ray, deflecting it
a little. The
heavier atoms will deflect it most. This effect
has been very
successfully investigated by Geiger. It is known
as "scattering."
The angle of scattering increases rapidly with the
decrease of
velocity. Now the effect of the scattering will be

to cause some
of the rays to complete their ranges

217

or, more accurately, to leave their direct line of
advance a
little sooner than others. In the beautiful
experiments of C. T.
R. Wilson we are enabled to obtain ocular
demonstration of the
scattering. The photograph (Fig. 14.), which I owe
to the
kindness of Mr. Wilson, shows the deflection of
the ray towards
the end of its path. In

{Fig. 14}

this case the path of the ray has been rendered
visible by the
condensation of water particles under the
influence of the
ionisation; the atmosphere in which the ray
travels being in a
state of supersaturation with water vapour at the
instant of the
passage of the ray. It is evident that if we were
observing the
ionisation along a sheaf of parallel rays, all
starting with
equal velocity,

218

the effect of the bending of some of the rays near
the end of
their range must be to cause a decrease in the
aggregate

ionisation near the very end of the ultimate range. For, in fact, some of the rays complete their work of ionising at points in the gas before the end is reached. This is the cause, or at least an important contributory cause, of the decline in the ionisation near the end of the range, when the effects of a bundle of rays are being observed. The explanation does not suggest that the ionising power of any one ray is actually diminished before it finally ceases to be an alpha ray.

The full line in Fig. 13 gives the ionisation curve which it may be expected would be struck out by a single alpha ray. In it the ionisation goes on increasing till it abruptly ceases altogether, with the entire loss of the initial kinetic energy of the particle.

A highly remarkable fact was found out by Bragg. The effect of the atom traversed by the ray in checking the velocity of the ray is independent of the physical and chemical condition of the atom. He measured the "stopping power" of a medium by the distance the ray can penetrate into it compared with the distance to which it can penetrate in air. The less the ratio the greater

is the stopping power. The stopping power of a substance is proportional to the square root of its atomic weight. The stopping power of an atom is not altered if it is in chemical union with another atom. The atomic weight is the one quality of importance. The physical

219

state, whether the element is in the solid, liquid or gaseous state, is unimportant. And when we deal with molecules the stopping power is simply proportional to the sum of the square roots of the atomic weights of the atoms entering into the molecule. This is the "additive law," and it obviously enables us to calculate what the range in any substance of known chemical composition and density will be, compared with the range in air.

This is of special importance in connection with phenomena we have presently to consider. It means that, knowing the chemical composition and density of any medium whatsoever, solid, liquid or gaseous, we can calculate accurately the distance to which any particular alpha ray will penetrate. Nor have the temperature and pressure to which the medium is subjected any

influence save in
so far as they may affect the proximity of one
atom to another.
The retardation of the alpha ray in the atom is
not affected.

This valuable additive law, however, cannot be
applied in
strictness to the amount of ionisation attending
the ray. The
form of the molecule, or more generally its
volume, may have an
influence upon this. Bragg draws the conclusion,
from this fact
as well as from the notable increase of ionisation
with loss of
speed, that the ionisation is dependent upon the
time the ray
spends in the molecule. The energy of the ray is,
indeed, found
to be less efficient in producing ionisation in
the smaller
atomm.

220

Before leaving our review of the general laws
governing the
passage of alpha rays through matter, another
point of interest
must be referred to. We have hitherto spoken in
general terms of
the fact that ionisation attends the passage of
the ray. We have
said nothing as to the nature of the ionisation so
produced. But
in point of fact the ionisation due to an alpha
ray is sui

generis. A glance at one of Wilson's photographs (Fig. 14.)

illustrates this. The white streak of water particles marks the path of the ray. The ions produced are evidently closely crowded

along the track of the ray. They have been called into existence

in a very minute instant of time. Now we know that ions of

opposite sign if left to themselves recombine. The rate of

recombination depends upon the product of the number of each sign

present in unit volume. Here the numbers are very great and the

volume very small. The ionic density is therefore high, and

recombination very rapidly removes the ions after they are

formed. We see here a peculiarity of the ionisation effected by

alpha rays. It is linear in distribution and very local. Much of

the ionisation in gases is again undone by recombination before

diffusion leads to the separation of the ions. This "initial

recombination" is greatest towards the end of the path of the ray

where the ionisation is a maximum. Here it may be so effective

that the form of the curve is completely lost unless a very large

electromotive force is used to separate the ions when the

ionisation is being investigated.

We have now reviewed recent work at sufficient length to understand something of the nature of the most important advance ever made in our knowledge of the atom. Let us glance briefly at what we have learned. The radioactive atom in sinking to a lower atomic weight casts out with enormous velocity an atom of helium. It thus loses a definite portion of its mass and of its energy. Helium which is chemically one of the most inert of the elements, is, when possessed of such great kinetic energy, able to penetrate and ionise the atoms which it meets in its path. It spends its energy in the act of ionising them, coming to rest, when it moves in air, in a few centimetres. Its initial velocity depends upon the particular radioactive element which has given rise to it. The length of its path is therefore different according to the radioactive element from which it proceeds. The retardation which it experiences in its path depends entirely upon the atomic weight of the atoms which it traverses. As it advances in its path its effectiveness in ionising the atom rapidly increases and attains a very marked

maximum. In a gas the ions produced being much crowded together recombine rapidly; so rapidly that the actual ionisation may be quite concealed unless a sufficiently strong electric force is applied to separate them. Such is a brief summary of the climax of radioactive discovery:—the birth, life and death of the alpha ray. Its advent into Science has altered fundamentally our conception of

222

matter. It is fraught with momentous bearings upon Geological Science. How the work of the alpha ray is sometimes recorded visibly in the rocks and what we may learn from that record, I propose now to bring before you.

In certain minerals, notably the brown variety of mica known as biotite, the microscope reveals minute circular marks occurring here and there, quite irregularly. The most usual appearance is that of a circular area darker in colour than the surrounding mineral. The radii of these little disc-shaped marks when well defined are found to be remarkably uniform, in some cases four hundredths of a millimetre and in others three hundredths, about.

These are the measurements in biotite. In other minerals the measurements are not quite the same as in biotite. Such minute objects are quite invisible to the naked eye. In some rocks they are very abundant, indeed they may be crowded together in such numbers as to darken the colour of the mineral containing them. They have long been a mystery to petrologists.

Close examination shows that there is always a small speck of a foreign body at the centre of the circle, and it is often possible to identify the nature of this central substance, small though it be. Most generally it is found to be the mineral zircon. Now this mineral was shown by Strutt to contain radium in quantities much exceeding those found in ordinary rock substances.

223

Some other mineral may occasionally form the nucleus, but we never find any which is not known to be specially likely to contain a radioactive substance. Another circumstance we notice. The smaller this central nucleus the more perfect in form is the darkened circular area surrounding it. When the circle is very

perfect and the central mineral clearly defined at its centre we find by measurement that the radius of the darkened area is generally 0.033 mm. It may sometimes be 0.040 mm. These are always the measurements in biotite. In other minerals the radii are a little different.

We see in the photograph (Pl. XXIII, lower figure), much magnified, a halo contained in biotite. We are looking at a region in a rock-section, the rock being ground down to such a thickness that light freely passes through it. The biotite is in the centre of the field. Quartz and felspar surround it. The rock is a granite. The biotite is not all one crystal. Two crystals, mutually inclined, are cut across. The halo extends across both crystals, but owing to the fact that polarised light is used in taking the photograph it appears darker in one crystal than in the other. We see the zircon which composes the nucleus. The fine striated appearance of the biotite is due to the cleavage of that mineral, which is cut across in the section.

The question arises whether the darkened area surrounding the zircon may not be due to the influence of the

radioactive
substances contained in the zircon. The

224

extraordinary uniformity of the radial measurements of perfectly formed haloes (to use the name by which they have long been known) suggests that they may be the result of alpha radiation.

For in that case, as we have seen, we can at once account for the definite radius as simply representing the range of the ray in biotite. The furthest-reaching ray will define the radius of the

halo. In the case of the uranium family this will be radium C,

and in the case of thorium it will be thorium C. Now here we

possess a means of at once confirming or rejecting the view that

the halo is a radioactive phenomenon and occasioned by alpha

radiation; for we can calculate what the range of these rays will

be in biotite, availing ourselves of Bragg's additive law,

already referred to. When we make this calculation we find that

radium C just penetrates 0.033 mm. and thorium C 0.040 mm. The

proof is complete that we are dealing with the effects of alpha

rays. Observe now that not only is the coincidence of measurement

and calculation a proof of the view that alpha

radiation has occasioned the halo, but it is a very complete verification of the important fact stated by Bragg, that the stopping power depends solely on the atomic weight of the atoms traversed by the ray.

We have seen that our examination of the rocks reveals only the two sorts of halo: the radium halo and the thorium halo. This is not without teaching. For why not find an actinium halo? Now Rutherford long ago suggested that this element and its derivatives were

225

probably an offspring of the uranium family; a side branch, as it were, in the formation of which relatively few transforming atoms took part. On Rutherford's theory then, actinium should always accompany uranium and radium, but in very subordinate amount. The absence of actinium haloes clearly supports this view. For if actinium was an independent element we would be sure to find actinium haloes. The difference in radius should be noticeable. If, on the other hand, actinium

was always associated with uranium and radium, then its effects would be submerged in those of the much more potent effects of the uranium series of elements.

It will have occurred to you already that if the radioactive origin of the halo is assured the shape of a halo is not really circular, but spherical. This is so. There is no such thing as a disc-shaped halo. The halo is a spherical volume containing the radioactive nucleus at its centre. The true radius of the halo may, therefore, only be measured on sections passing through the nucleus.

226

In order to understand the mode of formation of a halo we may profitably study on a diagram the events which go on within the halo-sphere. Such a diagram is seen in Fig. 15. It shows to relatively correct scale the limiting range of all the alpha-ray producing members of the uranium and thorium families. We know that each member of a family will exist in equilibrium amount within the nucleus possessing the parent element. Each alpha ray leaving the nucleus will just attain its range and then cease to

affect the mica. Within the halosphere, there must be, therefore, the accumulated effects of the influences of all the rays. Each has its own sphere of influence, and the spheres are all concentric.

The radii in biotite of the several spheres are given in the following table

URANIUM FAMILY.

Radium C -	0.0330 mm.
Radium A -	0.0224 mm.
Ra Emanation -	0.0196 mm.
Radium F -	0.0177 mm.
Radium -	0.0156 mm.
Ionium -	0.0141 mm.
Uranium 1 -	0.0137 mm.
Uranium 2 -	0.0118 mm.

THORIUM FAMILY.

Thorium CE -	0.040 mm.
Thorium A -	0.026 mm.
Th Emanation -	0.023 mm.
Thorium Ci -	0.022 mm.
Thorium X -	0.020 mm.
Radiothorium -	0.119 mm.
Thorium -	0.013 mm.

In the photograph (Pl. XXIV, lower figure), we see a uranium and a thorium halo in the same crystal of mica. The mica is contained in a rock-section and is cut across the cleavage. The effects of thorium Ca are clearly shown

as a lighter border surrounding the accumulated inner darkening due to the other thorium rays. The uranium halo (to the right) similarly shows the effects of radium C, but less distinctly.

Haloes which are uniformly dark all over as described above are, in point of fact, "over-exposed"; to borrow a familiar photographic term. Haloes are found which show much beautiful internal detail. Too vigorous action obscures this detail just as detail is lost in an over-exposed photograph. We may again have "under-exposed" haloes in which the action of the several rays is incomplete or in which the action of certain of the rays has left little if any trace. Beginning at the most under-exposed haloes we find circular dark marks having the radius 0.012 or 0.013 mm. These haloes are due to uranium, although their inner darkening is doubtless aided by the passage of rays which were too few to extend the darkening beyond the vigorous effects of the two uranium rays. Then we find haloes carried out to the radii 0.016, 0.018 and 0.019 mm. The last sometimes show very beautiful outer rings having radial dimensions such as would be

produced by
radium A and radium C. Finally we may have haloes
in which
interior detail is lost so far out as the radius
due to emanation
or radium A, while outside this floats the ring
due to radium C.
Certain variations of these effects may occur,
marking,
apparently, different stages of exposure. Plates
XXIII and XXIV
(upper figure) illustrate some of these stages;

228

the latter photograph being greatly enlarged to
show clearly the
halo-sphere of radium A.

In most of the cases mentioned above the structure
evidently
shows the existence of concentric spherical shells
of darkened
biotite. This is a very interesting fact. For it
proves that in
the mineral the alpha ray gives rise to the same
increased
ionisation towards the end of its range, as Bragg
determined in
the case of gases. And we must conclude that the
halo in every
case grows in this manner. A spherical shell of
darkened biotite
is first produced and the inner colouration is
only effected as
the more feeble ionisation along the track of the
ray in course
of ages gives rise to sufficient alteration of the

mineral. This more feeble ionisation is, near the nucleus, enhanced in its effects by the fact that there all the rays combine to increase the ionisation and, moreover, the several tracks are there crowded by the convergency to the centre. Hence the most elementary haloes seldom show definite rings due to uranium, etc., but appear as embryonic disc-like markings. The photographs illustrate many of the phases of halo development.

Rutherford succeeded in making a halo artificially by compressing into a capillary glass tube a quantity of the emanation of radium. As the emanation decayed the various derived products came into existence and all the several alpha rays penetrated the glass, darkening the walls of the capillary out to the limit of the range of radium C in glass. Plate XXV shows a magnified section of the

229

tube. The dark central part is the capillary. The tubular halo surrounds it. This experiment has, however, been anticipated by some scores of millions of years, for here is the same effect in a biotite crystal (Pl. XXV). Along what are

apparently tubular passages or cracks in the mica, a solution, rich in radioactive substances, has moved; probably during the final consolidation of the granite in which the mica occurs. A continuous and very regular halo has developed along these conduits. A string of halo-spheres may lie along such passages. We must infer that solutions or gases able to establish the radioactive nuclei moved along these conduits, and we are entitled to ask if all the haloes in this biotite are not, in this sense, of secondary origin. There is, I may add, much to support such a conclusion.

The widespread distribution of radioactive substances is most readily appreciated by examination of sections of rocks cut thin enough for microscopic investigation. It is, indeed, difficult to find, in the older rocks of granitic type, mica which does not show haloes, or traces of haloes. Often we find that every one of the inclusions in the mica—that is, every one of the earlier formed substances—contain radioactive elements, as indicated by the presence of darkened borders. As will be seen presently the quantities involved are generally vanishingly

small. For example
it was found by direct determination that in one
gram of the
halo-rich mica of Co. Carlow there was rather less
than twelve
billionths of a gram of radium, We are

230

entitled to infer that other rare elements are
similarly widely
distributed but remain undetectable because of
their more stable
properties.

It must not be thought that the under-exposed halo
is a recent
creation. By no means. All are old, appallingly
old; and in the
same rock all are, probably, of the same, or
nearly the same,
age. The under-exposure is simply due to a lesser
quantity of the
radioactive elements in the nucleus. They are
under-exposed, in
short, not because of lesser duration of exposure,
but because of
insufficient action; as when in taking a
photograph the stop is
not open enough for the time of the exposure.

The halo has, so far, told us that the additive
law is obeyed in
solid media, and that the increased ionisation
attending the
slowing down of the ray obtaining in gases, also
obtains in
solids; for, otherwise, the halo would not

commence its development as a spherical shell or envelope. But here we learn that there is probably a certain difference in the course of events attending the immediate passage of the ray in the gas and in the solid. In the former, initial recombination may obscure the intense ionisation near the end of the range. We can only detect the true end-effects by artificially separating the ions by a strong electric force. If this recombination happened in the mineral we should not have the concentric spheres so well defined as we see them to be. What, then, hinders the initial recombination in the solid? The answer probably is that the newly formed

231

ion is instantly used up in a fresh chemical combination. Nor is it free to change its place as in the gas. There is simply a new equilibrium brought about by its sudden production. In this manner the conditions in the complex molecule of biotite, tourmaline, etc., may be quite as effective in preventing initial recombination as the most effective electric force we could apply. The final result is that we find the Bragg

curve

reproduced most accurately in the delicate shading of the rings making up the perfectly exposed halo.

That the shading of the rings reproduces the form of the Bragg curve, projected, as it were, upon the line of advance of the ray and reproduced in depth of shading, shows that in yet another particular the alpha ray behaves much the same in the solid as in the gas. A careful examination of the outer edge of the circles always reveals a steep but not abrupt cessation of the action of the ray. Now Geiger has investigated and proved the existence of scattering of the alpha ray by solids. We may, therefore, suppose with much probability that there is the same scattering within the mineral near the end of the range. The heavy iron atom of the biotite is, doubtless, chiefly responsible for this in biotite haloes. I may observe that this shading of the outer bounding surface of the sphere of action is found however minute the central nucleus. In the case of a nucleus of considerable size another effect comes in which tends to produce an enhanced shading. This will

result if rays proceed from different depths in the nucleus. If the nucleus were of the same density and atomic weight as the surrounding mica, there would be little effect. But its density and molecular weight are generally greater, hence the retardation is greater, and rays proceeding from deep in the nucleus experience more retardation than those which proceed from points near to the surface. The distances reached by the rays in the mica will vary accordingly, and so there will be a gradual cessation of the effects of the rays.

The result of our study of the halo may be summed up in the statement that in nearly every particular we have the phenomena, which have been measured and observed in the gas, reproduced on a minute scale in the halo. Initial recombination seems, however, to be absent or diminished in effectiveness; probably because of the new stability instantly assumed by the ionised atoms.

One of the most interesting points about the halo remains to be referred to. The halo is always uniformly darkened all round its circumference and is perfectly spherical. Sections, whether taken in the plane of cleavage of the mica or across it,

show the same exactly circular form, and the same radius. Of course, if there was any appreciable increase of range along or across the cleavage the form of the halo on the section across the cleavage should be elliptical. The fact that there is no measurable ellipticity is, I think, one which on first consideration would not be expected.

233

For what are the conditions attending the passage of the ray in a medium such as mica? According to crystallographic conceptions we have here an orderly arrangement of molecules, the units composing the crystal being alike in mass, geometrically spaced, and polarised as regards the attractions they exert one upon another. Mica, more especially, has the cleavage phenomenon developed to a degree which transcends its development in any other known substance. We can cleave it and again cleave it till its flakes float in the air, and we may yet go on cleaving it by special means till the flakes no longer reflect visible light. And not less remarkable is the uniplanar nature of its cleavage. There is little cleavage in any plane but the one,

although it is
easy to show that the molecules in the plane of
the flake are in
orderly arrangement and are more easily parted in
some directions
than in others. In such a medium beyond all others
we must look
with surprise upon the perfect sphere struck out
by the alpha
rays, because it seems certain that the cleavage
is due to lesser
attraction, and, probably, further spacing of the
molecules, in a
direction perpendicular to the cleavage.

It may turn out that the spacing of the molecules
will influence
but little the average number per unit distance
encountered by
rays moving in divergent paths. If this is so, we
seem left to
conclude that, in spite of its unequal and
polarised attractions,
there is equal retardation and equal ionisation in
the molecule
in whatever

234

direction it is approached. Or, again, if the
encounters indeed
differ in number, then some compensating effect
must exist
whereby a direction of lesser linear density
involves greater
stopping power in the molecule encountered, and
vice versa.

The nature of the change produced by the alpha rays is unknown.

But the formation of the halo is not, at least in its earlier

stages, attended by destruction of the crystallographic and optical properties of the medium. The optical properties are

unaltered in nature but are increased in intensity. This applies

till the halo has become so darkened that light is no longer

transmitted under the conditions of thickness obtaining in rock

sections. It is well known that there is in biotite a maximum

absorption of a plane-polarised light ray, when the plane of

vibration coincides with the plane of cleavage. A section across

the cleavage then shows a maximum amount of absorption. A halo

seen on this section simply produces this effect in a more

intense degree. This is well shown in Plate XXIII (lower figure),

on a portion of the halo-sphere. The descriptive name "Pleochroic

Halo" has originated from this fact. We must conclude that the

effect of the ionisation due to the alpha ray has not been to

alter fundamentally the conditions which give rise to the optical

properties of the medium. The increased absorption is probably

associated with some change in the chemical state

of the iron
present. Haloes are, I believe, not found in
minerals from which
this

235

element is absent. One thing is quite certain. The
colouration is
not due to an accumulation of helium atoms, _i.e._
of spent alpha
rays. The evidence for this is conclusive. If
helium was
responsible we should have haloes produced in all
sorts of
colourless minerals. Now we sometimes see zircons
in feldspars and
in quartz, etc., but in no such case is a halo
produced. And
halo-spheres formed within and sufficiently close
to the edge of
a crystal of mica are abruptly truncated by
neighbouring areas of
feldspar or quartz, although we know that the rays
must pass
freely across the boundary. Again it is easy to
show that even in
the oldest haloes the quantity of helium involved
is so small
that one might say the halo-sphere was a tolerably
good vacuum as
regards helium. There is, finally, no reason to
suppose that the
imprisoned helium would exhibit such a
colouration, or, indeed,
any at all.

I have already referred to the great age of the halo. Haloes are not found in the younger igneous rocks. It is probable that a halo less than a million years old has never been seen. This, *primâ facie*, indicates an extremely slow rate of formation. And our calculations quite support the conclusions that the growth of a halo, if this has been uniform, proceeds at a rate of almost unimaginable slowness.

Let us calculate the number of alpha rays which may have gone to form a halo in the Devonian granite of Leinster.

236

It is common to find haloes developed perfectly in this granite, and having a nucleus of zircon less than 5×10^{-4} cms. in diameter. The volume of zircon is 65×10^{-12} c.cs. and the mass 3×10^{-10} grm.; and if there was in this zircon 10^{-8} grm. radium per gram (a quantity about five times the greatest amount measured by Strutt), the mass of radium involved is 3×10^{-18} grm. From this and from the fact ascertained by Rutherford that the number of alpha rays expelled by a gram of radium in one second is 3.4×10^{10} , we find that three rays are shot from the

nucleus in a year. If, now, geological time since the Devonian is 50 millions of years, then 150 millions of rays built up the halo. If geological time since the Devonian is 400 millions of years, then 1,200 millions of alpha rays are concerned in its genesis. The number of ions involved, of course, greatly exceeds these numbers. A single alpha ray fired from radium C will produce 2.37×10^5 ions in air.

But haloes may be found quite clearly defined and fairly dark out to the range of the emanation ray and derived from much less quantities of radioactive materials. Thus a zircon nucleus with a diameter of but 3.4×10^{-4} cms. formed a halo strongly darkened within, and showing radium A and radium C as clear smoky rings. Such a nucleus, on the assumption made above as to its radium content, expels one ray in a year. But, again, haloes are observed with less blackened pupils and with faint ring due to radium C, formed round nuclei

237

of rather less than 2×10^{-4} cms. diameter. Such nuclei would expel one ray in five years. And even lesser nuclei will generate

in these old rocks haloes with their earlier characteristic features clearly developed. In the case of the most minute nuclei, if my assumption as to the uranium content is correct, an alpha ray is expelled, probably, no oftener than once in a century; and possibly at still longer intervals.

The equilibrium amount of radium contained in some nuclei may amount to only a few atoms. Even in the case of the larger nuclei and more perfectly developed haloes the quantity of radium involved is many millions of times less than the least amount we can recognise by any other means. But the delicacy of the observation is not adequately set forth in this statement. We can not only tell the nature of the radioactive family with which we are dealing; but we can recognise the presence of some of its constituent members. I may say that it is not probable the zircons are richer in radium than I have assumed. My assumption involves about 3 per cent. of uranium. I know of no analyses ascribing so great an amount of uranium to zircon. The variety cyrtolite has been found to contain half this amount, about. But even if we doubled our estimate of radium content,

the remarkable
nature of our conclusions is hardly lessened.

It may appear strange that the ever-interesting
question of the
Earth's age should find elucidation from the

238

study of haloes. Nevertheless the subjects are
closely connected.

The circumstances are as follows. Geologists have
estimated the

age of the Earth since denudation began, by
measurements of the

integral effects of denudation. These methods
agree in showing an

age of about rob years. On the other hand,
measurements have been

made of the accumulation in minerals of
radioactive _débris_—the

helium and lead—and results obtained which,
although they do not

agree very well among themselves, are concordant
in assigning a

very much greater age to the rocks. If the
radioactive estimate

is correct, then we are now living in a time when
the denudative

forces of the Earth are about eight or nine times
as active as

they have been on the average over the past. Such
a state of

things is absolutely unaccountable. And all the
more

unaccountable because from all we know we would
expect a somewhat

lesser rate of solvent denudation as the world

gets older and the
land gets more and more loaded with the washed-out
materials of
the rocks.

Both the methods referred to of finding the age
assume the
principle of uniformity. The geologist contends
for uniformity
throughout the past physical history of the Earth.
The physicist
claims the like for the change-rates of the
radioactive elements.
Now the study of the rocks enables us to infer
something as to
the past history of our Globe. Nothing is, on the
other hand,
known respecting the origin of uranium or thorium—
the parent
radioactive bodies. And while not questioning the
law

239

and regularity which undoubtedly prevail in the
periods of the
members of the radioactive families, it appears to
me that it is
allowable to ask if the change rate of uranium has
been always
what we now believe it to be. This comes to much
the same thing
as supposing that atoms possessing a faster change
rate once were
associated with it which were capable of yielding
both helium and
lead to the rocks. Such atoms might have been
collateral in

origin with uranium from some antecedent element. Like helium, lead may be a derivative from more than one sequence of radioactive changes. In the present state of our knowledge the possibilities are many. The rate of change is known to be connected with the range of the alpha ray expelled by the transforming element; and the conformity of the halo with our existing knowledge of the ranges is reason for assuming that, whatever the origin of the more active associate of uranium, this passed through similar elemental changes in the progress of its disintegration. There may, however, have been differences in the ranges which the halo would not reveal. It is remarkable that uranium at the present time is apparently responsible for two alpha rays of very different ranges. If these proceed from different elements, one should be faster in its change rate than the other. Some guidance may yet be forthcoming from the study of the more obscure problems of radioactivity.

Now it is not improbable that the halo may contribute directly to this discussion. We can evidently attack

the biotite with a known number of alpha rays and determine how many are required to produce a certain intensity of darkening, corresponding to that of a halo with a nucleus of measurable dimensions. On certain assumptions, which are correct within defined limits, we can calculate, as I have done above, the number of rays concerned in forming the halo. In doing so we assume some value for the age of the halo. Let us take the maximum radioactive value. A halo originating in Devonian times may attain a certain central blackening from the effects of, say, rob rays. But now suppose we find that we cannot produce the same degree of blackening with this number of rays applied in the laboratory. What are we to conclude? I think there is only the one conclusion open to us; that some other source of alpha rays, or a faster rate of supply, existed in the past. And this conclusion would explain the absence of haloes from the younger rocks; which, in view of the vast range of effects possible in the development of haloes, is, otherwise, not easy to account for. It is apparent that the experiment on the biotite has a direct bearing on the validity of the radioactive

method of
estimating the age of the rocks. It is now being
carried out by
Professor Rutherford under reliable conditions.

Finally, there is one very certain and valuable
fact to be
learned from the halo. The halo has established
the extreme
rarity of radioactivity as an atomic phenomenon.
One and all of
the speculations as to

241

the slow breakdown of the commoner elements may be
dismissed. The
halo shows that the mica of the rocks is
radioactively sensitive.
The fundamental criterion of radioactive change is
the expulsion
of the alpha ray. The molecular system of the mica
and of many
other minerals is unstable in presence of these
rays, just as a
photographic plate is unstable in presence of
light. Moreover,
the mineral integrates the radioactive effects in
the same way as
a photographic salt integrates the effects of
light. In both
cases the feeblest activities become ultimately
apparent to our
inspection. We have seen that one ray in each year
since the
Devonian period will build the fully formed halo:
an object
unlike any other appearance in the rocks. And we

have been able
to allocate all the haloes so far investigated to
one or the
other of the known radioactive families. We are
evidently
justified in the belief that had other elements
been radioactive
we must either find characteristic haloes produced
by them, or
else find a complete darkening of the mica. The
feeblest alpha
rays emitted by the relatively enormous quantities
of the
prevailing elements, acting over the whole
duration of geological
time—and it must be remembered that the haloes we
have been
studying are comparatively young—must have
registered their
effects on the sensitive minerals. And thus we are
safe in
concluding that the common elements, and, indeed,
many which
would be called rare, are possessed of a degree of
stability
which has preserved them un

242

changed since the beginning of geological time.
Each unaffected
flake of mica is, thus, unassailable proof of a
fact which but
for the halo would, probably, have been for ever
beyond our
cognisance.

THE USE OF RADIUM IN MEDICINE [1]

IT has been unfortunate for the progress of the radioactive treatment of disease that its methods and claims involve much of the marvellous. Up till recently, indeed, a large part of radioactive therapeutics could only be described as bordering on the occult. It is not surprising that when, in addition to its occult and marvellous characters, claims were made on its behalf which in many cases could not be supported, many medical men came to regard it with a certain amount of suspicion.

Today, I believe, we are in a better position. I think it is possible to ascribe a rational scientific basis to its legitimate claims, and to show, in fact, that in radioactive treatment we are pursuing methods which have been already tried extensively and found to be of definite value; and that new methods differ from the old mainly in their power and availability, and little, or not at all, in kind.

Let us briefly review the basis of the science. Radium is a metallic element chemically resembling barium. It

[1] A Lecture to Postgraduate Students of Medicine in connection with the founding of the Dublin Radium Institute, delivered in

the School of Physic in Ireland, Trinity College,
on October 2nd,
1914

244

possesses, however, a remarkable property which barium does not.

Its atoms are not equally stable. In a given quantity of radium a certain very small percentage of the total number of atoms

present break up per second. By "breaking up" we mean their

transmutation to another element. Radium, which is a solid

element under ordinary conditions, gives rise by transmutation to

a gaseous element—the emanation of radium. The new element is a

heavy gas at ordinary temperatures and, like other gases, can be

liquified by extreme cold. The extraordinary property of

transmutation is entirely automatic. No influence which chemist

or physicist can apply can affect the rate of transformation.

The emanation inherits the property of instability, but in its

case the instability is more pronounced. A relatively large

fraction of its atoms transmute per second to a solid element

designated Radium A. In turn this new generation of atoms breaks

up—even faster than the emanation—becoming yet

another element
with specific chemical properties. And so on for a
whole sequence
of transmutations, till finally a stable substance
is formed,
identical with ordinary lead in chemical and
physical properties,
but possessing a slightly lower atomic weight.

The genealogy of the radium series of elements
shows that radium
is not the starting point. It possesses ancestors
which have been
traced back to the element uranium.

Now what bearing has this series of transmutations

245

upon medical science? Radium or emanation, &c.,
are not in the
Pharmacopoeia as are, say, arsenic or bismuth. The
whole
medicinal value of these elements resides in the
very wonderful
phenomena of their radiations. They radiate in the
process of
transmuting.

The changing atom may radiate a part of its own
mass. The
"alpha"-ray (a-ray) is such a material ray. It is
an electrified
helium atom cast out of the parent atom with
enormous
velocity—such a velocity as would carry it, if not
impeded, all
round the earth in two seconds. All alpha-rays are
positively

electrified atoms of the element helium, which thereby is shown to be an integral constituent of many elements. The alpha-ray is not of much value to medical science, for, in spite of its great velocity, it is soon stopped by encounter with other atoms. It can penetrate only a minute fraction of a millimetre into ordinary soft tissues. We shall not further consider it.

Transmuting atoms give out also material rays of another kind: the β -rays. The β -ray is in mass but a very small fraction of, even, a hydrogen atom. Its speed may approach that of light. As cast out by radioactive elements it starts with speeds which vary with the element, and may be from one-third to nine-tenths the velocity of light. The β -ray is negatively electrified. It has long been known to science as the electron. It is also identical with the cathode ray of the vacuum tube.

246

Another and quite different kind of radiation is given out by many of the transmuting elements:—the γ -ray. This is not material, it is ethereal. It is known now with certainty that the γ -ray is in kind identical with light, but of very

much shorter wave length than even the extreme ultraviolet light of the solar spectrum. The γ -ray is flashed from the transmuting atom along with the β -ray. It is identical in character with the x-ray but of even shorter wave length.

There is a very interesting connection between the γ -ray and the β -ray which it is important for the medical man to understand—as far as it is practicable on our present knowledge.

When γ -rays or x-rays fall on matter they give rise to β -rays.

The mechanism involved is not known but it is possibly a result of the resonance of the atom, or of parts of it, to the short

light waves. And it is remarkable that the γ -rays which, as we

have seen, are shorter and more penetrating waves than the

x-rays, give rise to β -rays possessed of greater velocity and

penetration than β -rays excited by the x-rays. Indeed the β -rays

originated by γ -rays may attain a velocity nearly approaching

that of light and as great as that of any β -rays emitted by

transmuting atoms. Again there is demonstrable evidence that

β -rays impinging on matter may give rise to γ -rays. The most

remarkable demonstration of this is seen in the x-

ray tube. Here
the x-rays originate where the stream of β - or
cathode-rays

247

are arrested on the anode. But the first relation
is at present
of most importance to us—i.e.— that the γ -or x-
rays give rise to
 β -rays.

This relation gives us additional evidence of the
identity of the
physical effects of γ -, x-, and light-rays —using
the term light
rays in the usual sense of spectral rays. For it
has long been
known that light waves liberate electrons from
atoms. It has been
found that these electrons possess a certain
initial velocity
which is the greater the shorter the wave length
of the light
concerned in their liberation. The whole science
of
"photo-electricity" centres round this phenomenon.
The action of
light on the photographic plate, as well as many
other physical
and chemical phenomena, find an explanation in
this liberation of
the electron by the light wave.

Here, then, we have spectral light waves
liberating
electrons—i.e.— very minute negatively-charged
particles, and we

find that, as we use shorter light waves, the initial velocity of these particles increases. Again, we have x-rays which are far smaller in wave length than spectral light, liberating much faster negatively electrified particles. Finally, we have y-rays—the shortest nether waves of all-liberating negative particles of the highest velocity known. Plainly the whole series of phenomena is continuous.

We can now look closer at the actions involved in the therapeutic influence of the several rays and in

248

this way, also, see further the correlation between what may be called photo-therapeutics and radioactive therapeutics.

The β -ray, whether we obtain it directly from the transforming radioactive atom or whether we obtain it as a result of the effects of the y- or x-rays upon the atom, is an ionising agent of wonderful power. What is meant by this? In its physical aspect this means that the atoms through which it passes acquire free electric charges; some becoming positive, some negative. This can only be due to the loss of an electron by the

affected atom. The loss of the small negative charge carried in the electron leaves the atom positively electrified or creates a positive ion. The fixing of the wandering electron to a neutral atom creates a negative ion. Before further consideration of the importance of the phenomenon of ionisation we must fix in our minds that the agent, which brings this about, is the β -ray. There is little evidence that the γ -ray can directly create ions to any large extent. But the action of liberating high-speed β -rays results in the creation of many thousands of ions by each β -ray liberated. As an agent in the hands of the medical man we must regard the γ -ray as a light wave of extremely penetrating character, which creates high-speed β -rays in the tissues which it penetrates, these β -rays being most potent ionising agents. The β -rays directly obtained from radioactive atoms assist in the work of ionisation. β -rays do not

249

penetrate far from their source. The fastest of them would not probably penetrate one centimetre in soft tissues.

We must now return to the phenomenon of ionisation. Ionisation is revealed to observation most conspicuously when it takes place in a gas. The + and - electric charges on the gas particles endow it with the properties of a conductor of electricity, the + ions moving freely in one direction and the - ions in the opposite direction under an electric potential. But there are effects brought about by ionisation of more importance to the medical man than this. The chemist has long come to recognise that in the ion he is concerned with the inner mechanism of a large number of chemical phenomena. For with the electrification of the atom attractive and repulsive forces arise. We can directly show the chemical effects of the ionising β -rays. Water exposed to their bombardment splits up into hydrogen and oxygen. And, again, the separated atoms may be in part recombined under the influence of the radiation. Ammonia splits up into hydrogen and nitrogen. Carbon dioxide forms carbon, carbon monoxide, and oxygen; hydrochloric acid forms chlorine and hydrogen. In these cases, also, recombination can be partially effected by the rays.

We can be quite sure that within the complex structure of the living cell the ionising effects which everywhere accompany the β -rays must exert a profound influence. The sequence of chemical events which as yet seem

250

beyond the ken of science and which are involved in metabolism cannot fail to be affected. Any, it is not surprising that as the result of experiment it is found that the radiations are agents which may be used either for the stimulation of the natural events of growth or used for the actual destruction of the cell. It is easy to see that the feeble radiation should produce the one effect, the strong the other. In a similar way by a moderate light stimulus we create the latent image in the photographic plate; by an intense light we again destroy this image. The inner mechanism in this last case can be logically stated.[1]

_There is plainly a true physical basis here for the efficacy of radioactive treatment and, what is more, we find when we examine it, that it is in kind not different from that underlying treatment by spectral radiations. But in degree it

is very different and here is the reason for the special importance of radioactivity as a therapeutic agent._ The Finsen light is capable of influencing the soft tissues to a short depth only. The reason is that the wave length of the light used is too great to pass without rapid absorption through the tissues; and, further, the electrons it gives rise to—i.e._ the β -rays it liberates—are too slow-moving to be very efficient ionisers. X-rays penetrate in some cases quite freely and give rise to much faster and more powerful β -rays

[1] See _The Latent Image_, p. 202.

251

than can the Finsen light. But far more penetrating than x-rays are the γ -rays emitted in certain of the radioactive changes. These give rise to β -rays having a velocity approximate to that of light.

The γ -rays are, therefore, very penetrating and powerfully ionising light waves; light waves which are quite invisible to the eye and can beam right through the tissues of the body. To the mind's eye only are they visible. And a very

wonderful
picture they make. We see the transmuting atom
flashing out this
light for an inconceivably short instant as it
throws off the
 β -ray. And "so far this little candle throws his
beams" in the
complex system of the cells, so far atoms shaken
by the rays send
out β -rays; these in turn are hurled against other
atomic
systems; fresh separations of electrons arise and
new attractions
and repulsions spring up and the most important
chemical changes
are brought about. Our mental picture can claim to
be no more
than diagrammatic of the reality. Still we are
here dealing with
recognised physical and chemical phenomena, and
their description
as "occult" in the derogatory sense is certainly
not
justifiable.

Having now briefly reviewed the nature of the rays
arising in
radioactive substances and the rationale of their
influence, we
must turn to more especially practical
considerations.

The Table given opposite shows that radium itself
is responsible
for α - and β -rays only. It happens that

Period in which $\frac{1}{2}$ element is transformed.

URANIUM 1 & 2 { α } $\times 10^9$ years.

URANIUM X { α } 24.6 days.

IONIUM { α } $\times 10^4$ years.

RADIUM { α } 2×10^2 years.

EMANATION { α } 8.85 days.

RADIUM A { α } minutes.

RADIUM B { β } 26.7 minutes.

RADIUM C { α } 13.5 minutes.

RADIUM D { β } 15 years.

RADIUM E { β } 4.8 days.

RADIUM (Polonium) F { α } 140 days.

Table showing the successive generations of the elements of the Uranium-radium family, the character of their radiations and their longevity.

253

the β -rays emitted by radium are very "soft"—i.e. slow and easily absorbed. The α -ray is in no case available for more than mere surface application. Hence we see that, contrary to what is generally believed, radium itself is of little direct therapeutic value. Nor is the next body in succession—the emanation, for it

gives only α -rays. In fact, to be brief, it is not till we come to Radium B that β -rays of a relatively high penetrative quality are reached; and it is not till we come to Radium C that highly penetrative γ -rays are obtained.

It is around this element, Radium C, that the chief medical importance of radioactive treatment by this family of radioactive bodies centres. Not only are β -rays of Radium C very penetrating, but the γ -rays are perhaps the most energetic rays of the kind known. Further in the list there is no very special medical interest.

Now, how can we get a supply of this valuable element Radium C? We can obtain it from radium itself. For even if radium has been deprived of its emanation (which is easily done by heating it or bringing it into solution) in a few weeks we get back the Radium C. One thing here we must be clear about. With a given quantity of Radium only a certain definitely limited amount of Radium C, or of emanation, or any other of the derived bodies, will be associated. Why is this? The answer is because the successive elements are themselves decaying –

i.e. changing one
into the other. The atomic per-

254

centage of each, which decays in a second, is a
fixed quantity

which we cannot alter. Now if we picture radium
which has been

completely deprived of its emanation, again
accumulating by

automatic transmutation a fresh store of this
element, we have to

remember:— (i) That the rate of creation of
emanation by the

radium is practically constant; and (2) that the
absolute amount

of the emanation decaying per second increases as
the stock of

emanation increases. Finally, when the amount of
accumulated

emanation has increased to such an extent that the
number of

emanation atoms transmuting per second becomes
exactly equal to

the number being generated per second, the amount
of emanation

present cannot increase. This is called the
equilibrium amount.

If fifteen members are elected steadily each year
into a

newly-founded society the number of members will
increase for the

first few years; finally, when the losses by death
of the members

equal about fifteen per annum the society can get
no bigger. It

has attained the equilibrium number of members.

This applies to every one of the successive elements. It takes twenty-one days for the equilibrium quantity of emanation to be formed in radium which has been completely de-emanated; and it takes 3.8 days for half the equilibrium amount to be formed.

Again, if we start with a stock of emanation it takes just three hours for the equilibrium amount of Radium C to be formed.

255

We can evidently grow Radium C either from radium itself or from the emanation of radium. If we use a tube of radium we have an almost perfectly constant quantity of Radium C present, for as fast as the Radium C and intervening elements decay the Radium, which only diminishes very slowly in amount, makes up the loss.

But, if we start off with a tube of emanation, we do not possess a constant supply of Radium C, because the emanation is decaying fairly rapidly and there is no radium to make good its loss. In

3.8 days about one half the emanation is transmuted and the Radium C decreases proportionately and, of course, with the Radium C the valuable radiations also decrease. In another 3.8

days—that is in about a week from the start—the

radioactive value
of the tube has fallen to one-fourth of its
original value.

But in spite of the inconstant character of the
emanation tube
there are many reasons for preferring its use to
the use of the
radium tube. Chief of these is the fact that we
can keep the
precious radium safely locked up in the laboratory
and not
exposed to the thousand-and-one risks of the
hospital. Then,
secondly, the emanation, being a gas, is very
convenient for
subdivision into a large number of very small
tubes according to
the dosage required.

In fact the volume of the emanation is exceedingly
minute. The
amount of emanation in equilibrium with one gramme
of radium is
called the curie, and with one

256

milligramme the millicurie. Now, the volume of the
curie is only
a little more than one half a cubic millimetre.
Hence in dealing
with emanation from twenty or forty milligrammes
of radium we are
dealing with very small volumes.

How may the emanation be obtained? The process is
an easy one in
skilled and practised hands. The salt of radium—

generally the
bromide or chloride—is brought into acid solution.
This causes
the emanation to be freely given off as fast as it
is formed. At
intervals we pump it off with a mercury pump.

Let us see how many millicuries we will in future
be able to turn
out in the week in our new Dublin Radium
Institute.[1] We shall
have about 130 milligrammes of radium. In 3.8 days
we get 65
millicuries from this—_i.e._ half the equilibrium
amount of 130
millicuries. Hence in the week, we shall have
about 130
millicuries.

This is not much. Many experts consider this
little enough for
one tube. But here in Dublin we have been using
the emanation in
a more economical and effective manner than is the
usage
elsewhere; according to a method which has been
worked out and
developed in our own Radium Institute. The economy
is obtained by
the very simple expedient of minutely subdividing
the' dose. The
system in vogue, generally, is to treat the tumour
by inserting
into it one or two very active

[1] Then recently established by the Royal Dublin
Society.

tubes, containing, perhaps, up to 200 millicuries, or even more, per tube. Now these very heavily charged tubes give a radiation so intense at points close to the tube, due to the greater density of the rays near the tube, and, also, to the action of the softer and more easily absorbable rays, that it has been found necessary to stop these softer rays—both the γ and β —by wrapping lead or platinum round the tube. In this lead or platinum some thirty per cent. or more of the rays is absorbed and, of course, wasted. But in the absence of the screen there is extensive necrosis of the tissues near the tubes.

If, however, in place of one or two such tubes we use ten or twenty, each containing one-tenth or one-twentieth of the dose, we can avail ourselves of the softer rays around each tube with benefit. Thus a wasteful loss is avoided. Moreover a more uniform "illumination" of the tissues results, just as we can illuminate a hall more uniformly by the use of many lesser centres of light than by the use of one intense centre of radiation. Also we get what is called "cross-radiation," which is found to be beneficial.

The surgeon knows far better what he is doing by this method.

Thus it may be arranged for the effects to go on with approximate uniformity throughout the tumour instead of varying rapidly around a central point or—and this may be very important—the effects may be readily concentrated locally.

Finally, not the least of the benefit arises in the easy technique of this new method. The quantities of

258

emanation employed can fit in the finest capillary glass tubing and the hairlike tubes can in turn be placed in fine exploring needles. There is comparatively little inconvenience to the patient in inserting these needles, and there is the most perfect control of the dosage in the number and strength of these tubes and the duration of exposure.[1]

The first Radium Institute in Ireland has already done good work for the relief of human suffering. It will have, I hope, a great future before it, for I venture, with diffidence, to hold the opinion, that with increased study the applications and claims of radioactive treatment will increase.

[1] For particulars of the new technique and of some of the work already accomplished, see papers, by Dr. Walter C. Stevenson, British Medical Journal, July 4th, 1914, and March 20th, 1915.

259

SKATING [1]

IT is now many years ago since, as a student, I was present at a college lecture delivered by a certain learned professor on the subject of friction. At this lecture a discussion arose out of a question addressed to our teacher: "How is it we can skate on ice and on no other substance?"

The answer came back without hesitation: "Because the ice is so smooth."

It was at once objected: "But you can skate on ice which is not smooth."

This put the professor in a difficulty. Obviously it is not on account of the smoothness of the ice. A piece of polished plate glass is far smoother than a surface of ice after the latter is cut up by a day's skating. Nevertheless, on the scratched and torn ice-surface skating is still quite possible;

on the smooth
plate glass we know we could not skate.

Some little time after this discussion, the connection between skating and a somewhat abstruse fact in physical science occurred to me. As the fact itself is one which has played a part in the geological history of the earth,

[1] A lecture delivered before the Royal Dublin Society in 1905.

260

and a part of no little importance, the subject of skating, whereby it is perhaps best brought home to every one, is deserving of our careful attention. Let not, then, the title of this lecture mislead the reader as to the importance of its subject matter.

Before going on to the explanation of the wonderful freedom of the skater's movements, I wish to verify what I have inferred as to the great difference in the slipperiness of glass and the slipperiness of ice. Here is a slab of polished glass. I can raise it to any angle I please so that at length this brass weight of 250 grams just slips down when started with a slight shove. The angle is, as you see, about $12\frac{1}{2}$

degrees. I now transfer the weight on to this large slab of ice which I first rapidly dry with soft linen. Observe that the weight slips down the surface of ice at a much lower angle. It is a very low angle indeed: I read it as between 4 and 5 degrees. We see by this experiment that there is a great difference between the slipperiness of the two surfaces as measured by what is called "the angle of friction." In this experiment, too, the glass possesses by far the smoother surface although I have rubbed the deeper rugosities out of the ice by smoothing it with a glass surface. Notwithstanding this, its surface is spotted with small cavities due to bubbles and imperfections. It is certain that if the glass was equally rough, its angle of friction towards the brass weight would be higher.

261

We have, however, another comparative experiment to carry out. I made as you saw a determination of the angle at which this weight of 250 grams just slipped on the ice. The lower surface of the weight, the part which presses on the ice, consists of a light, brass curtain ring. This can be detached. Its mass

is only $6\frac{1}{2}$ grams, the curtain ring being, in fact, hollow and made of very thin metal. We have, therefore, in it a very small weight which presents exactly the same surface beneath as did the weight of 250 grams. You see, now, that this light weight will not slip on ice at 5 or 6 degrees of slope, but first does so at about 10 degrees.

This is a very important experiment as regards our present inquiry. Ice appears to possess more than one angle of friction according as a heavy or a light weight is used to press upon it. We will make the same experiment with the plate of glass. You see that there is little or no difference in the angle of friction of brass on glass when we press the surfaces together with a heavy or with a light weight. The light weight requires the same slope of $12\frac{1}{2}$ degrees to make it slip.

This last result is in accordance with the laws of friction. We say that when solid presses on solid, for each pair of substances pressed together there is a constant ratio between the force required to keep one in motion over the other, and the force pressing the solids together. This ratio is

called "the coefficient of friction." The coefficient is, in fact, constant or approximately

262

so. I can determine the coefficient of friction from the angle of friction by taking the tangent of the angle. The tangent of the angle of friction is the coefficient of friction. If, then, the coefficient is constant, so, of course, must the angle of friction be constant. We have seen that it is so in the case of metal on glass, but not so in the case of metal on ice. This curious result shows that there is something abnormal about the slipperiness of ice.

The experiments we have hitherto made are open to the reproach that the surface of the ice is probably damp owing to the warmth of the air in contact with it. I have here a means of dealing with a surface of cold, dry ice. This shallow copper tank about 18 inches (45 cms.) long, and 4 inches (10 cms.) wide, is filled with a freezing 'mixture circulated through it from a larger vessel containing ice melting in hydrochloric acid at a temperature of about -18° C. This keeps the tank

below the melting point of ice. The upper surface of the tank is provided with raised edges so that it can be flooded with water. The water is now frozen and its temperature is below 0° C. It is about 10° C. I can place over the ice a roof-shaped cover made of two inclined slabs of thick plate glass. This acts to keep out warm air, and to do away with any possibility of the surface of the ice being wet with water thawed from the ice. The whole tank along with its roof of glass can be adjusted to any angle, and a scale at the

263

raised end of the tank gives the angle of slope in degrees. A weight placed on the ice can be easily seen through the glass cover.

The weight we shall use consists of a very light ring of aluminium wire which is rendered plainly visible by a ping-pong ball attached above it. The weight rests now on a copper plate provided for the purpose at the upper end of the tank. The plate being in direct contact beneath with the freezing mixture we are sure that the aluminium ring is no hotter than the

ice. A light
jerk suffices to shake the weight on to the
surface of the ice.

We find that this ring loaded with only the ping-
pong ball, and
weighing a total of 2.55 grams does not slip at
the low angles. I
have the surface of the ice at an angle of rather
over $13\frac{1}{2}$, and
only by continuous tapping of the apparatus can it
be induced to
slip down. This is a coefficient of 0.24, and
compares with the
coefficient of hard and smooth solids on one
another. I now
replace the empty ping-pong ball by a similar ball
filled with
lead shot. The total weight is now 155 grams. You
see the angle
of slipping has fallen to 7° .

Every one who has made friction experiments knows
how
unsatisfactory and inconsistent they often are. We
can only
discuss notable quantities and broad results,
unless the most
conscientious care be taken to eliminate errors.
The net result
here is that ice at about -10° C. when pressed on
by a very light
weight possesses a

264

coefficient of friction comparable with the usual
coefficients of

solids on solids, but when the pressure is increased, the coefficient falls to about half this value.

The following table embodies some results obtained on the friction of ice and glass, using the methods I have shown you. I add some of the more carefully determined coefficients of other observers.

Wt. in On Ice Grams. at 10° C.		On Plate Glass.		On Ice at 0° C.	
Angle. Coeff.		Angle. Coeff.		Angle. Coeff.	
Aluminium	2.55	12½°	0.22	12°	
0.21	13½°	0.24			
Same	155	12½°	0.22	6°	0.11
	7°	0.12			
Brass	6.5	12½°	0.22	10°	
0.17	10½°	0.18			
Same	107	12½°	0.22	5°	0.09
	6°	0.10			
Steel on steel (Morin) - - - - 0.14					
Brass on cast iron (Morin) - - 0.19					
Steel on cast iron (Morin) - - 0.20					
Skate on ice (J. Müller) - - - 0.016–0.032					
Best-greased surfaces (Perry) - 0.03–0.036					

You perceive from the table that while the friction of brass or aluminium on glass is quite independent of the weight used, that of brass or aluminium on ice depends in some way

upon the weight,
and falls in a very marked degree when the weight
is heavy. Now,
I think that if we had been on the look out for
any abnormality
in the friction of hard substances on ice, we
would have rather
anticipated a variation in the

265

other direction. We would have, perhaps, expected
that a heavy
weight would have given rise to the greater
friction. I now turn
to the explanation of this extraordinary result.

You are aware that it requires an expenditure of
heat merely to
convert ice to water, the water produced being at
the temperature
of the ice, i.e. at 0° C., from which it is
derived. The heat
required to change the ice from the solid to the
liquid state is
the latent heat of water. We take the unit
quantity of heat to be
that which is required to heat 1 kilogram of water
 1° C. Then if
we melt 1 kilogram of ice, we must supply it with
80 such units
of heat. While melting is going on, there is no
change of
temperature if the experiment is carefully
conducted. The melting
ice and the water coming from it remain at 0° C.
throughout the
operation, and neither the thermometer nor your

own sensations
would tell you of the amount of heat which was
flowing in. The
heat is latent or hidden in the liquid produced,
and has gone to
do molecular work in the substance. Observe that
if we supply
only 40 thermal units, we get only one-half the
ice melted. If
only 10 units are supplied, then we get only one
eighth of a
kilogram of water, and no more nor less.

I have ventured to recall to you these
commonplaces of science
before considering a mode of melting ice which is
less generally
known, and which involves no supply of heat on
your part. This
method involves for its

266

understanding a careful consideration of the
thermal properties
of water in the solid state.

It must have been observed a very long time ago
that water
expands when it freezes. Otherwise ice would not
float on water;
and, what is perhaps more important in your eyes,
your water
pipes would not burst in winter when the water
freezes therein.
But although the important fact of the expansion
of water on
freezing was so long presented to the observation

of mankind, it was not till almost exactly the middle of the last century that James Thomson, a gifted Irishman, predicted many important consequences arising from the fact of the expansion of water on becoming solid. The principles he enunciated are perfectly general, and apply in every case of change of volume attending change of state. We are here only concerned with the case of water and ice.

James Thomson, following a train of thought which we cannot here pursue, predicted that owing to the fact of the expansion of water on becoming solid, pressure will lower the melting point of ice or the freezing point of water. Normally, as you are aware, the temperature is 0° C. or 32° F. Thomson said that this would be found to be the freezing point only at atmospheric pressure. He calculated how much it would change with change of pressure. He predicted that the freezing point would fall 0.0075 of a degree Centigrade for each additional atmosphere of pressure applied to the water. Suppose,

for instance, our earth possessed an atmosphere so heavy to as exert a thousand times the pressure of the existing atmosphere, then water would not freeze at 0° C., but at -7.5° C. or about 18° F. Again, in vacuo, that is when the pressure has been reduced to the relatively small vapour pressure of the water, the freezing point is above 0° C., i.e. at 0.0075° C. In parts of the ocean depths the pressure is much over a thousand atmospheres. Fresh water would remain liquid there at temperatures much below 0° C.

It will be evident enough, even to those not possessed of the scientific insight of James Thomson, that some such fact is to be anticipated. It is, however, easy to be wise after the event. It appeals to us in a general way that as water expands on freezing, pressure will tend to resist the turning of it to ice. The water will try to remain liquid in obedience to the pressure. It will, therefore, require a lower temperature to induce it to become ice.

James Thomson left his thesis as a prediction. But he predicted exactly what his distinguished brother, Sir William Thomson—later

Lord Kelvin—found to happen when the matter was put to the test of experiment. We must consider the experiment made by Lord Kelvin.

According to Thomson's views, if a quantity of ice and water are compressed, there must be _a fall of temperature_. The nature of his argument is as follows:

268

Let the ice and water be exactly at 0° C. to start with. Then suppose we apply, say, one thousand atmospheres pressure. The melting point of the ice is lowered to -7.5° C. That is, it will require a temperature so low as -7.5° C. to keep it solid. It will therefore at once set about melting, for as we have seen, its actual temperature is not -7.5° C., but a higher temperature, _i.e._ 0° C. In other words, it is 7.5° above its melting point. But as soon as it begins melting it also begins to absorb heat to supply the 80 thermal units which, as we know, are required to turn each kilogram of the ice to water. Where can it get this heat? We assume that we give it none. It has only two sources, the ice can take heat from itself, and it can take heat from the

water. It does both in this case, and both ice and water drop in temperature. They fall in temperature till -7.5° is reached. Then the ice has got to its melting point under the pressure of one thousand atmospheres, or, as we may put it, the water has reached its freezing point. There can be no more melting. The whole mass is down to -7.5° C., and will stay there if we keep heat from flowing either into or out of the vessel. There is now more water and less ice in the vessel than when we started, and the temperature has fallen to -7.5° C. The fall of temperature to the amount predicted by the theory was verified by Lord Kelvin.

Suppose we now suddenly remove the pressure; what will happen? We have water and ice at -7.5° C.

269

and at the normal pressure. Water at -7.5° and at the normal pressure of course turns to ice. The water will, therefore, instantly freeze in the vessel, and the whole process will be reversed. In freezing, the water will give up its latent heat, and this will warm up the whole mass till once again 0° C. is attained. Then there will be no more freezing, for

again the ice
is at its melting point. This is the remarkable
series of events
which James Thomson predicted. And these are the
events which
Lord Kelvin by a delicate series of experiments,
verified in
every respect.

Suppose we had nothing but solid ice in the vessel
at starting,
would the experiment result in the same way? Yes,
it assuredly
would. The ice under the increased pressure would
melt a little
everywhere throughout its mass, taking the
requisite latent heat
from itself at the expense of its sensible heat,
and the
temperature of the ice would fall to the new
melting point.

Could we melt the whole of the ice in this manner?
Again the
answer is "yes." But the pressure must be very
great. If we
assume that all the heat is obtained at the
expense of the
sensible heat of the ice, the cooling must be such
as to supply
the latent heat of the whole mass of water
produced. However, the
latent heat diminishes as the melting point is
lowered, and at a
rate which would reduce it to nothing at about
18,000
atmospheres. Mousson, operating on ice enclosed in

a conducting
cylinder and cooled to -18° at starting

270

appears to have obtained very complete
liquefaction. Mousson must
have attained a pressure of at least an amount
adequate to lower
the melting point below -18° . The degree of
liquefaction actually
attained may have been due in part to the passage
of heat through
the walls of the vessel. He proved the more or
less complete
liquefaction of the ice within the vessel by the
fall of a copper
index from the top to the bottom of the vessel
while the pressure
was on.

I have here a simple way of demonstrating to you
the fall of
temperature attending the compression of ice. In
this mould,
which is strongly made of steel, lined with
boxwood to diminish
the passage of conducted heat, is a quantity of
ice which I
compress when I force in this plunger. In the ice
is a
thermoelectric junction, the wires leading to
which are in
communication with a reflecting galvanometer. The
thermocouple is
of copper and nickel, and is of such sensitiveness
as to show by
motion of the spot of light on the screen even a

small fraction
of a degree. On applying the pressure, you see the
spot of light
is displaced, and in such a direction as to
indicate cooling. The
balancing thermocouple is all the time imbedded in
a block of ice
so that its temperature remains unaltered. On
taking off the
pressure, the spot of light returns to its first
position. I can
move the spot of light backwards and forwards on
the screen by
taking off and putting on the pressure. The
effects are quite
instantaneous.

271

The fact last referred to is very important. The
ice, in fact, is
as it were automatically turned to water. It is
not a matter of
the conduction of heat from point to point in the
ice. Its own
sensible heat is immediately absorbed throughout
the mass. This
would be the theoretical result, but it is
probable that owing to
imperfections throughout the ice and failure in
uniformity in the
distribution of the stress, the melting would not
take place
quite uniformly or homogeneously.

Before applying our new ideas to skating, I want
you to notice a
fact which I have inferentially stated, but not

specifically
mentioned. Pressure will only lead to the melting
of ice if the
new melting point, _i.e._ that due to the
pressure, is below the
prevailing temperature. Let us take figures. The
ice to start
with is, say, at -3° C. Suppose we apply such a
pressure to this
ice as will confer a melting point of -2° C. on
it. Obviously,
there will be no melting. For why should ice which
is at -3° C.
melt when its melting point is -2° C.? The ice is,
in fact,
colder than its melting point. Hence, you note
this fact: The
pressure must be sufficiently intense to bring the
melting point
below the prevailing temperature, or there will be
no melting;
and the further we reduce the melting point by
pressure below the
prevailing temperature, the more ice will be
melted.

We come at length to the object of our remarks I
don't know who
invented skating or skates. It is said that in the
thirteenth
century the inhabitants of

272

England used to amuse themselves by fastening the
bones of an
animal beneath their feet, and pushing themselves
about on the

ice by means of a stick pointed with iron. With such skates, any performance either on inside or outside edge was impossible. We are a conservative people. This exhilarating amusement appears to have served the people of England for three centuries. Not till 1660 were wooden skates shod with iron introduced from the Netherlands. It is certain that skating was a fashionable amusement in Pepys' time. He writes in 1662 to the effect: "It being a great frost, did see people sliding with their skates, which is a very pretty art." It is remarkable that it was the German poet Klopstock who made skating fashionable in Germany. Until his time, the art was considered a pastime, only fit for very young or silly people.

I wish now to dwell upon that beautiful contrivance the modern skate. It is a remarkable example of how an appliance can develop towards perfection in the absence of a really intelligent understanding of the principles underlying its development. For what are the principles underlying the proper construction of the skate? After what I have said, I think you will readily understand. The object is to produce such a

pressure under the
blade that the ice will melt. We wish to establish
such a
pressure under the skate that even on a day when
the ice is below
zero, its melting

273

point is so reduced just under the edge of the
skate that the ice
turns to water.

It is this melting of the ice under the skate
which secures the
condition essential to skating. In the first
place, the skate no
longer rests on a solid. It rests on a liquid. You
are aware how
in cases where we want to reduce friction—say at
the bearing of a
wheel or under a pivot—we introduce a liquid. Look
at the
bearings of a steam engine. A continuous stream of
oil is fed in
to interpose itself between the solid surfaces. I
need not
illustrate so well-known a principle by
experiment. Solid
friction disappears when the liquid intervenes. In
its place we
substitute the lesser difficulty of shearing one
layer of the
liquid over the other; and if we keep up the
supply of oil the
work required to do this is not very different, no
matter how
great we make the pressure upon the bearings.

Compared with the resistance of solid friction, the resistance of fluid friction is trifling. Here under the skate the lubrication is perhaps the most perfect which it is possible to conceive. J. Müller has determined the coefficient by towing a skater holding on by a spring balance. The coefficient is between 0.016 and 0.032. In other words, the skater would run down an incline so little as 1 or 2 degrees; an inclination not perceivable by the eye. Now observe that the larger of these coefficients is almost exactly the same as that which Perry found in the case of well-greased surfaces. But evidently no

274

artificial system of lubrication could hope to equal that which exists between the skate and the ice. For the lubrication here is, as it were, automatic. In the machine if the lubricant gets squeezed out there instantly ensues solid friction. Under the skate this cannot happen for the squeezing out of the lubricant is instantly followed by the formation of another film of water. The conditions of pressure which may lead to solid friction in the machine here automatically call the lubricant

into
existence.

Just under the edge of the skate the pressure is enormous.

Consider that the whole weight of the skater is born upon a mere knife edge. The skater alternately throws his whole weight upon

the edge of each skate. But not only is the weight thus

concentrated upon one edge, further concentration is secured in

the best skates by making the skate hollow-ground, i.e.

increasing the keenness of the edge by making it less than a

right angle. Still greater pressure is obtained by diminishing

the length of that part of the blade which is in contact with the

ice. This is done by putting curvature on the blade or making it

what is called "hog-backed." You see that everything is done to

diminish the area in contact with the ice, and thus to increase

the pressure. The result is a very great compression of the ice

beneath the edge of the skate. Even in the very coldest weather

melting must take place to some extent.

As we observed before, the melting is instantaneous,

Heat has not to travel from one point of the ice to another;
immediately the pressure comes on the ice it turns to water. It takes the requisite heat from itself in order that the change of state may be accomplished. So soon as the skate passes on, the water resumes the solid state. It is probable that there is an instantaneous escape, and re-freezing of some of the water from beneath the skate, the skate instantly taking a fresh bearing and melting more ice. The temperature of the water escaping from beneath the skate, or left behind by it, immediately becomes what it was before the skate pressed upon it.

Thus, a most wonderful and complex series of molecular events takes place beneath the skate. Swift as it passes, the whole sequence of events which James Thomson predicted has to take place beneath the blade Compression; lowering of the melting point below the temperature of the surrounding ice; melting; absorption of heat; and cooling to the new melting point, _i.e._ to that proper to the pressure beneath the blade. The skate now passes on. Then follow: Relief of pressure; re-solidification of the water; restoration of the borrowed heat from

the congealing
water and reversion of the ice to the original
temperature.

If we reflect for a moment on all this, we see
that we do not
skate on ice but on water. We could not skate on
ice any more
than we could skate on glass. We saw that with
light weights and
when the pressure

276

{Diagram}

Diagram showing successive states obtaining in
ice, before,
during, and after the passage of the skate. The
temperatures and
pressures selected for illustration are such as
might occur under
ordinary conditions. The edge of the skate is
shown in magnified
cross-section.

277

Was not sufficient to melt the ice, the friction
was much the
same as that of metal on glass. Ice is not
slippery. It is an
error to say that it is. The learned professor was
very much
astray when he said that you could skate on ice
because it is so
smooth. The smoothness of the ice has nothing to
do with the
matter. In short, owing to the action of gravity

upon your body,
you escape the normal resistance of solid on
solid, and glide
about with feet winged like the messenger of the
Gods; but on
water.

A second condition essential to the art of skating
is also
involved in the melting of the ice. The sinking of
the skate
gives the skater "bite." This it is which enables
him to urge
himself forward. So long as skates consisted of
the rounded bones
of animals, the skater had to use a pointed staff
to propel
himself. In creating bite, the skater again
unconsciously appeals
to the peculiar physical properties of ice. The
pressure required
for the propulsion of the skater is spread all
along the length
of the groove he has cut in the ice, and obliquely
downwards. The
skate will not slip away laterally, for the
horizontal component
of the pressure is not enough to melt the ice. He
thus gets the
resistance he requires.

You see what a very perfect contrivance the skate
is; and what a
similitude of intelligence there is in its
evolution. Blind
intelligence, because it is certain the true
physics of skating
was never held in view by

278

the makers of skates. The evolution of the skate has been truly organic. The skater selected the fittest skate, and hence the fit skate survived.

In a word, the possibility of skating depends on the dynamical melting of ice under pressure. And observe the whole matter turns upon the apparently unrelated fact that the freezing of water results in a solid more bulky than the water which gives rise to it. If ice was less bulky than the water from which it was derived, pressure would not melt it; it would be all the more solid for the pressure, as it were. The melting point would rise instead of falling. Most substances behave in this manner, and hence we cannot skate upon them. Only quite a few substances expand on freezing, and it happens that their particular melting temperatures or other properties render them unsuitable to skating. The most abundant fluid substance on the earth, and the most abundant substance of any one kind on its surface, thus possesses the ideally correct and suitable properties for the art of skating.

I have pointed out that the pressure must be such as to bring the temperature of melting below that prevailing in the ice at the time. We have seen also, that one atmosphere lowers the melting point of ice by the $\frac{1}{140}$ of a degree Centigrade;

more exactly by
 0.0075° . Let us now assume that the skate is so
far sunken in the
ice as to bear for a length of two inches, and for
a width of
one-hundredth of an inch. The skater weighs,

279

let us say—150 pounds. If this weight was borne on
one square
inch, the pressure would be ten atmospheres. But
the skater rests
his weight, in fact, upon an area of one-fiftieth
of an inch. The
pressure is, therefore, fifty times as great. The
ice is
subjected to a pressure of 500 atmospheres. This
lowers the
melting point to -3.75° C. Hence, on a day when
the ice is at
this temperature, the skate will sink in the ice
till the weight
of the skater is concentrated as we have assumed.
His skate can
sink no further, for any lesser concentration of
the pressure
will not bring the melting point below the
prevailing
temperature. We can calculate the theoretical bite
for any state
of the ice. If the ice is colder the bite will not
be so deep. If
the temperature was twice as far below zero, then
the area over
which the skater's weight will be distributed,
when the skate has
penetrated its maximum depth, will be only half

the former area,
and the pressure will be one thousand atmospheres.

An important consideration arises from the fact that under the very extreme edge of the skate the pressure is indefinitely great. For this involves that there will always be some bite, however cold the ice may be. That is, the narrow strip of ice which first receives the skater's weight must partially liquefy however cold the ice.

It must have happened to many here to be on ice which was too cold to skate on with comfort. The

280

skater in this case speaks of the ice as too hard. In the Engadine, the ice on the large lakes gets so cold that skaters complain of this. On the rinks, which are chiefly used there, the ice is frequently renewed by flooding with water at the close of the day. It thus never gets so very cold as on the lakes. I have been on ice in North France, which, in the early morning, was too hard to afford sufficient bite for comfort. The cause of this is easily understood from what we have been considering.

We may now return to the experimental results which we obtained early in the lecture. The heavy weights slip off the ice at a low angle because just at the points of contact with the ice the latter melts, and they, in fact, slip not on ice but on water. The light weights on cold, dry ice do not lower the melting point below the temperature of the ice, i.e. below -10° C., and so they slip on dry ice. They therefore give us the true coefficient of friction of metal on ice.

This subject has, more recently been investigated by H. Morphy, of Trinity College, Dublin. The refinement of a closed vessel at uniform temperature, in which the ice is formed and the experiment carried out, is introduced. Thermocouples give the temperatures, not only of the ice but of the aluminium sleigh which slips upon it under various loads. In this way we may be certain that the metal runners are truly at the temperature of the ice. I now quote from Morphy's paper

281

"The angle of friction was found to remain constant until a certain stage of the loading, when it suddenly fell to about half

of its original value. It then remained constant for further increases in the load.

"These results, which confirmed those obtained previously with less satisfactory apparatus, are shown in the table below. In the first column is shown the load, i.e. the weight of sleigh + weight of shot added. In the second and third columns are shown, respectively, the coefficient and angle of friction, whilst the fourth gives the temperature of the ice as determined from the galvanometer deflexions.

Load.	Tan μ .	μ .	Temp.
5.68 grams.	0.36±.01	20°±30'	-5.65° C.
10.39			-5.65°
11.96			-5.75°
12.74			-5.60°
13.53			-5.65°
14.31			-5.65°
15.10 grams.	0.17±.01	9°.30'±30'	-5.60°
16.67			-5.55°
19.81			-5.60°
24.52			-5.60°
5.68 grams.	0.36±.01	20°±30'	-5.60°

"These experiments were repeated on another occasion with the same result and similar results had been obtained with different apparatus.

"As a result of the investigation the following points are clearly shown:—

282

"(1) The coefficient of friction for ice at constant temperature may have either of two constant values according to the pressure per unit surface of contact.

"(2) For small pressures, and up to a certain well defined limit of pressure, the coefficient is fairly large, having the value $0.36 \pm .01$ in the case investigated.

"(3) For pressures greater than the above limit the coefficient is relatively small, having the value $0.17 \pm .01$ in the case investigated."

It will be seen that Morphy's results are similar to those arrived at in the first experimental consideration of our subject; but from the manner in which the experiments have been carried out, they are more accurate and reliable.

A great deal more might be said about skating, and the allied sports of tobogganing, sleighing, curling, ice yachting, and last, but by no means least, sliding—that unpretentious pastime of the million. Happy the boy who has nails in his

boots when
Jack-Frost appears in his white garment, and
congeals the
neighbouring pond. But I must turn away at the
threshold of the
humorous aspect of my subject (for the victim of
the street
"slide" owes his injured dignity to the abstruse
laws we have
been discussing) and pass to other and graver
subjects intimately
connected with skating.

James Thomson pointed out that if we apply
compressional stress
to an ice crystal contained in a vessel

283

which also contains other ice crystals, and water
at 0° C., then
the stressed crystal will melt and become water,
but its
counterpart or equivalent quantity of ice will
reappear elsewhere
in the vessel. This is, obviously, but a deduction
from the
principles we have been examining. The phenomenon
is commonly
called "regelation." I have already made the usual
regelation
experiment before you when I compressed broken ice
in this mould.
The result was a clear, hard and almost flawless
lens of ice. Now
in this operation we must figure to ourselves the
pieces of ice
when pressed against one another melting away

where compressed,
and the water produced escaping into the spaces
between the
fragments, and there solidifying in virtue of its
temperature
being below the freezing point of unstressed
water. The final
result is the uniform lens of ice. The same
process goes on in a
less perfect manner when you make—or shall I
better say—when you
made snowballs.

We now come to theories of glacier motion; of
which there are
two. The one refers it mainly to regelation; the
other to a real
viscosity of the ice.

The late J. C. M'Connell established the fact that
ice possesses
viscosity; that is, it will slowly yield and
change its shape
under long continued stresses. His observations,
indeed, raise a
difficulty in applying this viscosity to explain
glacier motion,
for he showed that an ice crystal is only viscous
in a certain
structural

284

direction. A complex mixture of crystals such, as
we know glacier
ice to be, ought, we would imagine, to display a
nett or
resultant rigidity. A mass of glacier ice when

distorted by application of a force must, however, undergo precisely the transformations which took place in forming the lens from the fragments of ice. In fact, regelation will confer upon it all the appearance of viscosity.

Let us picture to ourselves a glacier pressing its enormous mass down a Swiss valley. At any point suppose it to be hindered in its downward path by a rocky obstacle. At that point the ice turns to water just as it does beneath the skate. The cold water escapes and solidifies elsewhere. But note this, only where there is freedom from pressure. In escaping, it carries away its latent heat of liquefaction, and this we must assume, is lost to the region of ice lately under pressure. This region will, however, again warm up by conduction of heat from the surrounding ice, or by the circulation of water from the surface. Meanwhile, the pressure at that point has been relieved. The mechanical resistance is transferred elsewhere. At this new point there is again melting and relief of pressure. In this manner the glacier may be supposed to move down. There is continual flux of

conducted heat and converted latent heat, hither
and thither, to
and from the points of resistance. The final
motion of the whole
mass is necessarily slow; a few feet in the day
or, in winter,

285

even only a few inches. And as we might expect,
perfect silence
attends the downward slipping of the gigantic
mass. The motion
is, I believe, sufficiently explained as a skating
motion. The
skate is, however, fixed, the ice moves. The great
Aletsch
Glacier collects its snows among the highest
summits of the
Oberland. Thence, the consolidated ice makes its
way into the
Rhône Valley, travelling a distance of some 20
miles. The ice now
melting into the youthful Rhône fell upon the
Monch, the Jungfrau
or the Eiger in the days when Elizabeth ruled in
England and
Shakespeare lived.

The ice-fall is a common sight on the glacier. In
great lumps and
broken pinnacles it topples over some rocky
obstacle and falls
shattered on to the glacier below. But a little
further down the
wound is healed again, and regelation has restored
the smooth
surface of the glacier. All such phenomena are

explained on James Thomson's exposition of the behaviour of a substance which expands on passing from the liquid to the solid state.

We thus have arrived at very far-reaching considerations arising out of skating and its science. The tendency for snow to accumulate on the highest regions of the Earth depends on principles which we cannot stop to consider. We know it collects above a certain level even at the Equator. We may consider, then, that but for the operation of the laws which James Thomson brought to light, and which his illustrious brother,

286

Lord Kelvin, made manifest, the uplands of the Earth could not have freed themselves of the burthen of ice. The geological history of the Earth must have been profoundly modified. The higher levels must have been depressed; the general level of the ocean relatively to the land thereby raised, and, it is even possible, that such a mean level might have been attained as would result in general submergence.

During the last great glacial period, we may say the fate of the world hung on the operation of those laws which have concerned us throughout this lecture. It is believed the ice was piled up to a height of some 6,000 feet over the region of Scandinavia. Under the influence of the pressure and fusion at points of resistance, the accumulation was stayed, and it flowed southwards the accumulation was stayed, and it flowed southwards over Northern Europe. The Highlands of Scotland were covered with, perhaps, three or four thousand feet of ice. Ireland was covered from north to south, and mighty ice-bergs floated from our western and southern shores.

The transported or erratic stones, often of great size, which are found in many parts of Ireland, are records of these long past events: events which happened before Man, as a rational being, appeared upon the Earth.

287

A SPECULATION AS TO A PREMATERIAL UNIVERSE [1]

"And therefore...these things likewise had a birth; for things which are of mortal body could not for an infinite time back...

have been able to set at naught the puissant strength of immeasurable age."—LUCRETIUS, *_De Rerum Natura._*

"O fearful meditation! Where, alack! Shall Time's best jewel from Time's chest lie hid?" —SHAKESPEARE.

IN the material universe we find presented to our senses a physical development continually progressing, extending to all, even the most minute, material configurations. Some fundamental distinctions existing between this development as apparent in the organic and the inorganic systems of the present day are referred to elsewhere in this volume.[2] In the present essay, these systems as having a common origin and common ending, are merged in the same consideration as to the nature of the origin of material systems in general. This present essay is occupied by the consideration of the necessity of limiting material interactions in past time. The speculation originated in the difficulties which present themselves when we ascribe to these interactions infinite duration in the past. These difficulties first claim our consideration.

[1] Proc. Royal Dublin Soc., vol. vii., Part V, 1892.

[2] _The Abundance of Life._

288

Accepting the hypothesis of Kant and Laplace in its widest extension, we are referred to a primitive condition of wide material diffusion, and necessarily too of material instability.

The hypothesis is, in fact, based upon this material instability.

We may pursue the sequence of events assumed in this hypothesis

into the future, and into the past.

In the future we find finality to progress clearly indicated. The

hypothesis points to a time when there will be no more

progressive change but a mere sequence of unfruitful events, such

as the eternal uniform motion of a mass of matter no longer

gaining or losing heat in an ether possessed of a uniform

distribution of energy in all its parts. Or, again, if the ether

absorb the energy of material motion, this vast and dark

aggregation eternally poised and at rest within it. The action is

transferred to the subtle parts of the ether which suffer none of

the energy to degrade. This is, physically, a thinkable future.

Our minds suggest no change, and demand none. More than this,

change is unthinkable according to our present ideas of energy.

Of progress there is an end.

This finality _à parte post_ is instructive.

Abstract

considerations, based on geometrical or analytical illustrations,

question the finiteness of some physical developments. Thus our

sun may require eternal time to attain the temperature of the

ether around it, the approach to this condition being assumed to

be asymptotic in

289

character. But consider the legitimate _reductio ad absurdum_ of

an ember raked from a fire 1000 years ago. Is it not yet cooled

down to the constant temperature of its surroundings? And we may

evidently increase the time a million-fold if we please. It

appears as if we must regard eternity as outliving every

progressive change, For there is no convergence or enfeeblement

of time. The ever-flowing present moves no differently for the

occurrence of the mightiest or the most insignificant events. And

even if we say that time is only the attendant upon events, yet

this attendant waits patiently for the end,

however long
deferred.

Does the essentially material hypothesis of Kant and Laplace account for an infinite past as thinkably as it accounts for the infinite future? As this hypothesis is based upon material instability the question resolves itself into this:— Is the assumption of an infinitely prolonged past instability a probable or possible account of the past? There are, it appears to me, great difficulties involved in accepting the hypothesis of infinitely prolonged material instability. I will refer here to three principal objections. The first may be called a metaphysical objection; the second is partly metaphysical and partly physical, the third may be considered a physical objection, as it is involved directly in the phenomena presented by our universe.

The metaphysical objection must have presented itself to every one who has considered the question. It may

290

be put thus:—If present events are merely one stage in an infinite progress, why is not the present stage

long ago passed
over? We are evidently at liberty to push back any
stage of
progress to as remote a period as we like by
putting back first
the one before this and next the stage preceding
this, and so on,
for, by hypothesis, there is no beginning to the
progress.

Thus, the sum of passing events constituting the
present universe
should long ago have been accomplished and passed
away. If we
consider alternative hypotheses not involving this
difficulty, we
are at once struck by the fact that the future of
material
development is free of the objection. For the
eternity of
unprogressive events involved in the future on
Kant's hypothesis,
is not only thinkable, but any change is, as
observed,
irreconcilable with our ideas of energy. As in the
future so in
the past we look to a cessation to progress. But
as we believe
the activity of the present universe must in some
form have
existed all along, the only refuge in the past is
to imagine an
active but unprogressive eternity, the
unprogressive activity at
some period becoming a progressive activity—that
progressive
activity of which we are spectators. To the

unprogressive
activity there was no beginning; in fact,
beginning is as
unthinkable and uncalled for to the unprogressive
activity of the
past as ending is to the unprogressive activity of
the future,
when all developmental actions shall have ceased.
There is no
beginning or ending to the activity of the
universe.

291

There is beginning and ending to present
progressive activity.
Looking through the realm of nature we seek
beginning and ending,
but "passing through nature to eternity" we find
neither. Both
are justified; the questioning of the ancient poet
regarding the
past, and of the modern regarding the future,
quoted at the head
of this essay.

The next objection, which is in part metaphysical,
is founded on
the difficulty of ascribing any ultimate reality
or potency to
forces diminishing through eternal time. Thus,
against the
assumption that our universe is the result of
material
aggregation progressing over eternal time, which
involves the
primitive infinite separation of the particles, we
may ask, what

force can have acted between particles sundered by infinite distance? The gravitational force falling off as the square of the distance, must vanish at infinity if we mean what we say when we ascribe infinite separation to them. Their condition is then one of neutral stability, a finite movement of the particles neither increasing nor diminishing interaction. They had then remained eternally in their separated condition, there being no cause to render such condition finite. The difficulty involved here appears to me of the same nature as the difficulty of ascribing any residual heat to the sun after eternal time has elapsed. In both cases we are bound to prolong the time, from our very idea of time, till progress is no more, when in the one case we can imagine no mutual approximation of the

292

particles, in the other no further cooling of the body. However, I will not dwell further upon this objection, as it does not, I believe, present itself with equal force to every mind. A reason less open to dispute, as being less subjective, against the aggregation of infinitely remote particles as the

origin of our
universe, is contained in the physical objection.

In this objection we consider that the appearance presented by
our universe negatives the hypothesis of
infinitely prolonged
aggregation. We base this negation upon the
appearance of
simultaneity ~ presented by the heavens,
contending that this
simultaneity is contrary to what we would expect
to find in the
case of particles gathered from infinitely remote
distances.

Whether these particles were endowed with relative
motions or not
is unimportant to the consideration. In what
respects do the
phenomena of our universe present the appearance
of simultaneous
phenomena? We must remember that the suns in space
are as fires
which brighten only for a moment and are then
extinguished. It is
in this sense we must regard the longest burning
of the stars.

Whether just lit or just expiring counts little in
eternity. The
light and heat of the star is being absorbed by
the ether of
space as effectually and rapidly as the ocean
swallows the ripple
from the wings of an expiring insect. Sir William
Herschel says
of the galaxy of the milky way:— "We do not know
the rate of

progress of this mysterious chronometer, but it is nevertheless certain that it cannot

293

last for ever, and its past duration cannot be infinite." We do not know, indeed, the rate of progress of the chronometer, but if the dial be one divided into eternal durations the consummation of any finite physical change represents such a movement of the hand as is accomplished in a single vibration of the balance wheel.

Hence we must regard the hosts of glittering stars as a conflagration that has been simultaneously lighted up in the heavens. The enormous (to our ideas) thermal energy of the stars resembles the scintillation of iron dust in a jar of oxygen when a pinch of the dust is thrown in. Although some particles be burnt up before others become alight, and some linger yet a little longer than the others, in our day's work the scintillation of the iron dust is the work of a single instant, and so in the long night of eternity the scintillation of the mightiest suns of space is over in a moment. A little longer,

indeed, in duration than the life which stirs a moment in response to the diffusion of the energy, but only very little. So must an Eternal Being regard the scintillation of the stars and the periodic vibration of life in our geological time and the most enduring efforts of thought. The latter indeed are no more lasting than

"... the labour of ants In the light of a million million of suns."

But the myriad suns themselves, with their generations, are the momentary gleam of lights for ever after extinguished.

294

Again, science suggests that the present process of material aggregation is not finished, and possibly will only be when it prevails universally. Hence the very distribution of the stars, as we observe them, as isolated aggregations, indicates a development which in the infinite duration must be regarded as equally advanced in all parts of stellar space and essentially a simultaneous phenomenon. For were we spectators of a system in which any very great difference of age prevailed,

this very great
difference would be attended by some such
appearance as the
following:—

The appearance of but one star, other generations
being long
extinct or no others yet come into being; or,
perhaps, a faint
nebulous wreath of aggregating matter somewhere
solitary in the
heavens; or no sign of matter beyond our system,
either because
ungathered or long passed away into darkness.[1]

Some such appearances were to be expected had the
aggregation of
matter depended solely on chance encounters of
particles
scattered through infinite space.

For as, by hypothesis, the aggregation occupies an
infinite time
in consummation it is nearly a certainty that each
particle
encountered after immeasurable time, and then for
the first time
endowed with actual gravitational potential
energy, would have
long expended this energy

[1] It is interesting to reflect upon the effect
which an entire
absence of luminaries outside our solar system
would have had
upon the views of our philosophers and upon our
outlook on life.

before another particle was gathered. But the fact that so many fires which we know to be of brief duration are scattered through a region of space, and the fact of a configuration which we believe to be a transitory one, suggest their simultaneous aggregation here and there. And in the nebulous wreaths situated amidst the stars there is evidence that these actually originated where they now are, for in such no relative motion, I believe, has as yet been detected by the spectroscope. All this, too, is in keeping with the nebular hypothesis of Kant and Laplace so long as this does not assume a primitive infinite dispersion of matter, but the gathering of matter from finite distances first into nebulous patches which aggregating with each other have given rise to our system of stars. But if we extend this hypothesis throughout an infinite past by the supposition of aggregation of infinitely remote particles we replace the simultaneous approach required in order to account for the simultaneous phenomena visible in the heavens, by a succession of aggregative events, by hypothesis at intervals of nearly infinite duration, when the events of the universe had

consisted of fitful
gleams lighted after eternities of time and
extinguished for yet
other eternities.

Finally, if we seek to replace the eternal
instability involved
in Kant's hypothesis when extended over an
infinite past, by any
hypothesis of material stability, we at once find
ourselves in
the difficulty that from the known properties of
matter such
stability must have been

296

permanent if ever existent, which is contrary to
fact. Thus the
kinetic inertia expressed in Newton's first law of
motion might
well be supposed to secure equilibrium with
material attraction,
but if primevally diffused matter had ever thus
been held in
equilibrium it must have remained so, or it was
maintained so
imperfectly, which brings us back to endless
evolution.

On these grounds I contend that the present
gravitational
properties of matter cannot be supposed to have
acted for all
past duration. Universal equilibrium of
gravitating particles
would have been indestructible by internal causes.
Perpetual

instability or evolution is alike unthinkable and contrary to the phenomena of the universe of which we are cognisant. We therefore turn from gravitating matter as affording no rational account of the past. We do so of necessity, however much we feel our ignorance of the nature of the unknown actions to which we have recourse.

A prematerial condition of the universe was, we assume, a condition in which uniformity as regards the average distribution of energy in space prevailed, but heterogeneity and instability were possible. The realization of that possibility was the beginning we seek, and we today are witnesses of the train of events involved in the breakdown of an eternal past equilibrium. We are witnesses on this hypothesis, of a catastrophe possibly confined to certain regions of space, but which is, to the motions and configurations concerned, absolutely unique, reversible to

297

its former condition of potential by no process of which we can have any conception.

Our speculation is that we, as spectators of evolution, are witnessing the interaction of forces which have not always been acting. A prematerial state of the universe was one of unfruitful motions, that is, motions unattended by progressing changes, in our region of the ether. How extended we cannot say; the nature of the motions we know not; but the kinetic entities differed from matter in the one important particular of not possessing gravitational attraction. Such kinetic configurations we cannot consider to be matter. It was possible to construct matter by their summation or linkage as the configuration of the crystal is possible in the clear supersaturated liquid.

Duration in an ether filled with such motions would pass in a succession of mere unfruitful events; as duration, we may imagine, even now passes in parts of the ether similar to our own. An endless (it may be) succession of unprogressive, fruitless events. But at one moment in the infinite duration the requisite configuration of the elementary motions is attained; solely by the one chance disposition the stability of all must go, spreading from the fateful point.

Possibly the material segregation was confined to one part of space, the elementary motions condensing upon transformation, and so impoverishing the ether around till the action ceased. Again in the same sense as the

298

stars are simultaneous, so also they may be regarded as uniform in size, for the difference in magnitude might have been anything we please to imagine, if at the same time we ascribe sufficient distance sundering great and small. So, too;, will a dilute solution of acetate of soda build a crystal at one point, and the impoverishment of the medium checking the growth in this region, another centre will begin at the furthest extremities of the first crystal till the liquid is filled with loose feathery aggregations comparable in size with one another. In a similar way the crystallizing out of matter may have given rise, not to a uniform nebula in space, but to detached nebula, approximately of equal mass, from which ultimately were formed the stars.

That an all-knowing Being might have foretold the ultimate event at any preceding period by observing the motions

of the parts
then occurring, and reasoning as to the train of
consequences
arising from these nations, is supposable. But
considerations
arising from this involve no difficulty in
ascribing to this
prematerial train of events infinite duration. For
progress there
is none, and we can quite as easily conceive of
some part of
space where the same Infinite Intelligence,
contemplating a
similar train of unfruitful motions, finds that at
no time in the
future will the equilibrium be disturbed. But
where evolution is
progressing this is no longer conceivable, as
being contradictory
to the very idea of progressive development. In
this case
Infinite Intelligence

299

necessarily finds, as the result of his
contemplation, the
aggregation of matter, and the consequences
arising therefrom.

The negation of so primary a material property as
gravitation to
these primitive motions of (or in) the ether,
probably involves
the negation of many properties we find associated
with matter.
Possibly the quality of inertia, equally primary,
is involved

with that of gravitation, and we may suppose that these two properties so intimately associated in determining the motions of bodies in space were conferred upon the primitive motions as crystallographic attraction and rigidity are first conferred upon the solid growing from the supersaturated liquid. But in some degree less speculative is the supposition that the new order of motions involved the transformation of much energy into the form of heat vibrations; so that the newly generated matter, like the newly formed crystal, began its existence in a medium richly fed with thermal radiant energy. We may consider that the thermal conditions were such as would account for a primitive dissociation of the elements. And, again, we recall how the physicist finds his estimate of the energy involved in mere gravitational aggregation inadequate to afford explanation of past solar heat. It is supposable, on such a hypothesis as we have been dwelling on, that the entire subsequent gravitational condensation and conversion of material potential energy, dating from the first formation of matter to the stage of star formation

may be insignificant in amount compared with the conversion of
etherial energy attending the crystallizing out of
matter from
the primitive motions. And thus possibly the
conditions then
obtaining involved a progressively increasing
complexity of
material structure the genesis of the elements,
from an
infra-hydrogen possessing the simplest material
configuration,
resulting ultimately in such self-luminous nebula
as we yet see
in the heavens.

The late James Croll, in his Stellar Evolution,
finds objections
to an eternal evolution, one of which is similar
to the
"metaphysical" objection urged in this paper. His
way out of the
difficulty is in the speculation that our stellar
system
originated by the collision of two masses endowed
with relative
motion, eternal in past duration, their meeting
ushering in the
dawn of evolution. However, the state of
aggregation here
assumed, from the known laws of matter and from
analogy, calls
for explanation as probably the result of prior
diffusion, when,
of course, the difficulty is only put back, not
set at rest. Nor

do I think the primitive collision in harmony with the number of relatively stationary nebula visible in space.

The metaphysical objection is, I find, also urged by George Salmon, late Provost of Trinity College, in favour of the creation of the universe.—(_Sermons on Agnosticism_.)

A. Winchell, in _World Life_, says: "We have not

301

the slightest scientific grounds for assuming that matter existed in a certain condition from all eternity. The essential activity of the powers ascribed to it forbids the thought; for all that we know, and, indeed, as the _conclusion_ from all that we know, primal matter began its progressive changes on the morning of its existence."

Finally, in reference to the hypothesis of a unique determination of matter after eternal duration in the past, it may not be out of place to remind the reader of the complexity which modern research ascribes to the structure of the atom.

302

INDEX

A.

Abney, Sir Wm., on sensitisers, 210.

Abundance of life, numerical, 98-100.

Adaptation and aggressiveness of the organism, 80.

Additive law, the, with reference to alpha rays, 220.

Age of Earth, comparison of denudative and radioactive methods of finding, 23-29.

Aletsch glacier, 286.

Allen, Grant, on colour of Alpine plants, 104.

Allen, H. Stanley, on photo-electricity, 203.

Alpha rays, nature of, 214; velocity of, 214; effects of, on gases, 214; range of, in air, 215; visualised, 218; ionisation curve of, 216; number of, from one gram of radium, 237; number of ions made by, 237.

Alpine flowers, intensity of colour of, 102.

Alps, history of, 141; Tertiary denudation of, 148; depth of sedimentary covering of, 148; evidence of high pressures and temperatures in, 149; recent theories of formation of, 150 _et seq._; upheaval of, 147; age of, 147; volcanic phenomena attending elevation of, 147.

Andes, trough parallel to, 123; not volcanic in origin, 118.

Angle of friction on ice, 261-265, 281-283; on glass, 261-265.

Animate systems, dynamic conditions of, 67; and transfer of energy, 71; and old age, 72; mechanical imitation of, 76, 77.

Animate and inanimate systems compared, 73-75.

Appalachian range, formation of, 120.

Arrhenius, on elevation of continents, 17.

Aryan Era of India, 136.

Asteroids, probable origin of, 175; discovery of, 175; dimensions of, 176; orbits of, 176; Mars' moons derived from, 177.

B.

Babbage and Herschel, theory of mountain building, 123.

Babes (and Cornil), size of spores, 98.

Becker, G. F., age of Earth by sodium collection, 14; age of minerals by lead ratio, 20.

Berthelot, law of maximum work, 62.

Bertrand, Marcel, section of Mont Blanc Massif, 154.

Beta rays, nature of, 246; accompanied by gamma rays, 247; production of, by gamma rays, 247; as ionising agents, 249.

Biotite, containing haloes, 223; pleochroism of, 235; intensified pleochroism in halo, 235.

Body and mind, as manifestations of progressiveness of the organism, 86.

Boltwood, age of minerals by lead ratio, 20.

Bose, theory of latent image, 203.

Bragg and Kleeman, on path of the alpha ray, 215; stopping power, 219; laws affecting ionisation by alpha rays, 220; curve of ionisation and structure of the halo, 232.

Brecciendecke, sheet of the, 154.

Brdche, sheet of the, 154.

Burrard and Hayden on the Himalaya, 138; sections of the Himalaya, 139.

C.

Canals and "canali," 166; curvature of, and path of a satellite, 188 _et seq._; double and triple accounted for, 186, 187; doubling of, 195; disappearance and reappearance of, 196-198; photography of, 198; not due to cracks, 167; not due to rivers, 167; of Mars, double nature of, 166, 170; crossing dark regions of planet's surface, 168; of Mars, Lowell's views on, 168 _et

seq._; shown on Lowell's map, investigation of, 192 _et seq._;
radiating, explanation of, 193, 194; number of, 194; developed by
secondary disturbances, 194; nodal development of, due to raised
surface features, 195.

Chamberlin and Salisbury, the Laramide range, 121.

Clarke, F. W., estimate of mass of sediments, 9; age of Earth by
sodium collection, 14; average composition of sedimentary and
igneous rocks, 42; on average composition of the crust, 126;
solvent denudation of the continents, 17, 40.

Claus, protoplasm the test of the cell, 67; abortion of useless
organs, 69.

Coefficient of friction, definition of, 262; deduction of, from
angle of friction, 263; abnormal values on ice, 261-265, 282; for
various substances, 265.

Continental areas, movements of, 144.

Cornil and Babes, size of spores, 98.

Croll, James, dawn of evolution, 301.

Crust of the Earth, average composition of, 126; depth of
softening in, 128.

Curie, definition of the, 256.

D.

Dana, on mountain building, 120.

Dawson, reduction of surface represented by Laramide range, 123.

Deccan traps, 137

déferlement, theory of, 155; explanation of, 155
et seq.;
temperature involved in, 156.

Deimos, dimensions of, 177; orbit of, 577.

De Lapparent, exotic nature of the Préalpes, 150.

De Montessus and the association of earthquakes with
geosynclines, 142.

Denudation as affected by continental elevation, 17; factors
promoting, 30 _et seq._; relative activity in
mountains and on
plains, 35-40; solvent, by the sea, 40; the sodium
index of,
46-50; thickness of rock-layer removed from the
land, 51.

De Quincy, System of the Heavens, 200.

Dewar, Sir James, latent image formed at low
temperatures, 202.

Dixon, H. H., and AGnadance of Life, 60.

Double canals, formation by attraction of a
satellite, 585-187.

Douglass, A. E., observations on Mars, 167.

Dravidian Era of India, 135.

E.

Earth, early history of, 3, 4; dimensions of, relative to surface features, 117.

Earth's age determined by thickness of sediments, 5; determined by mass of the sediments, 7; determined by sodium in the ocean, 12; determined by radioactive transformations, 19; significance of, 2.

Earthquakes associated with geosynclines, 142.

Efficiency, tendency to maximum, in organisms, 113, 114.

Elements, probable wide diffusion of rare, 230; rarity of radioactive, 241.

Elster and Geitel, photo-electric activity and absorption, 207; photo-electric properties of gelatin, 212; Emanation of radium, therapeutic use of, 256-259; advantages of, in medicine, 256; volume of, 257; how obtained, 257; use of, in needles, 258.

Equilibrium amount, meaning of, 254, 255.

Evolution and acceleration of activity, 79; of the universe not eternal a pane ante, 298.

F.

Faraday and ionisation, 57.

Finality of progress a part, post, 289.

Flahault, experiments on colour of flowers, 108.

Fletcher, A. L., proportionality of thorium and uranium, 26,

G.

Galileo, discovery of Jupiter's moons, 162.

Gamma rays, nature of, 247: production of, by beta rays, 247; as ionising agents, 249.

Geddes and Thomson, hunger and living matter, 71.

Geiger, range of alpha rays in air, 215; ionisation affected by alpha rays in air, 216; on "scattering," 217; scattering and the structure of the halo, 232.

Geikie, Sir A., uniformity in geological history, 15.

Geosynclines, 119; association with earthquakes and volcanoes, 142; of the tethys, 142; radioactive heat in, due to sediments, 130; temperature effects due to lateral compression of, 131.

Glacial epoch, phenomena of, 287.

Glacier motion, cause of. 285.

Glossopteris and Gangamopteris flora, 136.

Gondwanaland, 136.

Gradient of temperature in Earth's surface crust, 126.

H.

Haimanta period of India, 135.

Halley, Edmund, finding age by saltiness of ocean, 13.

Hallwachs, photo-electric activity and absorption, 207.

Haloed, pleochroic, finding age of rocks by, 21; due to uranium and thorium families, 227; radii of, 227; over-exposed and underexposed, 228; intimate structure of, 229 _et seq._; artificial, 229; tubular, in mica, 230; extreme age of, 231; effect of nucleus on structure of, 232; inference from spherical form of, in crystals, 233; structure of, unaffected by cleavage, 235; origin of the name "pleochroic," 235; colouration due to iron, 235; colouration not due to helium, 236; age of, 236; slow formation of, 237, 238; number of rays required to build, 237; and age of the Earth, 238-241.

Hayden, H.H., geology of the Himalaya, 134, 138, 139.

Heat-tendency of the universe, 62.

Heat emission from the Earth's surface, 126; from average igneous

rock due to radioactivity, 126.

Helium and the alpha ray, 214, 222; colouration of halo not due to, 236.

Hering, E., and physiological or unconscious memory, 111.

Herschel and Babbage theory of mountain building, 123.

Herschel, Sir W., on galaxy of milky way, 293.

Hertz, negative electrification discharged by light, 204.

Himalaya, geological history of, 134-139.

Hobbs, on association of earthquakes and geosynclines, 143.

Holmes, A., original lead in minerals, 20; age of Devonian, 21.

Horst concerned in Alpine *_déferlement_*, objections to, 156.

Hyperion, dimensions of, 177.

I.

Ice, melting of, by pressure, 267 *_et seq._*; expansion of water in becoming, 267; lowering of melting-point by pressure, 267; fall of temperature under pressure, 268 *_et seq._*; viscosity of, 284.

Igneous rocks, average composition of, 43.

Inanimate actions, dynamic conditions of, 61.

Inanimate systems, secondary effects in, 63-65;
transfer of
energy into, 66.

Indian geology, equivalent nomenclature of, 139.

Initial recombination of ions due to alpha rays,
221, 222, 231;
and structure of the halo, 231.

Insect life in the higher Alps, 104, 105;
destruction of, on the
Alpine snows, 106.

Ionisation by alpha ray, density of, 221;
importance in chemical
actions, 250; in living cell, 250.

Ions, number of, produced by an alpha ray, 237.

Isostasy, 53; and preservation of continents, 53.

Ivy, inconspicuous blossoms of, 107; delay in
ripening seed,
107.

K.

Kant and Laplace, material hypothesis of, does not
account for
the past, 290.

Kelvin, Lord, experiment on effects of pressure on
ice, 268-270.

Kleeman and Bragg. See Bragg.

Klopstock introduces skating into Germany, 273.

L.

Lakes, cause of blue colour of, 55.

Land, movements of the, 53, 54.

Laukester, Ray, the soma and reproductive cells, 85.

Lapworth, structure of the Scottish Highlands, 153.

Latent heat of water, 266.

Latent image, formed at low temperatures, 202; Bose's theory of, 203; photo-electric theory of, 204, 209 _et seq._

Least action, law of, 66.

Lembert and Richards, atomic weight of lead, 27.

Length of life dependent on conditions of structural development, 93; dependent on rate of reproduction, 94.

Life-curves of organisms having different activities, 92.

Life, length of, 91.

Life waves of a cerial, 95; of AUSAEBIA, 87; of a species, 90.

Light, effects of, in discharging negative electrification, 204; chemical effects of, 205; experiment showing effect of, in discharging electrified body, 205.

Lindemann, Dr., duration of solar heat, 29.

Lowell, Percival, observations on Mars, 167 _et seq._; map of

Mars, reliability of, 198.

Lucretius, birth-time of the world, 1.

Lugeon, formation of the Préalpes, 171; sections in the Alps, 154.

Lyell, uniformity in geological history, 15.

M.

Magee, relative areas of deposition and denudation, 16.

Mars, climate of, 170; position in solar system, 174, 175;
dimensions of satellites of, 177; snow on, 169;
water on, 169;
clouds on, 169; atmosphere of, 170; melting of snow on, 170;
dimensions of canals, 171; signal on, 172; times of opposition, 164;
orbit of, 165; distance from the Earth, 165; eccentricity of his orbit, 165;
observations of, by Schiaparelli, 165, 166;
Lowell's observations on, 167 _et seq._

Maxwell, Clerk, changes made under constraints, 65; on conservation of energy, 61.

M'Connel, J. C., viscosity and rigidity of ice, 284.

Memory, physiological, 111, 112.

Metamorphism, thermal, in Alpine rocks, 132, 149

Millicurie, definition of, 256.

Molasse, accumulations of, 148.

Morin, coefficients of friction, 265.

Morphy, H., experiments on coefficient of friction of ice, 281.

Mountain-building and the geosynclines, 119-121; conditioned by radioactive energy, 125; energy for, due to gravitation, 122; reduction of surface attending, 123; depression attending, 123; instability due to thermal effects of compression, 132; igneous phenomena attending, 132; rhythmic character of, accounted for, 133; movements confined to upper crust, 122; movements due to compressive stresses in crust, 122; movements, rhythmic character of, 121.

Mountain ranges built of sedimentary materials, 118.

Müller, J., coefficient of friction of skate on ice, 265, 274.

Muth deposits of India, 135.

N.

Newton, Professor, of Yale, on origin of Mars' satellites, 177.

Nucleus, dimensions of, 237; amount of radium in, 238.

Nummulitic beds of Himalaya, 138.

O.

Ocean, amount of rock salt in, 50; cause of black colour of, 55; estimated mass of sediments in, 48; increase of bulk due to solvent denudation, 52; its saltiness due to denudation, 41.

Old age and death, 82-85; not at variance with progressive activity, 83.

Organic systems, origin of, 78.

Organic vibrations, 86 _et seq._

Organism and accelerative absorption of energy, 79; and economy, 109-111; and periodic rigour of the environment, 94, 95.

Organism and sleep, 95; ultimate explanation of rhythmic events in, 96, 97; law of action of, 68 _et seq._; periodicity of; and law of progressive activity, 82 _et seq._

P.

Penjal traps, 135.

Pepys and skating, 273.

Perry, coefficient of friction of greased surfaces, 265.

Phobos, dimensions of, 177; orbit of, 177.

Photoelectric activity and absorption, 207; persists at low

temperatures, 208, 209; not affected by solution, 213.

Photo-electric experiment, 205; sensitiveness of the hands, 207;
theory of latent image, 204, 209 _et seq._

Photographic reversal, experiments on, by Wood, 211; theory of, 210.

Piazzi, discovery of first Asteroid, 175.

Pickering, W. H., observations on Mars, 167.

Planet, slowing of axial rotation of, 189.

Plant, expectant attitude of, 109.

Pleochroic haloes, measurements of, 224; theory of, 224 _et seq._; true form of, 226; radius of, and the additive law, 225;
absence of actinium haloes, 225; see _also_ Haloes; mode of occurrence of, 223 _et seq._

Poole, J. H. J., proportionality of thorium and uranium, 26.

Poulton, uniformity of past climate, 17.

Pratt, Archdeacon, and isostasy, 53.

Préalpes, exotic nature of, 150, 151.

Prematerial universe, nature of a, 297, 300.

Prestwich and thickness of rigid crust, 128; history of the Pyrenees, 140.

Primitive organisms, interference of, 89; life-curves of, 88.

Proctor and orbits of Asteroids, 176.

Protoplasm, encystment of, 68.

Purana Era of India, 134.

Pyrenees, history of, 140.

R.

Radioactive elements concerned in mountain building, 125.

Radioactive layer, failure to account for deep-seated temperatures, 127; assumed thickness of, 128; temperature at base of, due to radioactivity, 129; in the upper crust of the Earth, 125; thickness of, 126-128.

Radioactive treatment, physical basis of, 251.

Radioactivity and heat emission from average igneous rock, 126; rarity of, established by haloes, 241, 243.

Radium, chemical nature and transmutation of, 244-245; emanation of, 245; rays from, 253, 254; table of family of, 253; period of, 253; small therapeutic value of, 254.

Radium C, therapeutic value of, 254; rays from. 254; generation of, 254.

Rationality, conditions for development of, 163.

Rays, similarity in nature of gamma, X, and light rays, 248;
effects on living cell, 251; penetration of, 251.

Reade, T. Mellard, finding age of ocean by calcium sulphate, 13.

Recumbent folds, formation of, 155 _et seq._

Regelation, 284; affecting glacier motion, 285.

Reversal, photographic, explanation of, 211.

Richards and Lemberg, atomic weight of lead, 27.

Richter, Jean Paul, Dream of the Universe, 200.

Rock salt in the ocean, amount of, 13.

Rocks, average composition of, 43; radioactive heat from, 126;
rate of solution of, 36.

Russell, I. C., river supply of sediments, 10.

Rutherford, Sir E., determination of age of minerals, 19, 20; age of rocks by haloes, 22; derivation of actinium, 226; artificial halo, 229; number of alpha rays from one gram of radium, 237.

S.

Salt range deposits of India, 134. 135.

Saltness of the ocean due to denudation, 41-46.

Salisbury (and Chamberlin), the Larimide range, 121.

Salmon, Rev. George, on creation, 301.

Satellite, velocity of, in its orbit, 191; method of finding path of, over a rotating primary, 189 _et seq._; direct and retrograde, 178; ultimate end of, 178; path of, when falling into primary, 179; effect of Mars' atmosphere on infalling satellite, 179; stability of close to primary, 180; effects of, on crust of primary, 180 _et seq._

Schiaparelli, observations on Mars, 165 166.

Schmidt, C., original depth of Alpine layer, 131-148; structure of the Alps, 152.

Schmidt, G. C., on photo-electricity, 207, 208; effect of solution on photo-electric activity, 213.

Schuchert, C., average area of N. America during geological time, 16.

Sedimentary rocks, average composition of, 43; mass of, determined by sodium index, 47.

Sedimentation a convection of energy, 133.

Sediments, average river supply of, 11; on ocean floor, mass of, 48; average thickness of, 49; precipitation of, by dissolved salts, 56-58; radioactivity of 130; radioactive heat of, influential in mountain building, 130, 131; rate of collecting,

7; determination of mass of, 8; river supply of, 10; total thickness of, 6.

Semper, energy absorption of vegetable and animal systems, 78.

Sensitisers, effects of low temperature on, 210.

Simplon, radioactive temperature in rocks of, before denudation, 132.

Skates, early forms of, 273; principles of construction of, 273 _et seq._; action of, on ice, 276; bite of, 278-280.

Skating not dependent on smoothness of ice, 260; history of, 273.

Skating only possible on very few substances, 279.

Soddy, F., on isotopes, 24.

Sodium, deficiency of, in sediments, 44; discharge of rivers, 14.

Soils, formation of, 37-39; surface area exposed in, 39.

Sollas, W. J., age of Earth by sodium in ocean, 14; thickness of sediments, 6.

Spencer, on division of protoplasm, 67.

Spores, number of molecules in, 97.

Stevenson, Dr. Walter C., and technique of radioactive treatment, 259.

Stoletow, photo-electric activity anal absorption, 207.

Stopping power of substances with reference to alpha rays, 219.

Struggle for existence, dynamic basis of, 80.

Strutt, Prof. the Hon. R. J., age of geological periods, 20;
radioactivity of zircon, 223.

Sub-Apennine series of Italy, 148.

Suess, nature of earthquakes. 143.

Survival of the fittest and the organic law, 80.

T.

Talchir boulder-bed, 136.

Temperature gradient in Earth's crust, 126.

Termier, section of the Pelvoux Massif, 254.

Tethys, early extent of, 135-137; geosynclines of, 142.

Thermal metamorphism in Alpine rocks, 132, 149.

Thomson, James, prediction of melting of ice by pressure, 267.

Thorium and uranium, proportionality of, in older rocks, 26.

Triple canals, formation of, by attraction of a satellite, 187.

Tyndall, colour of ocean water, 55.

U.

Uniformitarian view of geological history, 15-18.

Universe, simultaneity of the, 293-295.

Uranium-radium family of elements, table of, 253.

V.

Val d'Hérens, earth pillars of, 33.

Van Tillo, nature of continental rock covering, 9.

Vegetable and animal systems, relative absorption of energy of, 78.

Vegetative organs, struggle between, 105, 106.

Volcanoes and mountain ranges, 118; associated with geosynclines, 142; Oligocene and Miocene of Europe, 147.

W.

Weinschenk and thermal metamorphism, 132, 149.

Weismaun, encystment of protoplasm, 68; length of life and somatic cells, 96; origin of death, 83; tendency to early reproductiveness, 98.

Wilson, C. T. R., visualised alpha rays, 218.

Winchell, progressive changes of matter not eternal, 302.

Wood, R. W., on photographic reversal, 211.

Z.

Zircon, radioactivity of, 223; as nucleus of halo, 223.

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